Chapter 12
Congenital surgery

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Overview of congenital surgery

Key facts

Congenital cardiac defects are infinitely variable and fascinating, and provide the surgeon with continuing intellectual challenges. The surgical therapy of these defects is technically enjoyable, and the postoperative care rewarding. There is no opportunity for complacency, as each operation brings new and often unexpected demands on the surgeon's abilities.



- Congenital heart disease is one of the commonest congenital defects with an incidence of 0.8%.
- Away from the operating room and the ICU, the results of surgery on very heterogeneous groups of patients must be analysed to improve procedures and decision-making pathways.
- In order to compare surgical results on small numbers of patients, a common nomenclature and risk stratification tools are required in order to pool the small groups of patients from various units.

Anatomical nomenclature

Congenital heart disease involves abnormalities in form, spatial relationships, interconnections, and physiology. It can be classified in terms of presentation (📖 p527), physiology (📖 p527), anatomy (📖 p526), outcome (📖 p523), aim (e.g., palliative or corrective), by type of correction (anatomical or physiological), and by technical aspects (first time, redo primary correction).

Controversies

Early primary repair versus staged palliation and repair

The principle of palliating a small infant to allow them to grow enough to reduce the risk of corrective or definitive surgery has been increasingly superseded by early primary repair, where children undergo the definitive repair at the first operation, reducing the number of operations needed, and eliminating morbidity associated with palliative procedures, e.g., pulmonary artery stenosis after systemic-pulmonary artery shunt. Other advantages include normalization of volume load on the heart, elimination of cyanosis, normal organ development in infancy, reduced mortality from one operation compared to two or three. But a staged approach is used in many types of anomaly, e.g., early primary repair in cyanotic tetralogy of Fallot may be associated with stormier postoperative course.



Other controversies in congenital cardiac surgery

- Long-term mechanical support as bridge to recovery or transplantation.
- Fenestrations connecting the systemic venous circulation to the atrium in the Fontan circulation.
- Type of shunt that is optimal for use in the Norwood stage I operation.
- Cardiac MRI as a substitute for cardiac catheterization.
- Use of selective cerebral perfusion and low flow cardiopulmonary bypass instead of deep hypothermic circulatory arrest.

Risk stratification

This enables clinicians to predict outcomes and compare results between surgeons and centers. It is difficult in congenital cardiac surgery because:

- The defects treated are rare.
- Patients with similar diagnoses may be treated in a variety of ways.
- Several techniques may be used in one operation.
- A single 'repair' may require several admissions/operations.
- Large variation in age and weight.
- Confounding variables, e.g., chromosomal abnormalities, prematurity.

Basic Aristotle Complexity Score

This reflects complexity as a composite of mortality, morbidity and technical difficulty, of 145 procedures rated by 50 surgeons who were required to arrive at a consensus 'complexity score' from 3 to 14.5. If a further 248 variables are rated, both procedure-dependent and independent, a 'comprehensive' score from 3 to 25 may be derived. It is a subjective score, and was not designed or validated to predict the three main outcomes.

Risk Adjustment in Congenital Heart Surgery (RACHS-1)

A consensus panel of 11 experts subjectively rated the risk of postoperative mortality for 207 surgical procedures, arriving at 6 risk categories. This was validated against 2 datasets and found to be effective.

Morphology

Key facts

- Sequential segmental analysis is a morphological approach to the heart that allows cardiac malformations to be described in a logical and straightforward way.
- It is based on the observation that all hearts are built from three segments: the atria, the ventricular mass, and the arterial trunks, with the atrioventricular and ventriculo-arterial junctions joining them.
- Sequential segmental analysis describes these segments and their relationships to each other.

Clinical nomenclature

Congenital heart disease is normally described in the following order:

- ***Cardiac position.***
- ***AV connection (segmental notation):***
  - Atria and systemic venous drainage (visceroatrial).
  - Ventricles.
  - Great arteries.
- Chamber, valve, and vessel abnormalities including coronary arteries.
- Physiological diagnoses including arrhythmias.
- Surgical corrections and persistent problems.

Describing chambers and great vessels

In the normal heart, there are left- and right-sided versions of each segment. In abnormal hearts, segments and structures may not be in the expected position, or may be absent or duplicated, so it is necessary to recognize the chambers. When cardiac chambers are described, the terms 'right' and 'left' refer to normal ***morphological*** characteristics, e.g., the right atrium is the atrium with a wide-based appendage. When describing structures according to where they lie within the body the terms 'right-sided' and 'left-sided' are used.

Morphologically right atrium

- Broad triangular appendage.
- Terminal crest (crista terminalis).
- Extensive pectinate muscles.
- Muscular rim of fossa ovalis.
- Sinus node.

Morphologically left atrium

- Narrow tubular appendage.
- No terminal crest.
- Limited pectinate muscles—smooth internal surface.
- Flap valve of fossa ovalis.
- No sinus node.

Morphologically right ventricle

- Coarse apical trabeculation.
- Septal leaflet of the tricuspid valve attached to the septum.
- Moderator band.

Morphologically left ventricle
- Fine apical trabeculations.
- Mitral valve leaflets all attached to the papillary muscles.
- No moderator band.

Pulmonary trunk
- Bifurcates into left and right.
- No coronary arteries.

Aorta
- Arch giving rise to head and neck vessels.
- Coronary arteries.

Atrial arrangement

- Four possible atrial arrangements:
- ***Situs solitus***: usual atrial arrangement. Morphologically right atrium on the right, morphologically left atrium on the left.
- ***Situs inversus***: mirror-image arrangement. Morphologically left atrium on the right, morphologically right atrium on the left. Associated with mirror-imagery of other organs, such as liver, spleen.
- ***Right atrial isomerism***: both atrial appendages are morphologically right. Associated with isomerism of bronchi, and variable arrangement of abdominal organs, often with ***asplenia*** (heterotaxy).
- ***Left atrial isomerism***: both atrial appendages are morphologically left. Associated with isomeric left bronchi and variable arrangement of abdominal organs, often with multiple spleens (heterotaxy).

Atrioventricular connections

Atrioventricular connections may be ***biventricular*** or ***univentricular***:
- Biventricular: each atrium is connected to its own ventricle:
  - ***Concordant:*** right atrium to right ventricle, left atrium to left ventricle.
  - ***Discordant:*** right atrium connected to left ventricle, left atrium connected to right ventricle.
  - ***Ambiguous:*** in the case of atrial isomerism, e.g., right atrium to right ventricle, and right atrium to left ventricle.
- Univentricular connections: one of the ventricles has a double inlet, with both atria opening into it. The right-sided or left-sided atrioventricular connection is absent, leaving only one remaining atrioventricular connection.

Ventriculo-arterial junction

- ***Concordant***: right ventricle to pulmonary artery, left ventricle to aorta.
- ***Discordant***: right ventricle to aorta, left ventricle to pulmonary artery.
- ***Double outlet***: both great arteries, or >50% of both great arteries arise from one of the ventricles, e.g., in double outlet right ventricle: the pulmonary artery and >50% of the aortic circumference arise from the right ventricle.
- ***Single outlet***: only one arterial trunk is connected to the ventricular mass.

Classification systems

Key facts

- This physiologically and anatomically heterogeneous group of conditions is difficult to classify.
- Physiologically the lesions can also be classified according to whether they are cyanotic or not.

Morphological classification

Anomalies with normal segmental connections:

- Septal defects.
- ASDs.
- VSDs.
- AVSDs

Malformations of the atrioventricular valves:

- Over-riding of the AV junction and straddling mitral or tricuspid valves.
- Mitral stenosis.
- Ebstein malformation of the tricuspid valve.
- True 'clefts' of anterior mitral leaflet.
- Parachute mitral valve.

Malformations of the arterial valves and outflow tracts:

- Aortic valve stenosis, bicuspid aortic valve, left ventricular outflow tract obstruction.
- Aortic valve regurgitation.
- Right ventricular outflow tract obstruction.
- Tetralogy of Fallot, absent pulmonary valve syndrome.
- Pulmonary atresia with VSD.
- Pulmonary atresia with intact ventricular septum.
- Congenital pulmonary valvular insufficiency.

Anomalies with abnormal segmental connections

- Double inlet ventricle.
- Tricuspid and mitral atresia/absent right or left atrioventricular connection.
- Transposition of the great arteries/discordant ventriculo-arterial connection.
- Congenitally corrected transposition of the great arteries/ discordant atrioventricular and ventriculo-arterial connections.
- Double outlet ventricle.
- Truncus arteriosus/common arterial trunk.

Anomalies of the great vessels

- Anomalous systemic venous drainage.
- Anomalous pulmonary venous connection.
- Anomalies of the aorta:
  - Coarctation and hypoplasia.
  - Interruption.
  - Vascular rings.
  - Sinus of Valsalva aneurysm.

- Pulmonary arterial abnormalities:
  - Anomalous origin, vascular sling.
  - Persistent arterial duct.
  - Aortopulmonary window.

Anomalies of the coronary arteries

- Anomalous origin.
- Anomalous course.
- Anomalous communications.

Physiological classification

Cyanotic lesions

- Tetralogy of Fallot.
- Transposition of the great arteries.
- Congenitally corrected transposition of the great arteries.
- Tricuspid atresia/absent right atrioventricular connection.
- Total anomalous pulmonary venous connection.
- Common arterial trunk.

Obstructive lesions

- Pulmonary stenosis with intact ventricular septum.
- Pulmonary atresia with intact ventricular septum.
- Left ventricular outflow tract obstruction.
- Coarctation of the aorta/hypoplastic aortic arch.
- Interrupted aortic arch.

Left-to-right shunt defects

- ASD.
- VSD.
- Patent ductus arteriosus.
- Partial anomalous pulmonary venous connection.
- Aortopulmonary window.

Other

- Congenital mitral valve disease.
- Cor triatriatum.
- Double inlet left ventricle.
- Double outlet right ventricle.
- Coronary artery anomalies.
- Hypoplastic left heart syndrome.
- Ebstein anomaly.
- Vascular rings/slings.

Embryology of heart and lung

The highly specialized four-chambered heart develops in a complex, set sequence of events (Fig. 12.1). Embryonic development begins with fertilization between a ***spermatocyte*** and a secondary ***oocyte*** within the fallopian tube, resulting in a ***zygote***: a diploid cell with 46 chromosomes. The zygote undergoes rapid mitotic cell divisions as it passes into the uterine cavity, to produce a solid sphere of 12–16 cells to form the ***morula***. Fluid spaces start to appear and coalesce inside the morula, which is now known as a ***blastocyst***. This implants into the uterine wall by day 7. The spaces are reorganized into a larger space on the uterine wall side which will become the ***yolk sac***, and a smaller space which becomes the ***amniotic sac***, separated by a ***trilaminar disc***. The ***vascular plate*** consists of a germ layer of mesenchymal tissue on the yolk sac of the disc.

Cardiogenic mesenchyme

The cardiovascular system arises from the vascular plate. The mesodermal layer forms into blood islands which evolve into four simple tubes (two anterior and two posterior). The two anterior tubes become the ***ventral aortas*** which fuse to form the ***cardiac tube***—a single-chambered heart. The two posterior tubes become the two ***dorsal aortas***. ***Six paired arches*** communicate between ventral and dorsal aortas. The cells between the dorsal and ventral aortas eventually form lung tissue, pulmonary veins, and foregut.

Chamber formation

By 23 days the ***cardiac tube*** lies within the embryo's pericardial cavity. The tube has three cell layers: an inner structureless mass of cells known as the ***cardiac jelly***, the second layer which will give rise to the endocardium, and a third layer which gives rise to the myocardium known together as the ***cardiac mantle***. The heart tube pumps blood in two parallel circulations by week 4. The heart tube is divided into three regions: the cranial portion (***conus truncus***), the caudal portion (***sinus venosus***), and the mid portion (***bulbus cordis***).

Folding of the heart tube

As the heart grows longitudinally it bends to the right: this is called ***d-looping***. The cardiac jelly acts as a valve controlling direction of blood flow. By day 24 the cardiac jelly contains ***trabeculae***. Differential growth of the proximal ventricle causes the folded heart tube to rotate counterclockwise. The conotruncal region rotates centrally to produce the curves of the aorta and PA. Bulboventricular looping is essential in establishing normal hemodynamic patterns.

Embryology of the lung

Development begins early in week 4 of embryonic development and continues up to around 8 years old. There are four phases of development.

Embryonic phase: weeks 4–8

The tracheo-esophageal groove appears in the cranial ventral foregut by invaginating ridges of the endoderm known as tracheoesophageal folds. These unite in the midline to form the dorsal esophageal lumen and the ventral laryngotracheal primordial. The intervening tissue degenerates

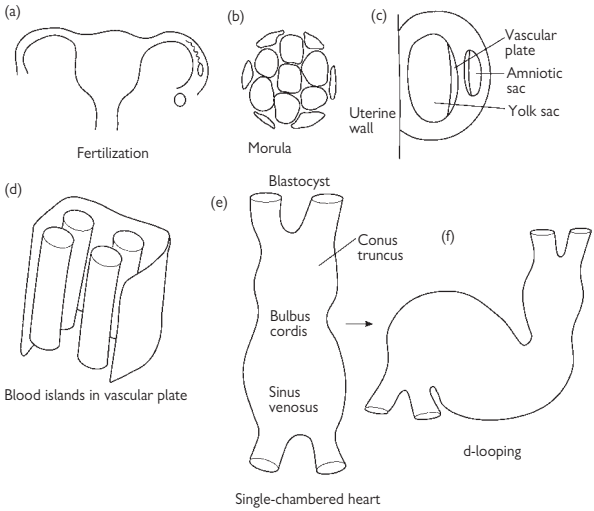


Fig. 12.1 Simple representation of the sequence of main events in the formation of the primitive heart during days 1–24 of gestation.

forming two separate tubes, the esophagus and the trachea, surrounded by mesenchyme derived from the splanchnic mesoderm. The primordial lung bud develops from the caudal end of the trachea. Shortly afterward right and left bronchial buds form. Axial growth and dichotomous division of the airways continue to produce the secondary bronchi.

Pseudoglandular phase: weeks 8–17

Bronchi continue to grow and branch forming the bronchopulmonary segments. Mesenchyme develops into vascular and stromal connective tissue elements. Cartilaginous rings develop.

Canalicular phase: weeks 17–26

Capillarization and morphogenesis of the pulmonary acini (a terminal bronchiole, 2–4 respiratory bronchioles, and 4–7 saccules per respiratory bronchiole). The distal bronchioles terminate in closely arranged buds with air spaces surrounded by abundant connective tissue. This tissue thins and elastic tissue forms around air spaces. Capillaries move towards adjacent air spaces and acinal endothelium differentiates into type I (specialized for gas exchange) and II (production of surfactant) pneumocytes.

Terminal sac phase: weeks 24 to birth

Respiratory saccule development. Respiratory bronchi develop from each terminal bronchiole and develop into respiratory saccules. These saccules become septated, the walls thin, and they mature into alveoli. After birth total alveolar surface area continues to increase by means of maturation of diverticula found on terminal bronchioles until 8 years of age.

Cardiac embryology weeks 3–5

Endocardial cushion growth and fusion

The ***endocardial and conus cushions*** are areas of mesenchymal proliferation. The endocardial cushions form a partition between the atrium and ventricle in the heart tube, to create the ***AV canals***. Proliferation of the fibrous skeleton and fusion of the cushions forms the ***mitral and tricuspid valves***. The conus cushions divide the outflow region of the ventricles from the great vessels. The ***truncus cushions*** divide the aorta from the pulmonary artery.

Interatrial septum formation

The interatrial septum forms between days 27–37. The ***sinus venosus*** consists of a transverse portion with ***two sinus horns***. Each horn receives blood from three veins: the ***umbilical*** draining the chorionic villi, the common or ***cardinal veins*** draining the cranial region of the body, and the ***vitelline veins*** draining the yolk sac. The left sinus horn is obliterated leaving the ***oblique vein*** of the LA and the ***coronary sinus***. The junction between right sinus venosus and atrium folds to form the ***right sinus valve***, which is large in the embryo and directs blood from IVC across the interatrial septum.

- The ***septum primum*** grows downwards from the atrial wall.
- The hole between the growth edge of the septum primum and the fused endocardial cushions is the ***foramen primum***.
- The growth edge fuses to the endocardial cushions.
- Apoptosis in the superior part of the septum primum forms a new opening, the ***foramen secundum***.
- The left sinus valve grows down as a ***septum secundum***.
- Fusion of the septum secundum to the endocardial cushions leaves an opening known as the ***foramen ovale***, and the superior part apoptoses, creating a one-way flap valve (Fig. 12.2).

Interventricular septum formation

The interventricular septum begins to develop around day 27. It is the final event in separating aortic and pulmonary outflow, and it occurs at the same time as aorto-pumonary septum formation. The septum is characterized by two portions, made up of three structures. The ***muscular portion*** is formed from the opposing walls of the growing ventricles which fuse, and grow up towards the fused endocardial cushions. The ***membranous portion*** is composed of the fused endocardial cushions and the descending truncus cushions.

Aorto-pulmonary septum formation

The truncus arteriosus divides into aorta and pulmonary artery from distal towards the conus cordis. The septum results from the growth of the bulbar cushions. The truncal swellings develop into pulmonary and aortic valves cusps. The intercalated swelling also contributes to the cusps of both valves.

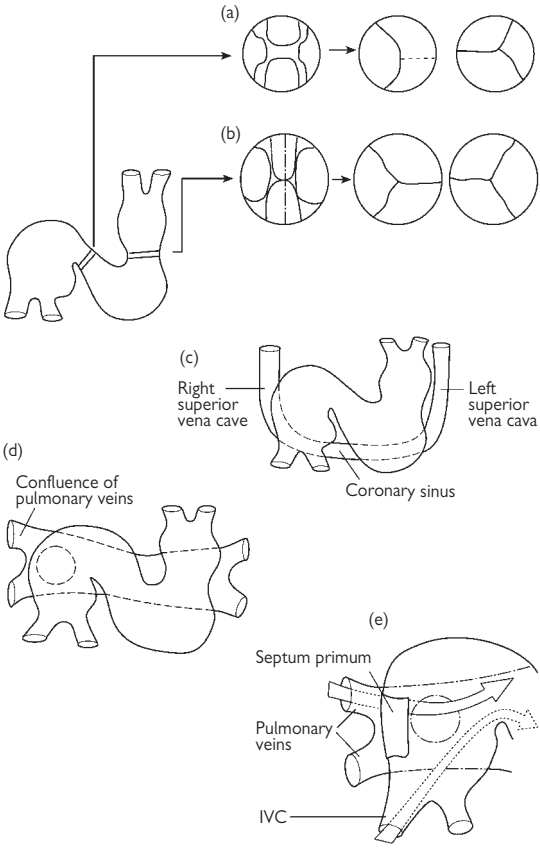


Fig. 12.2 Simple representation of the formation of the cardiac chambers and valves. (a) Endocardial cushions and formation of AV valves. (b) Conus cushions and formation of semilunar valves. (c) The left SVC disappears and the coronary sinus is absorbed into the AV groove. (d) The left atrium is formed in part from the confluence of pulmonary veins which are absorbed into the posterior wall of the heart. (e) Formation of the interatrial septum.

The fetal circulation

Key facts

- The fetal cardiac output is ~300mL/kg/min, with a heart rate of about 140bpm, and the systemic and placental circulations are in parallel.
- ***Hematopoiesis*** takes place in the yolk sac rather than bone marrow.
- ***Gas exchange*** takes place at the interface between fetus and placenta: so blood bypasses the developing lungs in the embryo.

Main differences in the fetal circulation

The main differences between fetal and postnatal vasculature arise as a result of three main factors:

Hematopoiesis

Hematopoiesis in the embryo takes place on the surface of the yolk sac. Blood enters the fetal circulation via the ***vitelline veins*** which drain into the sinus venosus. The yolk sac receives blood from the vitelline artery, which is a branch of the dorsal aorta.

Oxygenated blood supply

The embryo receives blood oxygenated at the placenta in the capillaries of the chorionic villi, via a single ***umbilical vein*** which enters the sinus venosus (📖 p530). Once the heart is partitioned into four chambers, oxygenated blood entering the right atrium from the umbilical veins is diverted by the sinus valve through the patent foramen ovale into the left atrium and hence systemic circulation, bypassing the RV (Fig. 12.3).

- About 40% of the cardiac output perfuses the placenta, returning to the fetus relatively oxygenated.
- Deoxygenated fetal blood passes to the placenta, where gas exchange occurs, via ***two umbilical arteries***, arising from the internal iliac arteries.
- After gas exchange, oxygenated blood returns to the fetus via a single ***umbilical vein***, which drains into the fetal IVC.
- The oxygen saturation in the umbilical vein is 80%, the highest found in the fetal circulation.
- About 50% of this blood supplies the ***liver***, the rest bypasses the liver via the venous duct, draining into the IVC, where it meets deoxygenated blood returning from the lower fetal extremities.
- The better-oxygenated blood streams posteriorly in the IVC, to a large extent not mixing with the deoxygenated blood.
- IVC blood enters the right atrium, where the ***eustachian valve*** directs the streams so that the posterior, leftward stream (more oxygenated) crosses the oval foramen into the left atrium. The oxygen saturation in the left atrium is 70%. This relatively oxygenated blood is ejected by the left ventricle into the ascending aorta and arch, thus supplying primarily the ***heart and brain***.
- The anterior, less-oxygenated rightward stream enters the right atrium, and thus the right ventricle, along with the coronary sinus blood and the blood returning from the upper body via the superior caval vein. RV oxygen saturation is ~50%.
- The RV ejects into the pulmonary artery, which is connected by the ***ductus arteriosus*** to the descending thoracic aorta and ***lower body***.

Only 8% of the right ventricular ejection passes to the pulmonary circulation, as the pressure of this vascular bed in the fetus is very high: the pressure in the pulmonary circulation is higher than the systemic circulation by about 1–2mmHg, and right atrial pressure >left atrial.

Proliferating tissue

Proliferating tissue (especially brain and liver) has a high O_2 demand: brain and liver see proportionately higher cardiac output with the highest O_2 saturation.

Differences in fetal hemoglobin

- 75% of fetal hemoglobin is hemoglobin F (HbF) with a higher overall hemoglobin concentration (18g/dL) than in the adult.
- HbF has lower affinity for 2,3-diphosphoglycerate (2,3-DPG).
- The oxygen dissociation curve for HbF is shifted leftward compared with that of adults, i.e., at a given PaO_2, the amount of oxygen bound to hemoglobin (O_2 saturation) is higher. This favours uptake of O_2 by the fetus at the placental interface with the mother.

Development of the venous circulation

The development of the venous system takes place via a complex system of shunts and folding, evolving from a symmetrical arrangement of paired veins to an asymmetrical arrangement that reflects visceral asymmetry.

Shunts

The brachiocephalic veins form between right and left cardinal veins. The ductus venosus forms between the left vitelline vein and the right umbilical vein. The portal vein is created by the fusion of the vitelline veins.

Involutions

- The SVC is formed from right anterior cardinal veins.
- The IVC is formed from the right common cardinal veins.
- The azygos vein is formed from the right posterior cardinal vein.
- The coronary sinus is formed from the left horn of the sinus venosus.
- The left posterior cardinal vein disappears.
- The left anterior cardinal vein disappears proximally, becomes the left first intercostal vein proximally, and contributes to the brachiocephalic.
- The umbilical veins mostly disappear into the ligamentum teres.
- The ductus venosus becomes the ligamentum venosum.

Development of the arterial system

Blood from the heart flows into paired aortic arches then the paired dorsal aortas. Tissues are supplied via segmental arteries. This symmetrical paired system evolves into the asymmetrical adult system:

- The first aortic arch becomes the maxillary artery and external carotid.
- The second aortic arch becomes the stapedial artery.
- The third aortic arch becomes the common and internal carotid.
- The fourth left arch becomes the ascending aorta and arch, and the 4th right contributes to the right subclavian.
- The fifth aortic arch involutes.
- The sixth left aortic arch becomes the left PA and the ligamentum arteriosum, the sixth right becomes the right PA.

Circulation after birth

Changes in the circulation at birth

Three main cardiopulmonary adaptations occur within a short time of birth to transfer gas exchange from the placenta to the lungs.

Change from placental to pulmonary gas exchange

- At birth the placental blood flow is interrupted. The decrease in oxygen saturation and the increase in carbon dioxide in the infant circulation triggers inspiration.
- When the lungs expand, there is a reduction in hypoxic vasoconstriction and pulmonary stretch receptors are stimulated, contributing to a dramatic fall in pulmonary vascular resistance and pulmonary arterial pressure.
- There is a resultant increase in blood flow into the pulmonary vasculature via the pulmonary arteries, and this in turn leads to a rise in the pulmonary venous return to the LA.

Closure of the foramen ovale

The increase in volume of blood returning to the LA means that LA pressure rises, forcing the flap covering the foramen ovale against the interatrial septum, finally completely separating the right and left atria.

Closure of fetal vessels

- The ductus arteriosus closes almost immediately after birth due to muscular contraction mediated by bradykinin released from the lungs after the first breath.
- The umbilical arteries close shortly after birth as a result of smooth muscle contraction triggered by the changes in oxygen tension. Fibrous proliferation results in their obliteration over the next few months.
- The umbilical vein and ductus venosus close after the umbilical arteries. The delay allows placental blood to flow into the infant circulation until the umbilical cord is divided. The ductus venosus forms the ligamentum venosum.
- The fall in blood flow through the IVC and venous duct causes the venous duct to close over the next 3–10 days.

Later changes

- Over the next few weeks, the RV wall thickness reduces, the LV wall thickness increases, and the muscle within the walls of the pulmonary arterial system reduces.
- Intimal hyperplasia obliterates any residual lumen of the ductus arteriosus.

Persistent fetal circulation

- This is a pathophysiological state that occurs when the neonate reverts to a fetal type circulation.
- It occurs in the presence of pulmonary arteriolar vasoconstriction, favouring a right-to-left shunt, and causing the oval foramen and the arterial duct to stay open.
- It may be caused by hypoxia, hypercarbia, acidosis, or hypothermia.

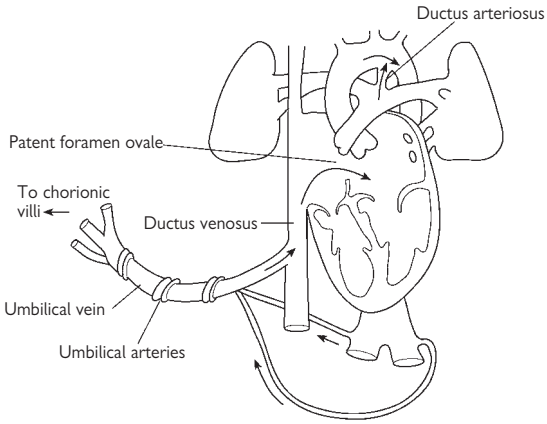


Fig. 12.3 Fetal circulation.

Preoperative management

Key facts

Surgeons, pediatric cardiologists, pediatricians, intensivists, neonatologists, anesthesiologists, and nurses are all involved in the pre-op review of these complex patients. This is usually formalized as an MDT meeting.

History and examination

By the time of admission for elective surgery the diagnosis and treatment plan are usually well established. The cardiac history and clinical findings are reviewed: change in severity may alter operative and anesthetic strategies. Non-cardiac anomalies, which may alter management, appear in up to 25% of patients with congenital cardiac disease. Acute non-cardiac pathology may affect timing of surgery.

Background

Note the primary cardiac condition, age, sex, and gestation of the child.

Symptoms

- Blue baby, pale.
- Failure to feed, failure to thrive, fatigue, poor exercise tolerance.
- Dyspnea, recurrent chest infections.
- Squatting, 'tet' spells (📖 p616), blackouts, mental obtundation.

History of presenting complaint

- Coexisting medical problems.
- Previous surgery, percutaneous procedures.
- Summary of known anomalies with relevant investigations.
  - EKG and CXR.
  - Echocardiography, cardiac catheterization, angiography, MRA.

Previous medical history

- Previous PICU/SCBU admissions and treatment.
- Meconium aspiration, pneumothorax, chest and upper respiratory tract infections, respiratory syncytial virus (RSV), asthma.
- Intestinal malrotation, herniation, atresia, necrotizing enterocolitis.
- Asplenia and athymia.
- Vitamin K deficiency, metabolic derangements.
- Exposure to infection.
- Allergies and current medication.

Signs

- Dysmorphic or syndromic features, small for age.
- Cyanosis, anemia, clubbing, mottled extremities, peripheral edema.
- Tachycardia, arrhythmia, hypertension, hypotension.
- Radio-femoral delay, reduced femoral pulses, upper limb hypertension.
- ↑JVP, hepatomegaly, ascites, added heart sounds, murmurs.
- Tachypnea, tracheal tug, wheeze, crackles.

Investigations

- Hematology: CBC, APTT, PT, (elevated in cyanosis), sickle cell.
- Biochemistry: Chem 7, BUN calcium, glucose, drug levels, urinalysis.

Preparation for surgery

Prostaglandin E1

Infants with cyanotic lesions may be dependent on a patent ductus arteriosus for survival. Prostaglandin E_1 (PGE_1) is given (5–100ng/kg/min iv) to prevent closure of the ductus. PGE_1 is a potent vasodilator, and volume resuscitation and inotropic support are frequently necessary. PGE_1 reduces the urgency of surgery, allowing more complete preoperative resuscitation. Lesions benefiting from pre-op PGE_1 infusion include:

- Lesions with inadequate pulmonary blood flow in neonates:
  - Tetralogy of Fallot.
  - Pulmonary atresia.
  - Tricuspid atresia.
- Lesions with inadequate systemic blood flow in neonates:
  - Coarctation.
  - Interrupted aortic arch.
  - Critical aortic stenosis.
  - Hypoplastic left heart syndrome.
- Lesions with inadequate mixing in neonates:
  - Transposition of the great arteries.

Ordering blood and products

- Neonates 2 units packed RBCs or fresh whole blood.
- Infants and children: 3 units packed RBCs or fresh whole blood, 2 units platelets, 2 units FFP, and 2 units cryoprecipitate.

Medication

- Preoperative cardiac medication is not given on the day of surgery.
- Vitamin K (0.25mg/kg by slow iv bolus) may be given in infants with deranged clotting.

Antibiotics

- Vancomycin 40mg/kg/24h iv on induction ***or***
- Cefotaxime 50mg/kg/24h iv on induction, 8-hourly until lines out.

Starving

- Stop solids or milk 6h, breast milk 4h, clear fluids 2h before surgery.
- Iv maintenance fluids (5% dextrose 2mL/kg/h) should be given if oral fluids are withheld >4h as cyanosis is associated with polycythemia, and dehydration can result in systemic thrombosis.

Consent

For most procedures the parents will have had the opportunity for discussing the options, risks, and benefits of surgery with the operating surgeon. Signing the consent form the night before surgery should therefore be a chance to review the important issues.

Adjuncts to surgery

Do any of the following need to be ordered in advance?

- Homograft: what type, what size?
- Synthetic conduit: what size, valved or unvalved?

Postoperative cardiac management

Key facts

- Principles of cardiovascular management after surgery for congenital heart disease are similar to those in Chapter 8, with several key differences, including the normal hemodynamic parameters (Table 12.1), dose ranges of vasoactive drugs (Table 12.2), and residual pathology.
- Management of low cardiac output and arrhythmias in pediatric population is dealt with in more detail on 📖 p542.

Hemodynamic monitoring

Monitoring

- Arterial lines are placed immediately pre-op.
- PA catheters are not used routinely because small hearts with highly abnormal anatomy make percutaneous insertion inappropriate.
- RA, LA, and PA catheters can be inserted at surgery and provide information on ventricular filling and function.
- SvO_2 and transoesophageal Doppler help assess cardiac output.
- Normal pediatric hemodynamic parameters are listed in Table 12.1.

Table 12.1 Pediatric reference table for hemodynamic variables

Age	Systolic (mmHg)	Diastolic (mmHg)	Heart rate (bpm)
Birth (<1kg)	40–60	15–35	85–205
Birth (3kg)	50–70	25–45	
Neonate (1–28 days)	60–80	20–50	
Infant (1–12 months)	80–105	50–65	100–190 (to 2 years)
Child and adolescent (1–18 years)	Median systolic BP = 90mmHg + (2 × age)		60–140 (to 10 years) 60–100 >10 years

Residual lesions

Many lesions are associated with specific cardiovascular problems postoperatively: these are dealt with in turn under the description of individual defects and operations. The following generalizations apply:

- Pulmonary hypertension and RV dysfunction, outflow tract obstruction, valvular incompetence, and intracardiac shunts, may occur postoperatively.
- Acute low-output states may result from thrombosed shunts, intracardiac shunts, valvular incompetence, or outflow tract obstruction.

Extracorporeal membrane oxygenation

Also known as extracorporeal life support (ECLS), ECMO can be used in children above 2.5kg as mechanical assistance during cardiac or pulmonary failure, for up to 30 days. It is indicated for acute, severe, reversible cardiac or respiratory failure where oxygenation and/or cardiac output cannot maintain life despite optimal conventional treatment. ECMO may

Table 12.2 Commonly used vasoactive drugs

Drug (range = micrograms/kg/min)		Comments
Epinephrine	0.01–0.1	Beta-adrenergic, first-line inotrope
	>0.1	Beta- and alpha-adrenergic, strong inotropic and vasoconstrictive effects
Dopamine	1.0–10.0	Dopaminergic inotrope, arrhythmogenic
	>10.0	Also vasoconstriction
Dobutamine	1.0–20.0	Beta-adrenergic inotropic, vasodilating, arrhythmogenic
Milrinone	0.3–0.9	Inhibits cAMP breakdown by phosphodiesterase, mild inotrope, synergy with beta-adrenergic, decreases SVR and PVR
Norepinephrine	0.01–0.5	Mainly alpha-adrenergic vasoconstriction, some beta-adrenergic inotropic effect
SNP	0.5–5.0	Smooth muscle dilator, easily titrated
Vasopressin	0.0003–0.002 units/kg/min	V1 receptor activation, potent pure vasoconstrictor

be venovenous (VV), (usually for pure respiratory failure), or venoarterial (VA), for cardiac failure or respiratory failure where there is cardiac compromise. VV ECMO does not provide cardiac support to assist the systemic circulation, and maintains pulmonary blood flow. The ECMO circuit consists of an oxygenator (silicone membrane oxygenator, hollow fibre oxygenator), a pump (centrifugal or roller pump), and a heat exchanger. There are bubble detectors, in-line blood gas and saturation monitors, and flow measurement devices.

Indications for ECMO for neonatal respiratory failure

- Congenital diaphragmatic hernia.
- Meconium aspiration syndrome.
- Primary pulmonary hypertension of the newborn.
- Group B streptococcal sepsis.
- Respiratory distress syndrome.

Venovenous ECMO cannulation

- Infants <10kg: double-lumen cannula in the internal jugular vein.
- Children >10kg: two separate cannulas to drain from femoral vein, and reinfuse into jugular vein (the reverse sometimes more efficient).

Venoarterial ECMO cannulation

Mostly cannulation is through the jugular vein and common carotid artery by cut-down in the neck. If VA ECMO is required for failure to wean from bypass, or soon after cardiac surgery, the CPB cannulation sites may be used. Venting of the left heart may be achieved through an ASD, or a vent placed through the RSPV or LAA.

Complications

- ***Mechanical***: clots or air in circuit, oxygenator failure, pump failure.
- ***Cannula***: mediastinal bleeding, carotid artery, and aortic dissection.
- ***Neurological***: thromboembolic, intracranial bleeds, seizures.
- Pneumothorax, tamponade, cardiac ischemia, overwhelming sepsis, coagulopathy, infection, renal failure, GI hemorrhage, or ischemia.

Outcomes

Current survival rate after ECMO for neonatal respiratory failure is 77%, pediatric respiratory failure 56%. eCPR is the use of ECMO to salvage patients with cardiorespiratory arrest. In children survival to hospital discharge is 40%.

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Low cardiac output

Key facts

- Low cardiac output state is when cardiac function is so poor that delivery of oxygen and substrates to the organs is impaired: the measured ***cardiac index <2L/min/m²***. In children this is associated with a elevated SVR and PVR, and can result in multiorgan failure.
- Clinical indicators include hypotension, slow capillary refill time (>2s), elevated core–peripheral temperature gradient (toe–core gap), oliguria (<2mL/kg/h), irritability or confusion, metabolic indicators, e.g., lactate >2mmol/L, BE <–2, arteriovenous O_2 difference >25% ($SaO_2 – MVO_2$).

General management of low cardiac output

Principles are similar to adults (📖 p184), i.e., optimizing preload (via measuring filling pressures and response to fluid boluses), afterload, contractility and rhythm, with continuous reassessment. Differences include:

- ***Pulmonary hypertension*** is more common cause of RV failure.
- Negative pressure ventilation (Hayek jacket) may help in RV dysfunction following Fallot repair, and in the Fontan circulation.
- Cooling is a useful adjunct to reduce O_2 consumption.
- Keep an open mind about possible residual pathology.

Low cardiac output scenarios

Bleeding

Excessive blood loss from surgical site may be due to inadequate hemostasis or clotting abnormalities, e.g., incomplete heparin reversal or rebound (📖 p116), dilution/consumption of clotting factors, platelet dysfunction (long CPB run, ECMO, sepsis, antiplatelet drugs). Bleeding may cause low cardiac output. If blood loss >10% of blood volume in 1h, or 20% in 4h, re-explore. Treat with transfusion, and correct coagulation (📖 p146). Tranexamic acid is useful during and after surgery (📖 p147). Children with cyanotic conditions, impaired coagulation, hepatic congestion, post deep hypothermic circulatory arrest, and post redo surgery are at increased risk of bleeding.

Cardiac tamponade

Hemorrhage may compress the RA and LA reducing filling and cardiac output (📖 p144). Even small collections may cause tamponade in small chests. Always exclude a pericardial collection in post-op LCOS by TEE. Worsening LCOS requires re-exploration of the chest.

Acute pulmonary hypertension

Patients with large, chronic left-to-right shunts frequently have hyperreactive pulmonary vasculature with a tendency to paroxysmal acute pulmonary hypertensive episodes characterized by rapid desaturation, arrhythmias, and hypotension.

- Triggers include hypoxia, hypercarbia, sympathetic activation, e.g., pain.
- Manually hyperventilate with 100% O_2.
- Give $NaHCO_3$ to induce an alkalosis which will reduce PVR.
- Sedate with opioids (fentanyl 2–5 micrograms/kg/h) and paralyze.
- Pulmonary vasodilators: NTG, PGE_1, milrinone, and inhaled nitric oxide.

Arrhythmias

- Occur in 10–20% of patients postoperatively.
- The aim is to restore a physiological heart rate and AV synchrony. Temporary epicardial pacing wires are placed on the RA and RV intraoperatively. When tachyarrhythmias are difficult to control and cause significant deterioration in cardiac output, a period of ***mechanical support*** (📖 p539) may be required.
- ***Junctional ectopic tachycardia***/His bundle tachycardia: a malignant arrhythmia originating near the AV node, associated with significant morbidity and increased ICU length of stay. Most commonly requires treatment postoperatively. Loss of AV synchrony leads to further deterioration in cardiac output.
  - Surface cooling to 34–35°C to lower metabolic rate and O_2 demand.
  - Intravenous amiodarone (25 micrograms/kg/min for 4h, then 5–15 micrograms/kg/min). Overdrive pacing.
- ***Atrial ectopic tachycardia***.
- ***Ventricular tachycardia*** (rare, associated with ALCAPA repair, Fallot repair, LVOTO relief):
  - DC cardioversion.
  - Amiodarone, lidocaine, Mg^{2+} if no hemodynamic compromise.
- ***Atrial flutter*** (rare), associated with RA dilatation, e.g., large ASD.
  - DC cardioversion. Overdrive pacing.
- ***AV node dysfunction***: occurs in 2–3% of congenital cardiac surgery, associated with LVOTO relief, tricuspid surgery, VSD closure, Fallot repair, in ccTGA. 70% recover conduction, by day 9 postoperatively.
  - Permanent pacemaker should be planned after this period.
- ***Sinus node dysfunction***: usually transient, associated with sinus venosus ASD repair, Mustard or Senning operations.
  - Atrial pacing resolves the bradycardia.

Management of life-threatening arrhythmias

Life-threatening arrhythmias in children are usually secondary to low cardiac output or respiratory compromise, not the primary event.

Bradycardia, asystole, pulseless EMD

- Bradycardia (<80bpm in infants, <60bpm in children) is a life-threatening arrhythmia, and is treated in the same way as asystole.
- Secure the airway and hyperventilate with 100% O_2.
- Single hand/two-finger cardiac massage at 80–100/min.
- Pacing at 100bpm if epicardial leads are present.
- Epinephrine (0.01mg/kg iv or 0.1mL/kg of 1:10,000 solution).
- Atropine 0.02mg/kg (maximum dose 0.5mg for child).

Ventricular fibrillation, pulseless ventricular tachycardia

- Defibrillate with 2J/kg once, then 4J/kg twice.
- Cardiac massage and ventilate as for bradycardia.
- Epinephrine (0.01mg/kg iv or 0.1mL/kg of 1:10,000 solution).
- Repeat defibrillation with 4J/kg after every 60s.
- Look for and treat airway compromise, tension pneumothorax, cardiac tamponade, metabolic derangements.

Postoperative fluid management

Key facts

- The inflammatory response to CPB causes fluid shifts and capillary leak. Neonates and infants are particularly at risk. Initial postoperative fluid intake from all sources is restricted to 50% of daily requirements.
- Pediatric patients are generally filtered on pump to try and avoid arriving on ICU in a state of fluid and salt overload.

Pediatric daily fluid and electrolyte requirements

- Fluids:
  - <10kg: 100mL/kg/day.
  - 10–20kg: 1000mL + (50mL/kg over 10kg).
  - >20kg: 1500mL + (20mL/kg over 20kg).

Target serum electrolytes

- Sodium: 135–140mmol/L.
- Potassium: 3.5–4.5mmol/L.
- Magnesium: >0.8mmol/L.

Types of fluid

- ***Blood products*** (including human albumin solution) should only be used for clearly defined indications. Cyanotic patients require a higher hemoglobin.
- ***Colloid solutions***: although there is little evidence to support the use of colloids over crystalloids, colloids may be given if there is a risk of severe capillary leak.
- ***Crystalloids***: higher volumes of crystalloids may be required to maintain plasma volume expansion.

Renal management

Oliguria

Oliguria is a urine output <1mL/kg/h. Management follows the same principles as outlined in adults (📖 p234), with the following differences:

- Furosemide is given at a dose of 0.5–1mg/kg iv, to a maximum of 3mg/kg.
- Mannitol is given at a dose of 0.5g/kg iv over 20min.

Renal failure

The management of renal failure follows the same principles as that outlined in 📖 pp236–237 for adults. Treatment of hyperkalemia and metabolic acidosis differs only in the doses of drugs used:

- $NaHCO_3$ mEq/kg iv over 15min, or 0.3 x wt (kg) x base deficit.
- Calcium gluconate 20mg/kg iv over 5min (0.2mL/kg of a 10% solution).
- 25% glucose 1–2g/kg iv (4–8mL/kg) followed by insulin 0.3U/kg iv.
- Hyperventilation to induce respiratory alkalosis.
- Dialysis (📖 pp236–237). In neonates peritoneal dialysis is efficient and readily available.

Metabolic management

Cardiac arrest can be caused by hypoglycemia, hypocalcemia, and hypomagnesemia in pediatric patients.

Hypoglycemia

Hypoglycemia is a BM <4.0 in neonates, or <6.0 in children.

- Give 0.25–0.5 g/kg (1–2mL/kg) 25% dextrose iv over 5min, and then maintenance 10% dextrose in 0.25% normal saline, with hourly blood glucose measurements.

Hpocalcemia

Hpocalcemia is an ionized calcium <1mmol/L.

- Give calcium gluconate (10mg/kg or 0.1mL/kg of a 10% solution iv slowly) or equivalent dose of calcium chloride.

Hypomagnesemia

Hypomagnesemia is a serum Mg <1.5mEq/L.

- Give magnesium sulphate 1mL/kg iv.

Bleeding and transfusion

Normal bleeding post operatively should be no more than 1mL/kg/h. Hemoglobin should be maintained above 9.0g/dL in patients with continued cyanotic lesions, and at above 8.0 g/dL with no cyanosis. In neonates pericardial collections as small as 10mL may cause hemodynamic compromise. Re-exploration should be considered for bleeding rates >5.0mL/kg/h. Children with cyanotic conditions, impaired coagulation, hepatic congestion, post-deep hypothermic circulatory arrest, and post redo-surgery are at increased risk of bleeding. Within the parameters listed, the principles of managing mediastinal bleeding are similar to those described on 📖 pp144–145 with the following differences:

- Administration of product is modified according to coagulation screens, including TEG.
- Transfusion of RBCs is either matched to the previous hour's mediastinal drainage, or guided by the difference between the actual and the desired hematocrit (δHCT) according to the formula:

  ML packed cells transfused = wt (kg) x δHCT

- FFP is given at a dose of 10mL/kg.
- Platelets are given at a dose of 0.2U/kg.
- Cryoprecipitate is given at a dose of 1U/kg.
- Desmopressin is given at a dose of 0.3 micrograms/kg iv in 50mL normal saline.
- Antifibrinolytics may be given (📖 p147).
- All products should be warmed before transfusion.
- Ionized calcium levels must be corrected: calcium gluconate 20mg/kg iv over 5min (0.2mL/kg of a 10% solution).

Postoperative considerations

Key facts

- Respiratory, cardiovascular, renal, and bleeding complications are outlined on 📖 pp542, 545 and 549.
- Most of the postoperative complications described in Chapters 5 and 6 occur after congenital heart surgery. The complications described here have certain features unique to congenital heart surgery.

Cyanosis

Cyanosis is blue discoloration of the skin due to increased amount of deoxygenated hemoglobin. Cyanosis may occur despite relatively high oxygen saturations in children with polycythemia: conversely in anemia, cyanosis may not be evident until oxygen saturations are extremely low.

- ***Peripheral cyanosis*** is limited to fingers and toes: arterial saturation is normal but sluggish flow through capillaries results in high oxygen extraction. Causes include:
  - Hypothermia.
  - Vasoconstrictors.
  - Low cardiac output.
- ***Central cyanosis*** includes fingers, toes, and mucous membranes. There are two categories:
  - ***Cardiac:*** caused by intracardiac right-to-left shunt (pulmonary hypertension, RVOT obstruction, aortopulmonary shunt occlusion), and usually not improved by ↑FiO_2.
  - ***Pulmonary:*** caused by inadequate oxygenation (usually VQ mismatch: pulmonary edema, atelectasis, pneumonia, PE), and improves with ↑FiO_2.

Shunt occlusion

This may be complete or partial. It may occur as a result of acute thrombosis. Occasionally a shunt will have no blood flow despite being patent: a non-functioning shunt. Signs include:

- Cardiopulmonary collapse, cyanosis.
- Decreased central and peripheral oxygen saturations.
- Loss of shunt murmur.
- TTE, TEE, or cardiac catheterization is diagnostic.
- Treatment includes:
  - Immediate iv heparinization which may reverse shunt thrombosis.
  - Cardiac catheterization with a view to percutaneous reopening.
  - Redo surgery may be necessary.

Chylothorax

This is seen as a result of extensive dissection in the region of the thoracic duct in the posterior mediastinum in, for example, formation or take-down of Blalock–Taussig shunts or coarctation of aorta repair. Diagnosis and management are discussed on 📖 p737, although intercostal drain technique differs in infants:

- Because there is less soft tissue, drains are often tunnelled to avoid pneumothorax on removal.

- The largest drains range from 12F in premature infants to 16F in infancy. The liver and spleen are relatively higher relative to the costal margin in children than they are in adults.
- Remove drain when drainage <2mL/kg/24h.
- Medium chain triglyceride (MCT) diet for 6–8 weeks, diuretics.
- Remove with weekly radiograph or US after finishing MCT diet.

Neonatal sepsis

Many infants have high temperatures postoperatively: this may be from an inflammatory response, atelectasis, or immature thermoregulation. Fever increases cardiac output requirements as a result of increased metabolic demand:

- Thermoregulation is an important feature of pediatric care:
  - Paracetamol 10–20mg/kg/dose reduces pyrexia.
  - Cooling blanket or surface cooling with tepid water may be used.
  - Glucocorticoids also reduce pyrexia.
- In the neonate signs if sepsis are subtle, and in addition to the usual signs described in adults (📖 p239) include:
  - Lethargy, agitation, seizures.
  - Apneic spells, tachypnea, tachycardia.
  - Decreased as well as increased temperature.
  - Persistent jaundice.
- Late features suggesting septic shock include:
  - ↓GCS.
  - Hypotension.
  - Reduced filling pressures.
  - Fall in cardiac output.
  - Decreasing urine output.
  - Coagulopathy.
- The following sites should be routinely cultured if sepsis is suspected (temperature after postoperative day 3–5):
  - Indwelling cannulas.
  - Umbilicus/umbilical lines (common cause).
  - Blood, urine, and stool.
  - Wounds.
  - Drain fluid.
  - Nasopharyngeal, CSF.

Nutrition and gut

The aim of ICU nutrition is to promote anabolism and growth and reduce the effects of catabolism and hypermetabolism due to stress. Feeds should commence as soon as the patient is hemodynamically stable, preferably enterally. Many children are nutritionally impaired before surgery. Enteral feeding should start within 12h of PICU admission. Contraindications to enteral feeding include:

- Necrotizing enterocolitis.
- Severe low cardiac output.
- Reduced splanchnic perfusion (e.g., coarctation of the aorta).
- Parenteral feeding is only used if enteral contraindicated or not tolerated, as it is expensive, requires central venous access, increases risk of line sepsis, thrombosis, gut mucosal atrophy, biochemical abnormalities.

Postoperative respiratory management

Key facts

- Normal respiratory parameters are different in infants and adults; and in all patients with cyanotic defects with persistent shunts (see Box 12.1).
- Choice of anesthetic agent and ventilatory management reflects the presence of shunts: changes in PVR and SVR affect oxygenation.
- Infants are more prone to both barotrauma and atelectasis (below).
- Hypoxic, pulmonary hypertensive crises are more common in patients with pulmonary hypertension.
- Nasal intubation is used routinely in infants (see below).
- Small infant ET tubes are more prone to obstruction (see opposite).
- After extubation, children commonly require a further period of non-invasive positive pressure support. Nasal CPAP may be delivered to neonates and infants via nasal cannulas or short nasopharyngeal tubes (short-tube CPAP). Older patients may have CPAP via face mask.

Modes of ventilation

Ventilatory management is about treading a fine line between barotrauma and atelectasis. Basic ventilator settings for SIMV (📖 p158) on admission to ICU are outlined in Box 12.1.

- Postoperatively, the most common ventilator strategy in children is positive pressure ventilation via an ET tube.
- Neonates and infants, who are susceptible to barotrauma, may be managed on pressure-limited ventilators which provide an inspiratory volume determined by a preset peak airway pressure.

PEEP in children

- PEEP opens collapsed alveoli, increasing functional and residual capacity, compliance, and PaO_2. PEEP can be used to treat pulmonary shunting. In certain circumstances PEEP may over-distend alveoli, decreasing compliance: the beneficial effects are then outweighed by the negative effects.
- Pulmonary vascular resistance is increased.
- Shunting of blood to underventilated alveoli is increased and PaO_2 falls.
- Cardiac output may be decreased by the fall in venous return.
- Pneumothoraces may result from barotrauma.

Box 12.1 Normal pediatric respiratory parameters

- ***Normal respiratory rates:*** neonates, 40–60rpm. 1–12 months, 30 rpm. 1–2 years, 26rpm. 2–6 years, 22–24rpm. 6–14 years, 20–18 rpm.
- ***ET tube*** is estimated by the formula (age + 16)/4 = tube size in mm. Size range: 2.5–3.0 uncuffed in 1kg, to 6.5–7.0 cuffed in 16-year-olds.
- ***Ventilator settings:*** initial FiO_2 = 1.0 reducing to 0.40, tidal volume = 6–8mL/kg, peak pressure <20cm H_2O, PEEP = 2–5cm H_2O, rate (see earlier), I/E ratio = 1:2 in children, 1:1 in neonates and infants.
- Normal pO_2 is >10.0kPa (75mmHg) in total physiological correction, and pO_2 6–8kPa (45–60mmHg) with oxygen saturations of 75–85% in cyanotic defects with persistent shunts.

Weaning

Children undergoing simple repairs such as PDA ligation or ASD closure may be extubated in the OR. Pre-term infants, children with large left-to right shunts, pulmonary hypertension, or complex cardiac repairs are usually weaned over several days using SIMV, progressing to CPAP in older but not younger children. Weaning is slower than in adults as small changes in ventilating volumes result in large changes in blood gases, and small reductions in PEEP or CPAP can result in massive atelectasis. Criteria are:

- Adequate level of consciousness, hemodynamic stability, normothermia.
- Mediastinal bleeding <1mL/kg/h.
- PaO_2 >10.0kPa (6.0kPa in cyanotic lesions) on FiO_2 0.40.
- $PaCO_2$ <6.0kPa.
- pH 7.35–7.45.
- Vital capacity >10–15mL/kg.

Extubation

- Stomach contents are emptied by aspiration of the NG tube.
- 0.5–2mL saline are instilled, followed by 2min of manual ventilation.
- ET suctioning and suctioning of the oropharynx.
- Ventilation and removal of the ET tube during peak inspiration.
- Repeat suctioning of the oropharynx.
- Steroids may be required to reduce tracheobronchial edema and stridor after traumatic or prolonged intubation.

Respiratory failure

In addition to ↓PaO_2, signs include cyanosis, hypotension, ↓↑BP, agitation, tachypnea, intercostal and subcostal retraction, and nasal flaring.
Causes of respiratory failure on the ventilator include:

- ET misplacement, obstruction, or kinking.
- The distal lumen of smaller (3.5–5mm) ET tubes may become obstructed by dried blood or secretions: this should be prevented by regular saline irrigation and suctioning.
- Ventilator malfunction.
- Increased oxygen demand, e.g., pain, agitation
- Pulmonary problems such as pneumothorax, hemothorax, chylothorax (📖 p737), atelectasis, pulmonary edema, aspiration, pneumonia.
- Pulmonary hypertension, shunt occlusion, low cardiac output.
- Phrenic nerve palsy, abdominal distension.

Management of acute respiratory failure

- Manually ventilate with 100% O_2.
- Auscultate for bilateral breath sounds: if there is doubt that the ET tube is patent or in the correct position it may be necessary to remove it and ventilate using a mask and oropharyngeal airway.
- Irrigate the ET tube with 1mL saline and suction.
- Get an immediate portable CXR.

Principles of bypass in children

Key fact

Children are not small adults.

Differences between children and adults

- ***Molecular and cellular***: immature myocytes and immature calcium metabolism lead to higher intracellular calcium concentrations after ischemia/reperfusion; this in turn leads to decreased levels of ATP, and also to abnormal activation of various enzyme systems causing cellular damage.
- ***Metabolic***: the immature myocardium can use anerobic glycolysis more efficiently than mature myocardium, making it more resistant to ischaemic injury.
- ***Physiological***: smaller circulating volume, thus more hemodilution by priming the circuit; increased edema, coagulopathy, transfusion requirements. Higher oxygen consumption rates of neonates and infants leads to higher flow requirements on bypass (Table 12.3).
- ***Pathology***: intra- and extracardiac shunts and more reactive pulmonary vascular bed lead to differences in bypass strategy and the response to bypass, presence of cyanosis.

Bypass circuit and techniques

- ***Prime***: composition varies and subject of debate and strong opinions.
  - Aim for hematocrit 20–35% depending on temperature.
  - Colloidal solutions: human albumin, FFP, synthetic colloids; avoid glucose.
  - Mannitol and bicarbonate.
  - Keep volume small (1m of 1/4 inch tubing = 32mL).
- ***Venous reservoirs:*** two main types:
  - Hard-shell with air–blood interface; allows vacuum assistance and easily handles entrained air.
  - Soft-shell without air–blood interface; may be associated with less bleeding and inflammatory response.
- ***Pumps***: either roller pumps or centrifugal pumps.
- ***Oxygenators***: hollow-fibre membrane oxygenators are used most commonly for routine bypass cases. Silastic oxygenator is used where longer-term perfusion is required, i.e., for ECMO.

Effects of CPB in children

Many effects are similar to those described in adults (📖 pp106–107).

- ***Inflammatory response***: this is more pronounced in children, leading to increased capillary permeability and interstitial edema, pulmonary dysfunction, and renal impairment. The inflammatory response is inversely proportional to the patient's age.
- ***Disordered glucose metabolism***: hyperglycemia may be implicated in adverse neurological outcome after bypass; hypoglycemia is a common complication due to decreased glycogen stores in small children/infants, and impaired hepatic function.

Table 12.3 Flow rates on cardiopulmonary bypass in children

Patient group	Pump flows
Neonates	120–200mL/kg/min
Infants <10kg	100–150mL/kg/min
Children	80–120mL/kg/min
Adults	2.4L/min/m^2

- ***Hematological effects***: platelet activation, tissue factor release, and thrombin release; this may lead to positive feedback activation of the coagulation and inflammatory cascades.
- ***Stress response***: catecholamine, cortisol, growth hormone, ACTH, TSH, endorphin release. ↓TSH up to 48h after bypass.
- ***Cardiac effects***: improved tolerance to ischemia.
- ***Neurological effects***: central injury after routine CPB is uncommon, but risk is increased if DHCA used, with up to 25% of infants manifesting some sign of CNS injury, e.g., seizures, changed tone, motor disorders, choreoathetosis. Higher intracellular calcium may promote generation of free radicals and thus cell injury/death.
- ***Lung injury***: leukocyte and complement activation cause an inflammatory response, while mechanical effects may lead to atelectasis and surfactant loss; these effects lead to lung edema and increased pulmonary vascular resistance.
- ***Renal effects***: 8% have acute renal insufficiency, defined as oliguria and increased serum creatinine.

Moderating the response to bypass in children

- ***Corticosteroids***: iv methylprednisolone pre- and postoperatively may reduce the inflammatory response to bypass—there is some evidence of reduced post-CPB edema, lung injury, and ICU stay.
- ***Serine protease inhibitors***: may reduce postoperative bleeding and inflammation; in children there is no association between use and increased rates of renal failure, neurological complications, or mortality.
- ***Leukocyte depleted blood***: may reduce the immunomodulatory effects of graft-vs.-host reactions of leukocytes in transfusion products.
- ***Leukocyte filters***: leukocyte filtration during bypass may reduce the inflammatory response post-bypass.
- ***Biocompatible circuits and oxygenators***: some evidence of improved postoperative outcomes with heparin-coated circuits, but these results have not been widely reproduced.
- ***Miniaturized circuits***: reduced length and diameter of tubing, pump designed to allow pump-heads to be closer to operating table; the aim is to reduce prime volume and thus hemodilution.
- ***Vacuum-assisted drainage***: this allows further miniaturization, as smaller diameter tubing may be used.
- ***Modified ultrafiltration (MUF)***: see 🕮 p553. Studies show decreased postoperative edema, intubation times, postoperative bleeding and pleural effusions, and increased ventricular systolic function, cerebral blood flow, pulmonary function; overall shorter hospital stay.

Conduct of bypass in children

Key facts

Anticoagulation: heparin at 3mg/kg, maintain ACT >450s. Reversed with protamine at 1.5mg per 1mg of circulating heparin.

Blood gas management: alpha stat versus pH stat

With body cooling, gas solubility increases, with lower $PaCO_2$, higher pH and leftwards shift of the oxygen-dissociation curve (📖 p152). During hypothermia or deep hypothermic circulatory arrest, blood gas measurements can either be corrected to patient temperature (***pH stat***) or not (***alpha stat***). pH stat is increasingly seen as beneficial in DHCA.

- With pH stat management, the pH is maintained at 7.4 by manipulating $PaCO_2$ to counteract some of the changes occurring due to cooling (Table 12.4).
- About 50% of centers use the pH stat strategy. In many centres, for DHCA, pH stat is used during cooling, converting to alpha stat during the arrest period and rewarming.

Table 12.4 Differences between pH stat and alpha stat

pH stat	Alpha stat
Temperature-corrected acid–base management	Non-temperature corrected acid–base management
CO_2 added to maintain temperature corrected pH at 7.4 and $PaCO_2$ at 40mmHg	pH allowed to drift up with cooling. pH neutral only at normothermia
Cerebral perfusion pressure autoregulation lost, cerebral blood flow becomes pressure dependent. Excess cerebral blood flow allows more uniform cooling but may increase microemboli in cerebral circulation	Cerebral perfusion pressure autoregulation maintained, even at lower pressures
Lower intracellular pH suppresses cellular function and metabolism—may delay cerebral metabolic recovery when rewarming, but may also reduce reperfusion injury	More normal transmembrane pH gradients, protein and enzyme functions, leading to better metabolic recovery
Leftward shift of O_2 dissociation curve counteracted—more O_2 uptake by tissues during cooling when brain still relatively warm	
Animal studies show better cerebral protection during DHCA, and there is evidence of improved short-term outcomes in infants undergoing DHCA	

Myocardial preservation

As in adults this is usually achieved by initiating diastolic arrest using high concentrations of potassium, with hypothermia and ventricular decompression. Neonatal hearts are more resistant to hypoxia, but less able to cope with increased myocardial edema from repeated doses of crystalloid cardioplegia, thus operations may be performed with a single dose of

crystalloid cardioplegia. 30mL/kg is a standard dose of crystalloid cardioplegia, usually delivered into the aortic root or directly into the coronary ostia. There is no evidence that blood cardioplegia is superior in infants and neonates; theoretical advantages are:

- Better oxygenation of myocytes during arrest.
- Improved buffering.
- Reduced risk of reperfusion damage.

Other adjuncts and strategies

Induced ventricular fibrillation without aortic cross-clamping may be used in some circumstances: care must be taken not to allow distension of the heart (e.g., if the aortic valve is incompetent, or there are extracardiac shunts, e.g., a PDA).

Neuroprotection

Deep hypothermic circulatory arrest (DHCA): See also 📖 p124. At 18°C, metabolic activity is reduced sufficiently to allow circulatory arrest for 40min without organ damage; it is thought that children may be protected for longer periods at equivalent temperatures. For DHCA, bypass with cooling should continue for at least 20min before arresting the circulation, perhaps longer in cyanotic patients with extensive aortopulmonary collaterals, to ensure uniform cooling.

Alternatives to DHCA

- ***Low-flow CPB***: low temperatures reduce cerebral metabolic activity, allowing periods of low flow. Alternatively, DHCA may be interrupted intermittently with periods of bypass, even at low flow, to improve cerebral recovery.
- ***Selective antegrade cerebral perfusion***: when DHCA is utilized for surgery on the aortic arch, selective antegrade cerebral perfusion may be achieved with direct cannulation of the carotid artery, cannulation of the innominate artery via a Gore-tex tube, or by advancing the aortic cannula up into the innominate artery. One-third to one-half of estimated full flow should maintain adequate cerebral blood flow, and often maintains a satisfactory femoral pressure as well (📖 p125).

Other aspects of bypass

- Mechanical ventilation is usually discontinued during bypass, and should be restarted as soon as possible during rewarming to allow time to reverse atelectasis, high pulmonary vascular resistance, and accumulated extravascular fluid, which are more problematic in children than adults
- ***Ultrafiltration***: two types:
  - Conventional ultrafiltration: a filter within the bypass circuit filter blood throughout bypass or during rewarming, thus removing excess water from the circuit.
  - ***Modified ultrafiltration:*** the circuit is modified slightly (or not at all) to allow blood to be removed from the arterial cannula at the end of bypass, filtered, and returned via the venous cannula. This can take 10–15min, or continued to a target hematocrit. It is used in small children and neonates.
- ***Hemodilution***: allows the use of hypothermia by reducing blood viscosity; minimizes exposure to donor blood.

Basic operative technique

Opening

- Spend time ensuring hemostasis at all stages: small amounts of blood loss have a much larger impact in children.
  - Arterial collaterals are enlarged in coarctation of the aorta.
  - In any cyanotic condition venous collaterals are prominent; they may require ligation rather than cautery.
- There is much less soft tissue, bone is less calcified, and cardiac structures and abdominal viscera may be disproportionately enlarged:
  - Set the cautery at half what you would use in adults.
  - Avoid subxiphisternal dissection prior to sternotomy.
- The thymus is much more prominent: a subtotal thymectomy improves exposure, and may facilitate subsequent resternotomy.
- If pericardium is needed for reconstruction, open the pericardium to one side so that you can harvest a large patch: some surgeons like this to be placed in 0.6% glutaraldehyde, fixing it, to reduce later shrinkage.
- Some surgeons open both pleura, to reduce the risk of postoperative cardiac tamponade by allowing mediastinal blood to enter the pleurae.

Cannulation and venting

- Cannulation must provide adequate flow, unobstructed venous drainage, perfusion of all organs, unobstructed operative field.
- Arterial: in small patients place longitudinal aortic purse-string sutures to avoid stenosis.
- Venous: usually bicaval venous cannulation; if circulatory arrest planned then atrial cannulation.
- Venting: decompressing the left heart during bypass; vent should be introduced after clamping the aorta to minimize the risk of entraining air into the left side of the heart.
  - RSPV/through the atrial septum—facile way to decompress the LA.
  - Pulmonary artery—decompresses both sides.
  - LV apex—rarely used.

Bypass

Pediatric heart surgery presents challenging problems to perfusionists.

- Circulating volume is much less:
  - Priming volume is kept to a minimum to avoid hemodilution.
  - Blood may be used to prime the circuit.
  - Changes in even small volumes can have profound consequences.
  - Albumin and FFP may be added to prime to counteract the reduction in colloid osmotic pressure and fibrinogen concentrations that results from hemodilution.
- In infants a larger percentage of venous return is via the SVC.
- Pediatric patients are often cooled to lower temperatures than adults: the brain receives 1/3 of cardiac output in children (much higher than in adults) and procedures are more complex and more likely to need circulatory arrest.
- Termination of bypass is challenging because of anatomic and physiological defects, as well as the small circulating volume.

Blood conservation

- Meticulous hemostasis is key. Tranexamic acid is increasingly used, and despite withdrawal from adult market aprotinin is also used in some countries as evidence of adverse sequelae in children not established.
- Cell salvage is routinely used in all patients.

Closure

Once again, meticulous hemostasis is key:

- If re-sternotomy is anticipated the pericardium is closed with 3/0 Prolene with an interrupted suture. A pericardial substitute, e.g., Gore-tex membrane, is used in the absence of pericardium.
- Pacing wires should be placed routinely in all except the simplest cardiac repairs and extracardiac defects as dysrhythmias are common.
- Extracardiac pressure lines may be placed in the RA, LA, and occasionally PA (which requires pledgetted purse-string):
  - The sternum is closed with wire, Ethibond, or Vicryl: sutures may be less likely then sternal wires to 'cheese-wire' through the sternum. Steel wires carry lower risk of stitch sinuses.
- In sick and unstable infants chest closure may cause unacceptable hemodynamic and respiratory compromise: it is not unusual in these cases to leave the chest open with a Bogota bag (📖 p151).

Special circumstances

Shunts

Shunts must be identified and mobilized pre-bypass, and controlled once on bypass in order to avoid massive run-off into the pulmonary circulation during bypass. Similarly, during repair of common arterial trunk, the pulmonary arteries must be controlled.

Persistent left superior vena cava (📖 pp121, 537)

These are present in perhaps 2% of congenital cardiac cases, and are usually identified on preoperative imaging. In adults they may not be spotted until the intraoperative TEE (classic features include an enormous coronary sinus on TEE, and the surgeon may find that the innominate vein is absent). In adults they can be ignored (it is not possible to give retrograde cardioplegia). In children they may be snared:

- If there is an innominate vein draining the LSVC into the RSVC, the LSVC may be snared without the need for cannulation.
- If there is no innominate vein, and the pressure in the LSVC on snaring it is >20mmHg, the LSVC must be cannulated to drain it. It may be cannulated directly to ensure a blood-free operative field, or may be drained via a pump sucker in the coronary sinus.

Redo surgery for congenital heart disease

It is vitally important to have a clear idea of what previous surgery has been carried out. Preoperatively identify the details of:

- Previous incisions and whether the pericardium was closed.
- Previous shunts and conduit, previous cannulation sites.
- Whether shunts and conduits are patent: exposing and previous shunts is an important part of redo surgery.
- Groin or neck (<2 years) should be prepared for peripheral cannulation.

Materials

Key facts

- Congenital cardiac surgery involves much surgical reconstruction, for which additional material may be required.
- The advantages and disadvantages of the available options are outlined as follows:

Patches and baffles

Autologous pericardium

Available, sterile, non-antigenic, hemostatic, little adjacent reaction; however, may thicken, fibrose, contract, or dilate. This is addressed to a certain extent by glutaraldehyde fixation which strengthens it helping to reduce the risk of aneurysmal dilatation and slowing the onset of calcification.

Cryopreserved homograft pericardium

Rarely used as it is expensive, with all the disadvantages of fresh autologous pericardium.

Glutaraldehyde-treated xenograft/heterograft pericardium

Usually bovine or equine; rapidly available, but develops severe calcification.

Cryopreserved homograft arterial wall

Hemostatic, handles nicely and conforms to contours; however, risk of disease transmission, expensive, requires time to thaw, may stretch/dilate, risk of calcification.

Dacron (polyethylene terephthalate)

Synthetic polymer. Stable, but stimulates inflammatory reaction causing adjacent fibrosis. Inelastic, does not conform to shape required; probably not suitable for complex baffles in small spaces.

Gore-Tex/Impra (expanded poytetrafluoroethylene, ePTFE)

This microporous synthetic polymer, derived from PTFE, was developed specifically as a biomaterial. The pores allow ingrowth of pseudointima. Less fibrous reaction than Dacron, thus advantageous for baffles, though as a VSD patch may allow the persistence of needle-hole leaks. Conforms better than Dacron, thus a good material for pulmonary artery reconstruction. There may be problems with needle-hole bleeding when used in high-pressure settings.

Valves

These are divided into biological and mechanical prostheses (📖 p364).

Autografts

The autograft is the patient's own pulmonary valve. The commonest use is in the ***Ross procedure*** as an aortic valve replacement (📖 p369, 📖 p388). The autograft grows with the child. However, there is a tendency for the neoaortic root to dilate when the pulmonary autograft is implanted as a root; this may be reduced with the early technique of subcoronary implantation.

Homografts

Aortic and pulmonary roots are available; in the pediatric population there is aggressive calcification when used in the aortic position, leading to early failure/stenosis. In the pulmonary position they may last longer.

Bioprosthetic valves

Stented porcine, stented bovine pericardial, stentless porcine. These are unsuitable for pediatric patients due to the accelerated calcification that occurs up to the age of about 25 years.

Mechanical valves

Bileaflet pyrolytic carbon valves—most commonly used, available down 19mm, newer modified versions 'equivalent' to 17mm.

Valved conduits

- Homograft—pulmonary or aortic.
- Bioprosthetic conduit e.g.:
  - Porcine valve within Dacron tube—glutaraldehyde-fixed; (Hancock).
  - Bovine jugular vein—glutaraldehyde fixed bovine jugular vein with valve (Contegra).
- Composite mechanical valved conduit—bileaflet mechanical valve within a Dacron tube.

Atrial septal defects

Key facts

- ASDs are defects at atrial level leading to shunting of blood between left and right atria. They constitute 10–15% of congenital cardiac defects, and up to 40% of congenital defects presenting in adulthood.
- The commonest defects are of the septum ***secundum***, the 'true' septum, these constitute >80% of ASDs. Female:male=2:1.

Anatomy

There are four types of ASD:

- ***Ostium secundum*** defects (70–80%): commonest type of ASD. They involve the fossa ovalis. Should not be confused with a patent foramen ovale which occurs in 25% of adults, where septum primum lying to the left of the foramen ovale fails to fuse with its sides postnatally, leading to 'probe patency'. Secundum ASDs may extend to the IVC.
- ***Ostium primum*** defects (20%): form of AV defect (see 📖 p 568) and as such are by definition associated with cleft mitral valve.
- ***Sinus venosus defects*** (10%): majority are superior sinus venosus ASDs, at the mouth of the SVC, associated in 90% of cases with ***partial anomalous pulmonary venous drainage (PAPVD)*** of anomalous right upper pulmonary vein from the upper ± middle lobe draining into this junction.
- ***Coronary sinus defects*** (<1%): varying communications between the coronary sinus and the LA, also known as ***unroofed coronary sinus***, almost always associated with persistent left SVC.

Pathophysiology

ASDs permit left-to-right shunting, causing volume overload of the right heart and increased pulmonary blood flow. The magnitude and direction of flow through any ASD depends on the size of the ASD and the relative diastolic filling properties/compliance of the ventricles. In children, the shunt in isolated ASD is left-to-right. With time (rarely in childhood), pulmonary overcirculation leads to pulmonary vascular occlusive disease, with raised PVR, pulmonary hypertension, and eventually decreased RV compliance and reversal of the shunt. When blood shunts from right to left secondary to RV failure, causing cyanosis, this is called ***Eisenmenger syndrome***, and has a very poor prognosis. A significant shunt has a Qp:Qs >1.5:1, or is associated with right heart dilatation. The chronic atrial dilatation leads to the development of atrial arrhythmias, usually after the third decade.

Natural history/long-term sequelae

In patients operated on before the age of 25, long-term survival is normal, but after this age, mortality is higher than in healthy controls. When ASDs are closed in childhood, the long-term risk of arrhythmias is not much higher than the general population, but after the age of 11 this risk increases, so that about 50% of 60-year-olds with ASD have flutter or fibrillation or sinus node dysfunction.

Presentation and initial management

ASDs rarely cause symptoms in children, and are usually found incidentally. The majority of children with PAPVC are asymptomatic throughout childhood, presenting in their third/fourth decade with fatiguability and reduced exercise tolerance. Rarely, if the defect is very big, there may be heart failure and failure to thrive (1% of patients). In older children and adolescents, there may be reduced exercise tolerance in comparison to peers. Patients are pink, with normal respiratory rate. (Infants may be tachypneic.) If desaturated, indicates right-to-left shunt, and should be further investigated before surgery. Parasternal heave at LLSE suggests RVH. PA systolic impulse at ULSE. Systolic ejection murmur over left second intercostal space—pulmonary flow murmur. Second heart sound is split, with no respiratory variation (***fixed splitting***). Diastolic murmur LLSE—tricuspid flow murmur.

Investigations

- EKG normal in 20–40% of children. May show right axis deviation, rSR′ or rsR′, RA enlargement (tall p wave, prolonged PR).
- CXR shows ↑RA, RV, PA, and pulmonary vascular markings.
- Echo is diagnostic and identifies RA, RV, and PA enlargement, septal defect and associated anomalies. Sinus venous ASDs may be difficult to visualize, requiring TEE or MRI to delineate the associated pulmonary venous anomalies.
- Catheterization is not indicated in the routine management of secundum ASD. It is necessary if there are concerns about pulmonary hypertension, to measure pulmonary vascular resistance and its reversibility with vasodilators. Older adult patients require coronary angiography.

Percutaneous options

Defects up to 40mm within the fossa ovalis, with a 5mm rim between the defect and AV valves/major veins, can be closed percutaneously in the cath lab. The device is placed across the atrial septum via a catheter in the femoral vein, under TEE guidance. Success depends on the anatomy of the defect, but this is becoming the standard treatment for suitable defects. ***Complications:*** incomplete defect closure, device migration or erosion, local vascular complications.

Surgical management

Up to 40% of small and moderate sized ASDs close spontaneously by age 4, although this becomes rare after age 2. Secundum ASDs still present after this time should be closed.

Indications

- In adults ASDs should be closed to prevent pulmonary hypertension.
- In older patients with pulmonary hypertension, if PVR <10U/m^2, the outcomes are good; but with 15U/m^2 or higher, the mortality is high.
- In infants, a large shunt with failure to thrive, recurrent respiratory tract infections or signs of heart failure. Evidence of RV or LV volume overload. Secundum ASDs should be closed electively in 3–5-year-olds.
- Paradoxical embolus causing TIA/stroke.

Atrial septal defect repairs

Key facts

- ASDs account for 10% of open congenital heart defect surgery. The classification of ASDs and indications for repair are outlined on 📖 p558 (see also Fig. 12.4).
- ASDs are usually repaired with a patch closure technique through a median sternotomy (or right anterolateral thoracotomy), via a right atriotomy, on bypass. Associated defects that require surgical repair include ***PAPVD***.
- PAPVD involves anomalous connection of at least one but not all four pulmonary veins with the right atrium or its systemic tributaries.

Set-up

- Asymmetrical pericardiotomy so that a large patch of pericardium can be harvested: this is placed in 0.6% glutaraldehyde.
- Median sternotomy, heparinization at 300U/kg.
- Standard CPB with distal ascending aortic cannula, bicaval venous cannula (for sinus venosus ASDs), SVC cannula must be placed high, at the junction with the innominate vein). Cool to 32–34°C.
- Antegrade cardioplegia with cross-clamp: or if fibrillatory arrest planned, care must be taken to leave the LA filled with blood.

Procedure

Repair of secundum atrial septal defects

The anatomy of the SVC, IVC, tricuspid valve, coronary sinus are noted. If the hole is small–moderate, it can be directly sutured with continuous 5/0 or 4/0 Prolene. If >1–2cm, a suitably sized pericardial patch may be sutured to the edges of the defect with continuous 5/0 or 4/0 Prolene.

Repair of superior sinus venosus atrial septal defects

The anatomy of the right pulmonary veins and their relation to the SVC and RA must be determined by careful mobilization of these structures. >90% of defects involve one or two right upper pulmonary veins draining into the RA or SVC. Most can be redirected when closing the ASD with a patch of Gore-tex/bovine pericardium through a slightly higher RA incision anterior to the insertion of the pulmonary veins, crossing into the SVC/RA junction. Care must be taken not to injure the SA node. The defect is patched to redirect the pulmonary veins to the LA. The incision is closed with a patch of autologous pericardium to prevent SVC stenosis. If a pulmonary vein drains high into the SVC (PAPVD), the ***Warden procedure*** may be used: the SVC is transected just distal to the highest PV, and proximally oversewn. Through a RA incision, the ASD is closed with a patch, redirecting PV flow to the LA. The distal SVC is now anastomosed to the RA appendage, restoring normal systemic venous return.

Repair of inferior sinus venosus atrial septal defects

Rare. The pulmonary veins are inspected to exclude anomalous drainage, from the outside of the heart, and on opening the RA. The defect is closed with a patch of pericardium, taking care not to obstruct the IVC.

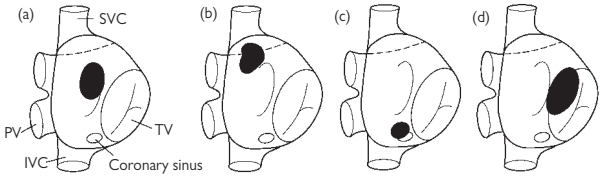


Fig. 12.4 Types of ASD. (a) Secundum ASD. (b) Sinus venosus ASD. (c) Coronary Sinus ASD. (d) primum ASD.

Repair of coronary sinus atrial sepal defects

The repair of unroofed coronary sinus is simple in the absence of a persistent LSVC: the septum between the CS and the oval fossa is excised and the defect closed with a patch of autologous pericardium, committing the CS to the LA. If there is a LSVC draining into the CS, then this flow must be rerouted to the RA. If there is a good-sized innominate vein accepting flow from the LSVC, the LSVC can be ligated distal to its insertion into the innominate. Even if there is no innominate vein, the LSVC can be clamped and pressure measured in the cranial end of the LSVC—if this is below 30cmH_2O, the LSVC can be safely ligated. However, in the absence of an innominate vein, most authorities advise the LSVC should be baffled into the RA and the defect closed. This is usually done using autologous pericardium, and can be complex. If complex repair of unroofed LSVC is high risk then LSVC may be anastomosed to the LPA.

Closure

- Just before the final closure of the ASD, the anesthesiologist should expand the lungs a few times, forcing blood into the LA, and air out of the LA into the RA, de-airing the left side of the heart.
- The RA can now be closed, and after further de-airing, the clamp removed, or the heart defibrillated.

Specific complications

- The aortic root lies just posterior to the superior end of a secundum ASD, so care must be taken when suturing the patch in this area.
- Residual leaks should not require surgery; if significant they can be closed with a device.
- Late pericardial effusions are common (15%), and tamponade (1.5%) can be fatal: routine opening of the right pleura may reduce risk.
- ASD predisposes to development of atrial arrhythmias, whether operated on or not. Atrial flutter and re-entrant tachycardias may occur around the patch or atriotomy scar. AF is the dominant arrhythmia in later life, and in untreated ASDs, sick sinus syndromes may develop.

Outcomes

Survival after interventional or surgical closure of all ASDs is now >99%. In older age groups, results are good if PVR<10U/m^2 at time of surgery.

PAPVD

Key facts

- Partial anomalous pulmonary venous drainage (PAPVD) is due to failure of one to three of the four pulmonary veins to connect with the LA during development.
- Most commonly the right superior pulmonary vein connects to SVC near the cavoatrial junction.
- 90% of PAPVD are associated with an ASD.

Set-up

- Median sternotomy.
- Standard aorto-atrial cardiopulmonary bypass.
- The ductus arteriosus is ligated before initiation of CPB.

Repair of drainage of right pulmonary veins to right atrium

The RA is opened (below and parallel to the AV groove), and the ASD closed with a patch baffling the right pulmonary veins into the LA. If the ASD is restrictive it can be enlarged by extending its inferior margin towards the IVC orifice (dashed line). The atrial septum can be opened in such a way as to create a flap of septal tissue, which can then be sutured to the lateral wall to baffle the pulmonary veins posteriorly into the LA. This flap can be augmented, or replaced, with a patch of autologous pericardium. Damage to the conduction tissues is avoided by taking shallow sutures around the anterior rim of the coronary sinus.

Repair of drainage of the right pulmonary veins to the IVC

The pulmonary veins form a curved channel that drains to the IVC above or below the diaphragm (also known as ***Scimitar syndrome***). This is often associated with hypoplasia of the right lung, which may be supplied by MAPCAs from the descending aorta, and other major cardiac anomalies. There is usually pulmonary venous obstruction. The anomalous venous channel is exposed by lifting the heart out of the pericardial well and incising the posterior pericardium. There are two techniques to correct the anomalous venous drainage:

- ***Tunnelling***: the RA is opened and the ASD inspected and, if necessary, enlarged inferiorly. During a period of DHCA, the IVC cannula is removed and the atrial incision extended into the IVC. The hepatic vein, IVC, and anomalous pulmonary vein can now be identified, and a channel created with a patch to baffle the pulmonary venous flow across the ASD to the LA.
- ***Reimplantation***: if the anomalous vein joins the IVC very low below the diaphragm then the vein can be ligated and divided, and then reanastomosed to the LA. The vein may be anastomosed to the LA in a side-to-side fashion for a long anastomosis, or end-to-side with a wide, spatulated anastomosis.

Pulmonary venous anastomoses have a tendency to shrink and stenose, therefore the tunnelling technique is preferred.

Repair of drainage of left pulmonary veins into the innominate vein

If the left upper lobe alone drains anomalously, this does not have to be corrected. However, when the whole left lung drains into the innominate vein, usually in the context of mixed type of total anomalous pulmonary venous drainage, it must be repaired. This is done through a median sternotomy using cardiopulmonary bypass: the vertical vein is ligated close to its insertion into the innominate vein, and then anastomosed to the LA or the LA appendage (Fig. 12.5). This procedure can be performed through a left thoracotomy without cardiopulmonary bypass.

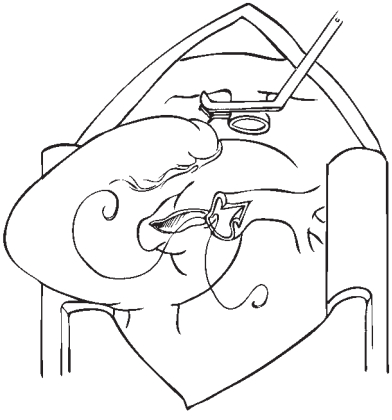


Fig. 12.5 Anastomosing vertical vein to left atrial appendage in repair of drainage of left pulmonary veins into the innominate vein.

Ventricular septal defects

Key facts

- A VSD is a congenital, abnormal defect in the ventricular septum allowing communication of blood between the ventricles.
- They occur as isolated lesions, or in combination with other anomalies constituting 25% of all defects: the commonest congenital heart defect.
- Prevalence increasing over the last three decades with the improvement of imaging technology, and associated with maternal drug and alcohol abuse.

Morphology

From a surgical standpoint, the location of a VSD is important as it defines the approach to closing it, and alerts the surgeon as to the location of conduction fibres that may be damaged when closing the VSD. VSDs can most simply be classified as follows:

Perimembranous (70–80%)

Perimembranous VSDs: where a margin of the VSD consists of fibrous continuity between the tricuspid and aortic valves. The conduction bundles run along the inferior rim of the defect. They may be ***inlet, outlet, or inlet–outlet VSDs***.

Muscular (5–10%)

- Also called ***trabecular*** and including central, mid-muscular, apical, marginal, and multiple or 'Swiss cheese' VSDs.
- They may be single or multiple, associated with other types of VSD and may occur anywhere within the muscular septum.
- The rim of the VSD is entirely muscular. The conduction bundles are remote from the defect, and in the case of the inlet VSD, they run near the superior margin of the defect. They may be located in the inlet, apicotrabecular, or outlet portions of the RV.

Juxta-arterial (5–10%)

- Also called ***conal septal, supracristal, infundibular, subpulmonary, doubly committed sub-arterial (DCSA) VSDs.***
- The conjoined leaflets of aortic and pulmonary valves form rim of VSD.
- The conduction bundles are remote from the defect.
- They extend up to the aortic and pulmonary annuli, and result in aortic valve prolapse in up to 50% of cases.

Low infundibular pulmonary stenosis and VSD

This is also known as double-chamber RV. It consists of an intraventricular stenosis separating the inflow from the outflow tract, usually associated with a VSD. The obstructing muscle band is excised, taking care to preserve tricuspid papillary muscle attachments, and the VSD is repaired, either through a right ventriculotomy or right atriotomy.

Pathophysiology

May be ***restrictive defect***: small defects, presenting resistance to flow across the defect, with Qp:Qs <1.5, or ***non-restrictive defect***: where the cross-sectional area of the defect is equivalent or larger than that of the

aortic annulus—there is no resistance to flow across the VSD, RV, and LV pressures approach parity, and Qp:Qs is inversely proportional to the ratio of PVR to SVR.

- Three hemodynamic consequences: (1) LV volume overload, (2) increased pulmonary blood flow, (3) reduced systemic cardiac output.
- The magnitude and direction of shunting across the defect depends on the size of the defect and the compliance of the distal vascular bed.
- Unrestricted left-to-right shunting leads to pulmonary overcirculation and LV volume overload, causing CHF acutely and irreversible pulmonary hypertension and pulmonary vascular disease in the long term.
- In untreated patients, raised PVR and high right-sided pressures can lead to shunt reversal and cyanosis in about 10% of patients in the third and fourth decades (***Eisenmenger syndrome***).

Presentation

Larger defects present in infancy with CHF, recurrent chest infections, failure to thrive; Qp;Qs >2 is poorly tolerated. Smaller defects may be entirely asymptomatic. Bacterial endocarditis is a risk. DCSA and outlet VSDs may present with AI with the RCC prolapsing through the VSD. Examination shows failure to thrive (FTT), tachypnea, and recession.

Investigations

- EKG shows LV and RV hypertrophy.
- CXR may show increased vascular markings.
- Echo is diagnostic, and defines the location and size of the defect, the extent of the hemodynamic consequences, and associated anomalies.
- In cases where pulmonary vascular disease is suspected, or there are multiple VSDs, cardiac catheterization may be required.

Natural history

- 30–40% of all VSDs close spontaneously. Up to 70% of small VSDs close spontaneously (50% of small PM VSDs close spontaneously in the first 2 years). Inlet perimembranous VSDs and DCSA VSDs do not close spontaneously, and are candidates for early closure.
- 10% of infants with large untreated VSDs die within first year of life.
- Eisenmenger syndrome may be complicated by fatal hemoptysis, polycythemia, cerebral abscess or infarction, and RV failure.

Percutaneous closure

VSDs can be closed by percutaneously placed devices—these are mainly used to close muscular defects, and smaller perimembranous defects. In the latter there is a high incidence of AV node block, up to 10%.

Surgical closure

Operative mortality for VSD closure in pediatric setting is <1%.

Indications

- Moderate to large VSD after age 1 year.
- Congestive cardiac failure resistant to medical therapy in infants.
- Elevated pulmonary vascular resistance after the age of 6 months.
- Evolving aortic regurgitation in outlet VSDs.
- Multiple muscular VSDs (Swiss cheese defects) with significant shunt undergo PA band (📖 p598).

Ventricular septal defect repair

Key facts

- VSDs are usually repaired with a patch closure technique via the tricuspid valve through a right atriotomy, and occasionally via a right ventriculotomy depending on location.
- VSD repair is indicated for severe symptoms during the first 6 months, or electively after the first 6 months based on the assessment of pulmonary vascular disease (📖 p565).
- Associated defects that commonly need surgical correction include PDAs (📖 p572), coarctation of the aorta (📖 p578), aortic regurgitation, straddling, and over-riding AV valve.

Set-up

- Standard invasive monitoring and general anesthesia.
- Median sternotomy, heparinization at 300U/kg.
- Standard aorto-bicaval venous cannulation, and patient cooled to 30–32°C. Antegrade cardioplegia.

Procedure

- Incisions used to approach VSDs are shown in Fig. 12.6.

Repair of perimembranous ventricular septal defects

- These are best repaired via a right atriotomy across the tricuspid valve.
- If the defect cannot be fully seen the septal leaflet is retracted or occasionally detached.
- A Gore-tex, Dacron, or bovine pericardium patch is sewn using continuous or interrupted pledgetted sutures parachuting the patch down, depending on how friable the tissues are.
- Sutures are placed around the rim of the defect avoiding the ***conduction tissue*** at the inferior rim (the right-hand side of the VSD from the surgeon's view: Fig. 12.6).

Repair of muscular ventricular septal defects

- Inlet and trabecular muscular defects are approached as previously described through the RA: apical trabecular muscular defects may need to be approached through a right ventriculotomy, in extreme cases with multiple apical VSDs through a left ventriculotomy; and outlet VSDs may be approached through the MPA, through a right ventriculotomy, through the RA or through a combination of approaches.
- Trabeculations that may obscure the lower rim may be divided.
- Multiple, small muscular VSDs may be closed by a singled pledgetted mattress suture.
- A single patch may be used to close multiple defects, e.g., when a muscular VSD coexists with a perimembranous VSD, or when there is a cluster of muscular VSDs.
- 'Swiss cheese' septum may only be correctable via a left ventriculotomy and single patch placed over the entire muscular septum.
- The conduction bundle in inlet VSDs lies along the superior margin (left hand of surgeon's view).

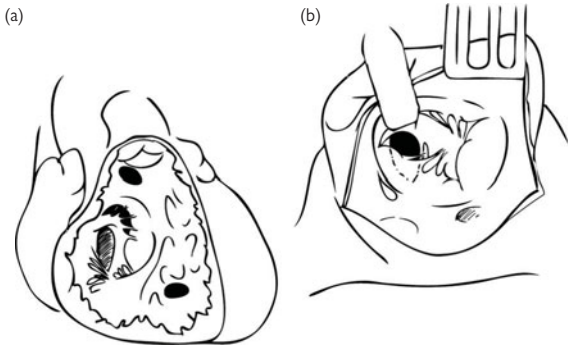


Fig. 12.6 Ventricular septal defects. (a) Types of VSDs. (b) Surgical approach through the tricuspid valve via right atriotomy to a perimembranous VSD showing retraction of the septal leaflet, and location of the conduction tissue.

Repair of juxta-arterial or DCSA defects

- These are approached through a pulmonary arteriotomy, across the pulmonary valve, with some sutures anchoring the patch through the base of pulmonary valve leaflets.
- A patch is used to close these defects.
- The conduction axis is remote from the margins.

Closure

- The right atriotomy is closed and the heart de-aired in the usual way.
- Cross-clamp is removed, caval tapes are released, and the heart is filled.
- TEE is used to assess residual shunts, and any aortic or tricuspid regurgitation requiring correction.
- If the result is satisfactory bypass is discontinued.
- The chest is closed in the usual manner.

Specific complications

- Mortality is <1% in isolated VSD; up to 8% in Swiss cheese defects or with other anomalies.
- Residual VSD (1/2%), may be due to missed defects, incomplete closure, reopening of defect.
- Conduction disturbance:
  - RBBB common; complete heart block 1%.
  - In the long term, 4% get sick sinus syndrome, 6% need reoperations, 15% develop aortic regurgitation.
- Persistent PAH and poor RV now unusual, due to earlier intervention.
- Acute pulmonary hypertensive crises (📖 p542).

Atrioventricular septal defects

Key facts

- AVSDs, also called ***endocardial cushion defects, AV canal defects,*** or ***atrioventricular communis***, are a spectrum of defects resulting from incomplete development of the atrial septum, the inflow portion of the ventricular septum, and the atrioventricular valves.
- These defects make up about 5% of congenital cardiac anomalies.
- Congenital heart disease is present in 45% of children with Down's syndrome, and 45% of these have an AVSD.

Anatomy

In all AVSDs the atrioventricular valves are abnormal, and cannot be termed tricuspid or mitral valves, rather they are called the right and left atrioventricular (AV) valves. If the common valve is predominantly centered over 1 ventricle, the other ventricle may be ***hypoplastic***, and the AVSD is '***unbalanced***' and may not be amenable to biventricular repair. There is a wide spectrum of lesions involving atrial, ventricular, and valvar components of the AV canal, but they are broadly classified as:

Partial AVSD (primum ASD)

Inferior ASD immediately superior to the separate left and right AV valves; the left AV valve is trileaflet, and in 10% it is incompetent.

Complete AVSD

Single AV valve common to the right and left atrioventricular junction; an ***ostium primum defect*** (📖 p561) above the valve leaflets and a ***ventricular septal defect*** (📖 p564) below. The common AV valve has four leaflets: the superior bridging leaflet, lying across the crest of the ventricular septum, the inferior bridging leaflet, also lying across the crest of the septum, the left mural leaflet, and the right mural leaflet. A bare area of ventricular septum can be seen lying under the bridging leaflets.

Intermediate AVSD

In this form there is a ***primum ASD*** (📖 p561) the superior and inferior bridging leaflets may meet and be attached to the underlying ventricular septum thus forming two AV valves, and there is a variable VSD.

Further characteristics of hearts with AVSD

- Aortic valve displaced anteriorly, superiorly, and rightward.
- Inlet of LV deficient relative to outlet, leading to long, horizontally oriented outflow tract, the '***gooseneck' deformity.***
- The AV node is displaced inferiorly and posteriorly toward the coronary sinus; the His bundle courses along the leftward aspect of the crest of the VSD.
- There are several commonly associated lesions:
  - In unbalanced AVSD, ***RV hypoplasia*** may be associated with tetralogy of Fallot (📖 p616), pulmonary stenosis/atresia (📖 p622). When there is non-dominant LV or ***LV hypoplasia***, associated lesions include LVOTO (5%), aortic coarctation (📖 p576).
  - 10% have tetralogy of Fallot (📖 p616), 10% PDA (📖 p572).

Pathophysiology and natural history

Partial AVSD

As with ASD, the long-term complications arise due to decades of right-sided volume overload, with dilatation leading to arrhythmias, ventricular dysfunction, and potential pulmonary vascular disease.

Complete AVSD

There is left-to-right shunting across ASD and VSDs with RV and LV pressures equivalent. This leads to pulmonary overcirculation and CHF. Irreversible pulmonary vascular disease may develop rapidly due to the large shunt: up to 90% of untreated patients develop pulmonary vascular disease by 1 year, with life expectancy <2years. Pulmonary disease develops earlier in Down's syndrome.

Presentation

Routine echo screening in Down's syndrome.

Partial AVSDs

In partial AVSD (as with ASD) in the 10% with LAV valve regurgitation, there is so-called LV–RA shunting, which may lead to earlier presentation with congestive heart failure. The latter may also occur in the presence of associated left-sided lesions such as coarctation, LVOTO. Partial AVSD may present very late, e.g., in adulthood with recurrent chest infections, a murmur, or congestive cardiac failure secondary to the shunt and left AV valve regurgitation. PVR >10 Wood units is a contraindication to surgery. Examination findings are similar to those in ASD (📖 p558): active precordium, pulmonary outflow murmur, fixed split second heart sound.

Complete AVSDs

Complete AVSD usually presents early with CHF. Examination reveals hyperactive precordium with thrill, systolic murmur LSE, apical systolic murmur of LAVVR; split first heart sound indicates elevated PVR.

Investigation

- EKG shows left axis deviation, prominent P waves demonstrating enlarged atria, prolonged PR interval.
- CXR shows cardiomegaly, increased pulmonary vascular markings.
- Echo is diagnostic and delineates the anatomy of the AV valves, the size of the septal defects, and the susceptibility of the LVOT to the characteristic gooseneck deformity that can cause obstruction. It is important to characterize the L AV valve, the size of the mural leaflet, so as to estimate the chances of successful repair.
- Cardiac catheterization is only required rarely, if AVSD is part of a more complex constellation of anomalies, or assess raised pulmonary vascular resistance, e.g., in children, >1 year with complete AVSD, or adults with partial AVSD.

Indications for surgery

- Partial AVSD: similar to ASD (📖 p558)—repair between ages of 2 and 4.
- Complete AVSD: surgical repair should take place between 3–6 months of age, but earlier if heart failure cannot be controlled: failure to thrive, intractable respiratory symptoms.

Atrioventricular defect repairs

Key facts

- A single or double patch technique is used to repair the VSD followed by the ASD.
- The size and shape of the patch is key in restoring competence to both AV valves, which may need to be replaced if significant regurgitation remains.
- ***PA band*** (📖 p598) is performed instead to palliate symptoms of heart failure and pulmonary overcirculation if the clinical condition precludes major cardiac surgery, in unbalanced VSD when the ultimate aim is a univentricular circulation, or if there are complex anomalies.

Set-up

- Standard invasive monitoring and general anesthesia.
- Median sternotomy, heparinization at 300U/kg.
- Asymmetrical pericardiotomy so that a large patch of pericardium can be harvested: this is placed in 0.6% glutaraldehyde.
- Standard CPB with distal ascending aortic cannula, bicaval venous cannula, and patient cooled to 28–32°C, antegrade cardioplegia.

Procedure

Access is via a right atriotomy Fig. 12.7b.

Partial AVSD

- Fill the ventricles with cold saline to inspect the valve.
- Pass buttressed mattress sutures through the base of the R AV valve.
- Pass these sutures through an autologous pericardial patch and tie it down securely, using the outer sutures to secure the patch to the remaining margin of the defect.
- Close the 'cleft' in the L AV valve with interrupted 6/0 Prolene sutures.
- The coronary sinus may be left behind the patch draining to the LA to protect the conduction system close to the CS (except if there is a persistent LSVC to coronary sinus).

Complete AVSD

- Fill the ventricles with cold saline to inspect the valve in the closed position, and identify the point of coaptation of the superior and inferior bridging leaflet above the crest of the ventricular septum—approximate them at this point with a 6/0 Prolene suture; this marks the point of separation of left and right AV valves.
- A hemi-oval patch is now sutured to the ventricular septal crest below the valves, either with continuous or interrupted sutures.
- Horizontal mattress sutures are placed through the superior straight edge of the patch, and then through the bridging leaflets, thus defining the separation of left and right AV valve leaflets.
- These sutures are then passed through the pericardial patch to be used to close the ASD, and tied down.
- Now the left AV valve is inspected, and the inferior and superior bridging leaflets approximated with 6/0 Prolene single sutures along their line of apposition up to the insertion of the chords.

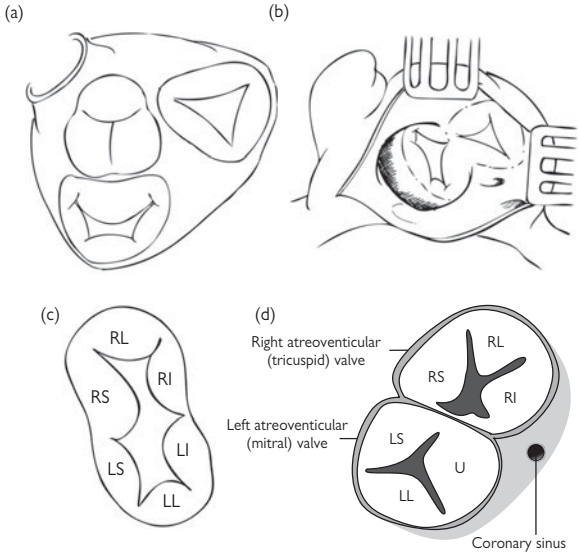


Fig. 12.7 Atrioventricular sepal defects (AVSDs). (a) Mitral tricuspid relationship. In the normal heart the valves are separated by the fibrous skeleton of the heart. (b) Surgical approach and patch closure of ostium primum ASD. The LS and RS are the superior bridging and the RI and LI are inferior bridging leaflet. (c) Complete AVSD. AL, Anterior; PL, posterior; SL, septal; LS, left superior; RS, right superior; RL, right lateral; RI, right inferior; LI, left inferior; LL, left lateral. (d) Partial AVSD.The mitral and tricuspid valves are fused but there is no interventricular communication.

- Finally, the pericardial patch is secured to the edges of the ASD with continuous 6/0 Prolene as for partial AVSD, leaving coronary sinus to drain into the RA or LA.
- Good results have also been reported with single patch techniques, where the VSD component is small.

Complications

- In those children repaired after 6 months, there is a risk of postoperative pulmonary hypertensive crises. Those at risk may be kept paralysed, treated with inhaled nitric oxide, milrinone, enoximone, high FiO_2; vasoconstricting agents and acidosis must be avoided.
- Significant AV valve regurgitation in 10–15%, reoperation in 10%.
- Permanent complete AV block in 1%, right bundle branch block in 20%.
- Down's syndrome patients may have lower mortality, lower reoperation rate, less pre-op AV valve regurgitation.

Patent ductus arteriosus

Key facts

- The arterial duct connects the pulmonary artery with the descending aorta, and in fetal life shunts blood from the pulmonary artery to the aorta, bypassing the immature fetal lungs.
- Most close within 2–3 weeks of birth, initially by contraction of the medial muscle of the duct, then by intimal hyperplasia. Persistence of the lumen after this time, with reversal of the flow, is called ***patent ductus arteriosus (PDA).***
- This anomaly makes up 12–15% of congenital heart defects, and up to 30% of defects in premature infants. 80% of premature infants <1200g present with this. Isolated PDA occurs 1 in 2000 live births.

Anatomy

- The arterial duct runs parallel to the arch of the aorta in close proximity to the recurrent laryngeal and vagus nerves (Fig. 12.8), and connects the main pulmonary artery with the descending aorta, just distal to the origin of the left subclavian artery.
- If there is a left aortic arch, it is usually left-sided; in right aortic arch it may be left- or right-sided. In some conditions, it may be absent, or rarely, bilateral. It varies in length and diameter.
- The media consists of spirally arranged smooth muscle, and the intima is thicker than the aorta.

Physiology

The PDA is closed by 8 weeks in 88% of children with a normal cardiovascular system (📖 p534). Closure occurs in two stages, and is mediated by increased oxygen tension and prostaglandin levels (PGE_1, PGE_2, and PGI_2). First, smooth muscle contraction within 10–15h of birth produces shortening, thickening, and functional closure. The second stage occurs over 2–3 weeks. It involves fibrous proliferation of the intima, associated with medial necrosis producing the fibrous ligamentum arteriosus.

- Pathophysiology depends on the size of the left-to-right shunt.
- When the PDA is large, aortic and PA pressures are equal, and the magnitude and direction of shunting depends on the PVR.
- As neonatal PVR falls the left-to-right shunt increases and congestive cardiac failure ensues.
- Pulmonary vascular changes may lead to irreversible pulmonary hypertension and reversal of the left-to-right shunt (Eisenmenger syndrome) which has a very poor prognosis, even though the initial reduction in the left-to-right shunt leads to a brief improvement in the clinical picture (see also 📖 p565).
- In moderate PDAs the size of the shunt is proportional to the size of the PDA, and LV hypertrophy usually compensates.
- A small PDA does not produce HF until the fourth or fifth decade.

Presentation

Presentation depends on the size of the PDA.

- Large PDAs present with symptoms of severe heart failure in preterm babies, a wide pulse pressure, ↑JVP, and a systolic murmur.

- The findings are similar in smaller PDAs but develop later; additionally the ductus may be calcified.
- Premature infants with this defect may be ventilator-dependent, or require long periods of non-invasive ventilation.
- Many children may be asymptomatic. Large shunts may present as congestive heart failure early in life.

Examination

- Bounding pulses due to wide pulse pressure.
- In premature infants, there is usually a systolic murmur at the second left intercostal space; with increasing size of the shunt, this murmur may extend into diastole and become the classic harsh, continuous 'machinery' murmur.

Investigations

- The EKG shows left and sometimes right ventricular hypertrophy.
- CXR may be normal, or show cardiomegaly, an enlarged PA, and pulmonary congestion in large shunts.
- Echo is diagnostic, demonstrating the shunt flow from descending aorta to pulmonary artery, and enlarged left-sided chambers. Associated anomalies must be carefully excluded.
- Cardiac catheterization is not usually required, but catheter closure is usually possible in infants >2kg.

Management

Indomethacin and ibuprofen are equally effective in closing the patent duct in pre-term infants, and improve outcomes even in asymptomatic pre-term infants. They are contraindicated in renal insufficiency or intracranial hemorrhage. The use of these drugs has reduced the need for surgical closure from 30% to 5% of hemodynamically significant ducts. If medical therapy fails, surgical ligation is considered for all symptomatic infants, and in all infants in whom the duct has not closed after 3 months. All ducts should be closed to prevent the risk of bacterial endocarditis. In patients with pulmonary vascular resistance >8U/m^2, duct closure is contraindicated. PDA surgical and catheter closure techniques are described on 📖 p574.

Indications for closure

- Persistent PDA is an indication for elective closure.
- Urgently if there is evidence of cardiac failure.

Contraindications to closure

PVR >8U/m^2.

Outcome

- Death rate falls and is 0.5% per annum after the first year, but increases after the third decade and 60% of patients are dead by the age of 45.
- Spontaneous closure occurs in 0.6% of patients per year.
- Operative mortality of isolated PDA closure is <1%.
- Recurrent laryngeal nerve palsy in 30%, permanent in 5%.

Closure of PDA

Key facts

- If medical therapy fails, surgical ligation is considered for all symptomatic infants, and in all infants in whom the duct has not closed after 3 months.
- All ducts should be closed to prevent the risk of bacterial endocarditis.
- In patients with pulmonary vascular resistance >8U/m^2, duct closure is contraindicated.

Set-up

- This procedure is often done on the ICU with pressure cuff monitoring of the right arm, and O_2 saturations monitored in the lower limb.
- Left side up with roll under ribs.

Operative technique

- Small left anterior thoracotomy through the 4th intercostal space.
- Lung gently retracted medially.
- Posterior pleura incised.
- Identify left vagus nerve (enters thorax between left subclavian and left common carotid arteries, crossing aortic arch and ductus arteriosus) and left recurrent laryngeal nerve (curves around ductus arteriosus and extends back up into the neck) to avoid injury. The vagus nerve may be reflected medially or laterally (Fig. 12.8).
- Dissect above and below the ductus, staying in plane near the aortic wall.
- When dissection is deep enough, right-angled instrument passed beneath the duct to allow passage of thick silk ligature.
- Trial occlusion of the duct (should note rise in systemic BP, no drop in lower limb O_2 saturations).
- Duct is doubly ligated.
- In premature infants, the duct may be ligated with titanium vascular clips. In some centers, this is achieved thoracoscopically. In larger patients, the duct may be divided between vascular clamps and both ends oversewn.
- The duct is often closed during median sternotomy for other procedures. This is usually achieved with a silk ligature just after commencement of cardiopulmonary bypass.

Coil occlusion

Transcatheter occlusion of patent ducts with coils are performed in patients >6kg in size, and can be done in >2kg.

Complications

Very rarely, hemorrhage and death, <1%, although in extremely premature infants 30-day mortality may be as high as 20%. Recurrent laryngeal nerve palsy may be as high as 30%, but is usually transient.

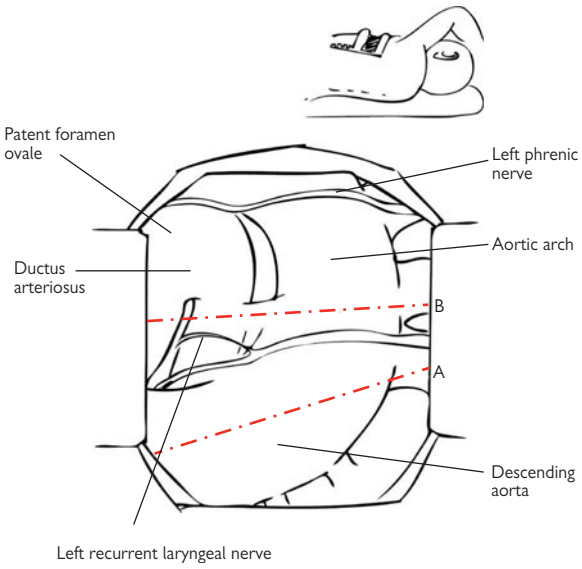


Fig. 12.8 Surgical anatomy of the ductus arteriosus. (A) This incision line is used if the vagus nerve is to be retracted medially. (B) This incision is used if the vagus nerve is to be retracted laterally.

Coarctation of the aorta

Key facts

- Coarctation of the aorta is defined as a congenital narrowing of the upper descending aorta, opposite the ductus arteriosus.
- This accounts for 5–8% of all congenital heart defects.
- It may be isolated, but is associated with bicuspid aortic valve and VSD. It is the most common cardiac defect in ***Turner syndrome***. (15–20%), and may be a feature of more complex defects such as TGA, Taussig–Bing, hypoplastic left heart syndrome. It may form part of the constellation of left-sided obstructive lesions known as the Shone complex.

Morphology

The constriction consists of localized medial thickening with infolding of medial and neointimal layers. It may form a ***shelf*** within the lumen of the aorta, or may be ***concentric***. The constriction may be discrete, or, more rarely, a long tubular segment. All congenital coarctations are ***juxtaductal***; however, inflammatory or autoimmune-mediated coarctations may develop in the descending thoracic aorta or abdominal aorta. There may be ***post-stenotic dilatation***, and there may be hypoplasia of the isthmus of the aorta, just proximal to the coarctation. The arch may be hypoplastic, and there may be ***collateral circulation*** from proximal to the coarctation to the lower part of the body (from the internal thoracic, vertebral, costocervical, and thyrocervical trunks, to the descending aorta distal to the obstruction).

Pathophysiology

- The hemodynamic consequences are high afterload on the LV, increasing LV wall stress and causing ***LV hypertrophy***.
- Systemic perfusion depends on ductal flow. In neonates with severe coarctation, there may be circulatory collapse when the duct closes.
- In less severe coarctation, there is a slower evolution, with LV failure, mitral incompetence, left-to-right shunt, and pulmonary congestion.
- After infancy, CHF rarely occurs before the age of 20; these children may be asymptomatic until hypertension is noted.
- ***Hypertension*** develops due to the mechanical obstruction and possibly, renin–angiotensin-mediated pathways.

Presentation and natural history

- Depends on the existence of coexisting abnormalities, as well as the location and severity of the location:
- ***Neonates***: collapse, acidosis, hypotension, heart failure; absent femoral pulses on routine review.
- ***Infancy***: upper extremity hypertension with absent/reduced femoral pulses; congestive heart failure causing dyspnea and failure to thrive.
- ***Children/young adults***: headaches, lower extremity weakness, exertional dyspnea, fatigue; hypertension.
- Examination findings also depend on the age and presentation:
  - In the shocked neonate, all pulses may be weak; however, absent femoral pulses should not be disregarded.

  - There may be differential cyanosis, with the duct supplying the lower body, and the aorta supplying the upper body, demonstrated with preductal and postductal oxygen saturation readings.
  - Systolic murmur in the left infraclavicular or infrascapular area.
  - In older children, BP reading should be made in all limbs, and discrepancies of >20mmHg between upper and lower are significant. There may be a systolic murmur in the left infrascapular region, or a continuous murmur in the presence of collaterals.
- Uncorrected coarctation leads to a curtailed life expectancy of 30–40 years, with causes of death including aortic rupture, intracranial hemorrhage, cardiac failure, bacterial endocarditis. Beyond infancy, even after correction there is a lifetime risk of hypertension and its sequelae. After correction, freedom from death or complication or hypertension is only 20% at 25 years.
- Patients can die from cardiac failure, aortic rupture, infective endocarditis, or intracranial hemorrhage. Untreated isolated coarctation has a 1-month mortality of 10%, 1-year mortality around 30%.

Investigation

- EKG: LV hypertrophy.
- CXR: in sick children, cardiomegaly, pulmonary congestion. Later, there may be a classic '3' hourglass sign of coarctation along the upper descending aorta, and rib-notching from collaterals in children >10 years.
- Echo: shows the arch, and the coarctation shelf is usually visible just distal to the subclavian origin. Doppler measures velocity across the coarctation, enabling estimation of pressure gradient. Colour flow mapping demonstrates the diastolic 'tail' of flow in the descending aorta.
- MRI/CT useful in older patients or reoperations to plan approach.

Management

Neonates

In the shocked neonate, the initial management is supportive, improving peripheral perfusion by ***reopening the duct*** if possible, correcting the acid and electrolyte disturbances, and supporting the circulation and ventilation as necessary prior to undertaking corrective surgery. In neonates who are stabilized within 24h of presentation, surgical repair can be carried out urgently. In those not responding to treatment of acidosis/anuria, earlier surgery may be required. In infants not critically ill, but with hypertension and cardiomegaly, repair is the nonetheless urgent, but can be performed semi-electively on the same admission.

Older children

Balloon angioplasty of the coarctation is now the first-line treatment. Hypertension should be treated, usually with β-blockers, but intervention should not be delayed while waiting for normotension.

Indications for surgery

Isolated coarctation is an indication for surgical repair, once critically ill infants have been stabilized, and within 4–6 weeks of presentation in stable children.

Coarctation of the aorta repair

Key facts

- Symptomatic neonates and infants should undergo surgery as soon as they are stabilized. Older children and adults may undergo surgery electively, even if they are asymptomatic. It is preferable to treat this lesion before the age of 5, to reduce the risk of residual hypertension.
- Isolated coarctation are usually approached via a left thoracotomy and repaired without bypass.
- Coarctation with associated defects, e.g., VSD, interrupted aortic arch are approached through a median sternotomy using DHCA (📖 p552).
- A small ***concomitant VSD*** will probably close spontaneously and does not need intervention. There are three main options for moderate or large concomitant VSDs: (1) PA band at the time of coarctation repair via a thoracotomy, (2) repair via median sternotomy on bypass during coarctation repair, (3) a staged procedure.

Set-up

- PGE_1 infusion.
- Left lateral thoracotomy for isolated coarctation.
- Collateral vessels temporarily ligated with clips.
- Spinal cord protection: surface cooling, short clamp time, avoid acidosis, measure distal aortic pressure after clamping to avoid paraplegia.

Procedure

There are several ways of repairing the coarctation:

End-to-end anastomosis

- This is used for discrete coarctations.
- The descending aorta, the LSCA, LCCA, and the arch are mobilized.
- Vascular occlusion clamps are placed either side of the coarctation.
- If the coarctation extends into the arch clamps are placed across the LSCA and LCCC, and the ductus arteriosus is ligated.
- The coarctation segment is completely excised, and the two ends are anastomosed together (Fig. 12.9). If the transverse arch is hypoplastic the incision can be extended into the proximal arch with a counter incision in the descending aorta in order to form an extended end-to-end anastomosis.

Subclavian flap repair

- This once popular technique has been largely superseded.
- The subclavian artery is divided and turned down as a flap, and used instead of a prosthetic patch to repair the coarctation.
- If there is a hypoplastic arch, the flap can be used to augment the arch.
- The repair is associated with the same frequency of recurrent coarctation, as well as problems due to division of the subclavian artery including long term weakness, subclavian steal and occasionally ischemia.

Prosthetic patch aortoplasty

- This is rarely used due to aneurysm formation at the suture site.
- Vascular occlusion clamps are placed either side of the coarctation.
- The ductus arteriosus is ligated, an incision is made longitudinally across the coarctation and a patch of Dacron or Gore-tex is used to augment the diameter of the aorta at the site of coarctation.

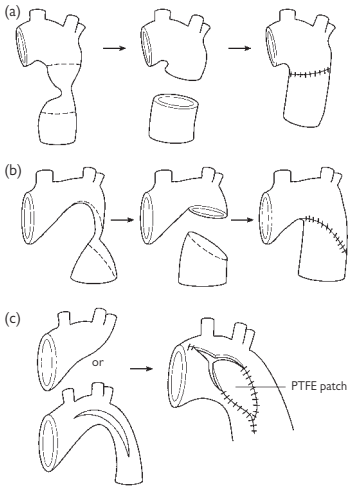


Fig. 12.9 Coarctation of the aorta repair. (a) End to end anastomosis. (b) Bevelled end-to-end anastomosis. (c) Prosthetic patch aortoplasty.

Interposition graft

This technique is rarely used as it results in a fixed stenosis.

Complications

- ***Early***: rebound hypertension in 30%, bleeding, paraplegia 0.4% (associated with long cross-clamp times and division of collaterals), renal failure in sick infants, abdominal symptoms from mesenteric arteritis including pain, ileus, rarely necrotizing enterocolitis—risk reduced if hypertension controlled, chylothorax, recurrent laryngeal nerve injury.
- ***Late***: aneurysm—this is most common after patch repair, which is therefore rarely performed now.
- ***Recoarctation*** is the main long-term complication and is reported in 10–45% of patients. It may be due to technically inadequate repair, hypoplasia of the arch proximal to the repair, or anastomotic stricture. Reintervention is usually transcatheter in the first instance, with freedom from reintervention after balloon dilatation up to 75% at 12 years and is indicated if resting gradient >20mmHg. Balloon dilatation is method of choice to abolish gradient, and in children older than 12, stenting may be considered.

Outcomes

Coarctation of the aorta, although apparently a simple technical problem, has continuing long-term sequelae in spite of repair. Only 20% of patients are alive and free from hypertension or a complication at 25 years. Repair before the age of 5 years may reduce the risk of hypertension later; however, even with early repair the life expectancy is reduced.

Vascular rings

Key facts

- These constitute <1% of congenital cardiac defects.
- Abnormal development of the embryonic aortic arches produces vascular rings or slings that partially or completely encircle the trachea and/or esophagus, causing compressive symptoms.

Morphology

These anomalies occur early in the development of the aortic arch and vessels: they result from incomplete regression of one of the six embryonic branchial arches. Rings usually surround the trachea and esophagus. There are two main categories:

Complete rings

- ***Double aortic arch*** (50%) is due to failure of involution of distal right 4th arch—the right and left arches encircle the trachea and esophagus. One is usually smaller, less dominant, and in 30% this is atretic. The RCCA and RSCA come off the right arch, the LCCA and LSCA come from the left arch.
- ***Right aortic arch with left ligamentum arteriosum*** (45%). Left 4th arch involutes, right 4th arch remains—in combination with left arterial ligament or aberrant LSCA. The right arch gives off the LCCA first, then the RCCA, the LSCA, with the duct originating at the base of the LSCA to the LPA.
- ***Right aortic arch with mirror image branching*** (<1%). Partial resorption of the left 4th arch.
- ***Left aortic arch*** with retroesophageal right subclavian artery, right-sided descending aorta and right ligamentum arteriosum (<1%). Left aortic arch with right descending aorta and right arterial ligament
- Anomalous left pulmonary artery or ***pulmonary sling***. Abnormal 6th arch development—anomalous origin of the LPA from posterior aspect of RPA—LPA swings over the RMB, around the trachea between trachea and esophagus, to the left lung. 50% are associated with tracheal stenosis due to complete tracheal rings and 20% have other cardiac anomalies.

Incomplete vascular rings

- 'Anomalous' innominate artery: the innominate artery arises more leftward on the arch than usual possibly causing tracheal compression.
- Retro-oesosophageal right subclavian artery (***arteria lusoria***).

Pathophysiology

These anomalies are associated with chromosome 22q11 deletion, anomalies of the aortic arch, and other cardiac defects. 30% of patients with tetralogy of Fallot have a right aortic arch.

Presentation

Symptoms and signs are primarily those of airway and/or esophageal compression. The majority present in infancy. Symptoms include:

- May be asymptomatic.

- Stridor, respiratory distress or apnea.
- Characteristic high-pitched 'brassy' cough.
- Asthma, recurrent chest infections.
- Dysphagia and difficulty in feeding are more common in late presentations.

Investigations

- CXR commonly performed because of presentation.
  - Identifies pulmonary complications, e.g., atelectasis, hyperinflation.
  - May identify position of aortic arch, tracheal compression.
- TTE and TEE help identify any concomitant cardiac defects, but are not good at identifying the ligamentum arteriosum, and compressed structures in the midline, and are diagnostic of double aortic arch.
- Angiography, MRA, or CT angio are mandatory to define anatomy.
- Bronchoscopy is mandatory where presentation is with respiratory symptoms, and is needed to diagnose tracheomalacia or stenosis (📖 p750).
- Barium swallow may delineate esophageal compression, which is more common in adults.

Surgical management

Left posterolateral thoracotomy (or occasionally median sternotomy) and cardiopulmonary bypass are used for surgical division of most arches. Sternotomy and bypass are required for pulmonary artery sling and tracheal reconstruction. Although configurations associated with a right descending thoracic aorta may be approached via a right thoracotomy.

Indications for surgery

Symptoms, particularly those of upper airway obstruction.

Contraindications

Asymptomatic patients may be managed expectantly.

Surgical techniques

- Double aortic arch: division of the non-dominant arch.
- Right aortic arch with ligamentum arteriosum: division of the ligament.
- Right aortic arch with mirror image: division of the duct and Kommerll's diverticulum if present.
- Left aortic arch: aortopexy (median sternotomy), division and reimplantation of RSCA.
- Pulmonary artery sling: division and translocation of the LPA ± repair of the trachea via left thoracotomy/median sternotomy with CPB.

Complications

- Injury to the vagus, phrenic, and recurrent laryngeal nerves.
- Chylothorax from injury to the thoracic duct.
- Tracheomalacia and bronchomalacia.
- Late vascular complications such as subclavian steal.
- The risk of death is low if surgery is for isolated vascular ring, i.e., <1%. If combined with tracheal surgery, the results are not as good.
- There is evidence of persistent airway obstruction in about 50% of patients postoperatively.

Interrupted aortic arch repair

Key facts

- Interrupted aortic arch is defined as an interruption of lumenal continuity between the ascending and descending aorta. There are three types.
- It is rare, comprising <1% of congenital cardiac defects.
- The PDA is the only supply of blood to the descending aorta and neonates present in circulatory shock when the duct closes.
- Repair is indicated as a single or staged procedure: under DHCA, the descending aorta is transected from its anomalous origin from the PA trunk, and anastomosed directly to the ascending aorta.

Anatomy

- Interrupted arch is commonly associated with other defects: VSD 80–90%, bicuspid aortic valve 60%, subaortic stenosis 20%, common arterial trunk 10%. There are three possible points of interruption, correlating with the different embryonic derivations of the aortic arch:
- Type A (25–35%): at the isthmus, just distal to the LSCA origin.
- Type B (60–70%): between the LCC and LSCA: often associated with aberrant RSCA and subaortic stenosis.
- Type C (5%): between the innominate and LCC arteries.

Pathophysiology and natural history

The patent arterial duct is the only source of blood supply distal to the interruption. Ductal closure thus leads to systemic hypoperfusion, acidosis, renal failure, necrotizing enterocolitis, hepatic ischemia, heart failure.

Presentation

Neonates present when the duct closes with circulatory shock and congestive cardiac failure. They must be resuscitated, with prostaglandin infusion, inotropes if necessary, and intubation and ventilation to allow easy manipulation of the pulmonary vascular bed, and thus ductal flow. There may be abnormal pulse patterns depending on the site of interruption.

Investigations

- CXR: cardiomegaly and pulmonary vascular congestion.
- EKG: RV hypertrophy may be evident.
- Echo is diagnostic, showing narrow ascending aorta, absent arch, descending aorta continuing from the duct.

Surgery

Single-stage complete repair

The commonest presentation of interrupted arch with VSD is most commonly repaired completely in the neonatal period through a median sternotomy using cardiopulmonary bypass. This approach can also be used for interrupted arch in conjunction with anomalous pulmonary artery, A-P window, common arterial trunk, TAPVC, TGA, Taussig–Bing.

- Median sternotomy, and cannulation of both the ascending aorta and the duct if temperature >28°C at onset of bypass
- Cool to 20°C, mobilize the head–neck vessels and arch while cooling.
- Arch repair during DHCA, or more frequently now with antegrade cerebral perfusion during aortic arch exclusion for repair.

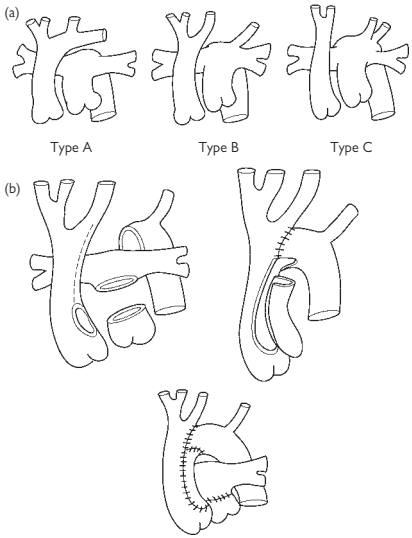


Fig. 12.10 Interrupted aortic arch repair. (a) Three types of interrupted aortic arch. (b) Repair of type B interrupted aortic arch.

- Duct is divided, ductal tissue excised, descending aorta anastomosed to the distal ascending aorta (7/0 Prolene).
- The left common carotid artery or left subclavian artery may be turned down as a flap to reconstitute the arch, with the distal left carotid anastomosed to the RCC to secure antegrade flow.
- Prosthetic tube grafts have been used in the past to achieve tensionless anastomosis, but this inevitably leads to the need for reoperation to upsize the graft, and is usually unnecessary.

Staged repair

The interrupted arch may be repaired through a thoracotomy, and the main PA banded to prevent pulmonary overflow in cases such as double inlet LV. Again, it is best to avoid prosthetic tube grafts. The left subclavian artery or left common carotid artery may be used to augment the repair, with or without a pericardial patch.

Specific complications

Mortality for single-stage repair is 10–20%. Risk of reoperation is no higher than with staged repair. 5-year survival is 65–70%; late survival is less with LVOTO. 10-year freedom from reoperation is 50%. Mortality for repair via thoracotomy with PA banding is low, but long-term reinterventions are as previously described. Anastomotic obstruction is common, with 25–40% requiring further arch intervention. Recurrent laryngeal and phrenic nerve damage and tracheal compression are also potential complications.

Tricuspid atresia

Key facts

- Absent right AV connection or tricuspid atresia comprises 1–3% of congenital cardiac defects: it is the third most common cyanotic heart defect.
- It results from failure of the development of the atrioventricular valve between the right atrium and ventricle.

Anatomy

Most commonly, there is an atretic tricuspid valve, hypoplastic RV, and restricted pulmonary blood flow and an ASD, leading to right-to-left shunting and cyanosis. There is usually a VSD, often restrictive. The ventriculoarterial arrangement is usually normal, and these patients usually have obstructed pulmonary flow, but in about 20% the great arteries are related abnormally.

- Type I: normally related great arteries.
- Type II: D-transposition of the great arteries.
- Type III: L-transposition of the great arteries.

Blood flows from the RA across the ASD to the LA. The RA is hypertrophied. The RV is hypoplastic and muscle bound, with a VSD communicating with the RVOT. There is usually, in 70%, pulmonary outflow obstruction, with a restricted infundibulum, and also pulmonary valvar stenosis/atresia or MPA hypoplasia. Left-sided obstructive lesions such as coarctation of the aorta, subaortic stenosis, and interrupted aortic arch are not uncommon.

Pathophysiology

The LA and LV support both systemic and pulmonary circulations and are therefore chronically volume overloaded. In neonates with pulmonary atresia, severe ***cyanosis*** develops with closure of the arterial duct; in those with unobstructed pulmonary flow, there is pulmonary overcirculation, pulmonary hypertension and heart failure.

Presentation and natural history

This is usually in the neonatal period with cyanosis. There may be heart failure in those with unrestricted pulmonary blood flow. 50% of patients have symptoms on the first day of life. The VSD causes a harsh ejection systolic murmur at the left sternal border; a cardiac murmur is present in 80% of cases. Without treatment survival beyond infancy is 10–20%.

Investigation

- EKG: characteristic notched P wave. LV hypertrophy.
- CXR: this is variable. There may be a boot shaped heart if there is pulmonary atresia/hypoplasia, or 'egg-on-a-string' if there are transposed great arteries; pulmonary vascularity may be decreased or increased.
- Echo: this is diagnostic, revealing difference in ventricular sizes, absence of the tricuspid valve, ASD with right-to-left flow. Relationship of the

great arteries, VSD, PDA, coarctation must all be assessed. The size of the ASD must be determined.
- Cardiac catheterization: this is recommended for operative strategy, determining the source and adequacy of pulmonary supply, measurement of the pulmonary artery pressure, pulmonary vascular resistance, left ventricular pressure, shunt fractions, and assessment of the pulmonary artery tree.

Management

Neonates presenting with severe cyanosis should be treated with PGE_1 to maintain ductal patency, those in heart failure should be treated with diuretics. However, most infants require some form of palliative surgery within the first year of life.

Surgery

Surgical management consists of initial palliation to achieve unobstructed systemic outflow and appropriate pulmonary blood flow.
- If the ASD is restrictive, a balloon atrial septostomy may be performed
- If the pulmonary blood flow is restricted, a systemic-pulmonary shunt, usually a BT shunt, is performed.
- If there is pulmonary overcirculation, a PA band is performed.

Subsequent definitive procedures aim to establish systemic venous return directly to the pulmonary arteries, thus bypassing the right heart altogether. This is the ***Fontan circulation*** (📖 p592) and tricuspid atresia provides the paradigm for this type of circulation.
- Bidirectional superior cavopulmonary shunt (📖 p601) to direct superior caval return to the right pulmonary artery (at about 6 months of age).
- Completion of total cavopulmonary connection with extracardiac tube graft draining the inferior caval vein flow to the right pulmonary artery, thus redirecting all systemic venous return directly to the lungs, bypassing the RV (2–4 years of age).

Outcomes

The overall mortality through all stages to completion of the Fontan circulation is <10% for this condition, and the 1-year survival after Fontan completion is 85%; the procedure should eliminate cyanosis, polycythemia and ventricular volume overload, improving prognosis.

Congenital tricuspid valve stenosis

Isolated congenital tricuspid stenosis and incompetence are exceedingly rare. Congenital tricuspid stenosis is usually due to hypoplastic annulus and forms part of the spectrum of tricuspid atresia, hypoplastic RV. Isolated tricuspid incompetence may be due to agenesis of one or two leaflets, annular distension, or absence of papillary muscles; surgery may be undertaken in older children.

Hypoplastic left heart syndrome

Key facts

- Hypoplastic left heart syndrome (HLHS) is a spectrum of left-sided cardiac malformations characterized by marked hypoplasia of the LV and ascending aorta (Fig. 12.11).
- It comprises 7–10% of congenital cardiac defects of the neonate.
- More common in Turner, Noonan, and Smith–Lemli–Opitz syndromes. 25% of infants have chromosomal abnormalities on karyotyping.

Morphology

Highly variable (Fig. 12.1). Hallmarks include:

- Aortic valve hypoplasia or stenosis.
- Ascending aorta hyoplasia or atresia, arch hypoplasia, coarctation.
- LV absence, atresia or hypoplasia.
- Associated mitral valve atresia, hypoplasia or stenosis.
- Ventricular septum usually intact (ASD necessary for viability).
- Systemic circulation is by necessity supplied by a large PDA.

Physiology

- Fully saturated pulmonary venous return to the LA cannot flow to the LV, and must cross the ASD to the RA, where it mixes with systemic venous return. Any restriction in the PFO/ASD is equivalent to pulmonary venous obstruction.
- As a result both systemic and pulmonary venous return are directed through the RV. The RV pumps blood to:
  - The lungs via the pulmonary arteries.
  - The body via the PDA which supplies the descending aorta.
  - The coronary arteries are supplied by retrograde flow of blood from the PDA down the hypoplastic ascending aorta.
- The relative flow to each circulation will depend on the relative vascular resistances in each bed (because flow is inversely proportional to vascular resistance).

After birth

After birth, a ***'balanced' circulation*** and maintenance of life depends on:

- Adequate interatrial communication.
- Patent ductus arteriosus.
- Pulmonary vascular resistance.

After birth, as the pulmonary vascular resistance falls, the RV will pump preferentially to the pulmonary vascular bed, thus increasing oxygen saturation, but at the expense of systemic blood flow, cerebral blood flow, and coronary blood flow. Poor systemic perfusion leads to metabolic acidosis, oliguria, and eventually collapse. Coronary and cerebral perfusion are also decreased. Conversely, increased systemic blood flow at the expense of pulmonary blood flow leads to good peripheral perfusion. However, the reduced pulmonary perfusion leads to hypoxia. Thus, a delicate balance between the two vascular beds must be maintained to ensure adequate tissue perfusion and oxygenation, and the aim of neonatal management is a 'balanced' circulation.

Presentation

HLHS is uniformly fatal in the first weeks of life without intervention. Usually presents in the first 24–48h of life. Most are term infants after a normal pregnancy. If there has been pulmonary venous obstruction ***in utero*** due to a restrictive ASD, the child may be born in respiratory distress. Usually, as the duct closes within the first 48h postnatally, the child becomes cyanosed, tachycardic, tachypnoeic, oliguric, acidotic, and hypothermic. There is a normal first heart sound, loud, single second heart sound, usually no murmurs, and prominent RV impulse.

Investigations

- The EKG shows RA enlargement and RV hypertrophy.
- CXR shows cardiomegaly and increased pulmonary vascular markings.
- Echo is diagnostic. Shows anatomy of left-sided hypoplasia and arch, coarctation, and adequacy of any ASD. TR predicts poorer outcomes.
- Cardiac catheterization is not normally required but may be necessary to perform atrial septostomy to relieve pulmonary venous obstruction
- Intracranial bleeds and abdominal pathology are common: US of head and abdomen is standard preoperatively.

Management

This lesion is the most frequent congenital heart lesion associated with cardiac death in the newborn. Goals of initial management include:

- Maintaining the PDA with PGE_1 infusion 0.025–0.05 micrograms/kg/min.
- Maintaining a balance between PVR and SVR to maintain flows using either CO_2 or hypoxic inhaled mixtures to maintain ↑PVR.
- The infant can usually be stabilized for a few days prior to surgery but the only definitive intervention is surgical.
- The option of ***supportive therapy*** only, with discharge home and eventual death, must be offered to parents. This requires an honest discussion about the mortality rates of each stage and the interstage mortality rates as well as an estimate of long-term functional outcomes.

Overview of surgical options

- The three options available are:
- ***Norwood***
  - ***Norwood stage I procedure*** within a few days of birth (📖 p588).
  - Second stage is performed at 3–6 months to minimize the time the RV has to cope with two circulations, and consists of performing a bidirectional ***Glenn shunt*** (📖 p601), dividing the B-T shunt, augmenting the pulmonary arteries, addressing residual surgical issues, e.g., restrictive ASD, or recoarctation. Some units prefer to perform the ***hemi-Fontan procedure*** (📖 p591) at this stage instead.
  - Finally a ***Fontan*** (📖 pp594–595) at 2 years.
- Orthotopic cardiac transplantation.
- Stenting of the duct and bilateral PA band (hybrid procedure).

The first two of these options are the more established procedures, with equivalent long-term outcomes (75% at 5 years). The third is a recent innovation, largely restricted in its use to 'high-risk' neonates. Therefore, indications for its use have not been fully elucidated at this time.

Norwood procedure

Key facts

- The Norwood procedure is the first of a three-stage repair of ***HLHS*** (📖 p586) which results in a single-ventricle ***Fontan circulation*** (📖 p592).
- It is also used as a palliative procedure for conditions in which the morphological left heart is inadequate to maintain systemic circulation such as AV and ventriculoarterial (📖 p598) malconnections.
- The aim of the Norwood stage I is to secure systemic cardiac output and balance the pulmonary and systemic vascular beds. This is done by:
  - Atrial septectomy to ensure pulmonary venous return bypasses LV.
  - Securing the systemic output: augmenting the aortic arch, repairing coarctation, and converting the right-sided ventricle and outflow tract to a systemic ventricle and outflow tract.
  - Securing a controlled pulmonary blood supply with a B-T shunt (📖 p600) or RV-PA conduit.
- The second and third stages are described on 📖 pp590–595.

Setup

- Standard invasive monitoring, with antifibrinolytic, PGE_1 discontinued.
- Median sternotomy, heparinization at 300U/kg.
- PDA, and left and right PA, are dissected completely free, and the aorta is freed from root to head and neck vessels, mobilized and snared.
- Standard CPB with distal ascending aortic cannula (or via 3–3.5mm Gore-tex tube which may be used for later B-T shunt), single-stage venous cannula, and patient cooled to 18–20°C.
- Pulmonary blood flow may require cross-clamping or snaring the PA.

Procedure

Repair of the hypoplastic aorta and connection to the LVOT

- The main PA is transected just proximal to the main bifurcation.
- The distal PA is closed with a patch of cryopreserved homograft.
- The ductus arteriosus is transected at its entrance to the aorta, all ductal and any coarcted tissue is completely excised.
- The aorta and arch is enlarged during a period of DHCA or selective cerebral perfusion using a patch of homograft tissue, and the back wall of the aorta and distal arch are then anastomosed.
- The aorta and proximal margin of the homograft patch is anastomosed to the proximal main PA (Damus–Kaye–Stansel anastomosis Fig. 12.11d). may be made in several ways depending on the geometry of the arch. An easy way of making this anastomosis is to make an incision in the patch used to augment the arch, and anastomosing the MPA into this.
- During DHCA, the RA cannula is removed and atrial septectomy performed through the cannulation site; if necessary a separate incision may be made to achieve this.

Formation of shunt to restore pulmonary blood

- Pulmonary blood flow is secured with an RV-PA conduit, usually a 5mm Gore-Tex tube, from the right ventricle to the left (Sano) or right (modified Sano) pulmonary artery.

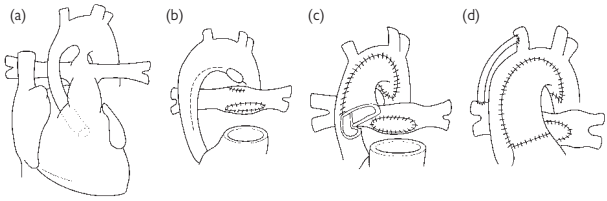


Fig 12.11 (a) Hypoplastic left heart. (b) Transected PDA, PA, and aortotomy. (c) Patch augmentation of aorta. (d) Anastomosis between aorta and PA, and B-T shunt.

- Alternatively a systemic to pulmonary artery shunt is constructed by anastomosing a 3–4mm PTFE graft between the innominate artery and the right PA, restoring flow to the pulmonary arteries.

Closure

Standard closure with epicardial pacing wires, LA catheter, meticulous hemostasis (blood products usually mandatory) and Ethibond to sternum.

Specific complications

- ↑PVR reduces pulmonary blood flow causing hypoxemia.
- ↓PVR increases pulmonary blood flow, reducing systemic cardiac output, and causing myocardial ischemia, acidosis, and cardiac arrest.
- Pulmonary venous hypertension.
- Right ventricular failure.

Outcomes

Mortality after the Norwood stage I varies from 5 to 30% between centers. Risk factors for poor outcomes include weight <2.5kg, no ASD at presentation, small ascending aorta, TR, late presentation, prematurity, mitral stenosis with aortic atresia. There is substantial mortality between stages, with mortality of 5–15% between stage I and II. The best centers achieve an 80% survival to completion of the Fontan circulation, usually performed at 4–5 years. 80% of children who reach school age are developmentally normal. Neonatal transplant is performed at a few centres worldwide for this condition, with a surgical mortality of 14%, and a further mortality of 20% of those waiting for transplant. The scarcity of donors makes this an option only in a few centers.

BT shunt vs. RV PA (Sano) conduit

Over the last 5 years, use of the RV to PA (Sano) shunt has increased due to apparently better outcomes. However, large centers with previously good results using B-T shunts have not shown reproducible benefit, and a multicenter trial from the USA has not shown benefit of one over the other. Postoperatively, RV-PA conduits may allow a more stable course and easier manipulation of competing vascular beds; however, an appropriately sized BT shunt should not lead to difficulties in this matter.

Glenn shunt and hemi-Fontan

Key facts

- These are both types of superior cavopulmonary anastomosis (see 📖 p600), and they are usually performed as an intermediate stage in staged surgical correction of a single ventricle (📖 p586), usually 3–6 months after the Norwood stage I (📖 p588) to minimize the time the RV has to cope with two circulations and considerable volume overload.
- The bidirectional superior cavopulmonary (bidirectional Glenn) shunt consists of an anastomosis between the SVC and the pulmonary circulation, with the proximal pulmonary artery oversewn.
- It can be performed without bypass off-pump, under cardioplegic arrest, or DHCA.
- The hemi-Fontan procedure, which involves anastomosis of the RA to the right PA, with the proximal PA oversewn and the SVC outlet oversewn, is rarely used today—it is a staging procedure for the lateral tunnel fenestrated Fontan which itself is not commonly used.

Set-up

- Triple lumen lines are not placed in the upper great veins because of the risk of thrombosis of the cavopulmonary anastomosis or PA; however, a single lumen SVC monitoring line is necessary.
- Standard preparation for resternotomy is used, including external defibrillator paddles and preparation of the groin or neck (<2 years).
- The bidirectional Glenn shunt and hemi-Fontan procedures are performed via a median sternotomy.
- Standard aorto-bicaval cannulation is used in the bidirectional Glenn procedure, and aorto-atrial cannulation is used in the hemi-Fontan.
- Right atrial cannulation with DHCA may be used to avoid trauma to the SVC.
- Bypass is commenced, the heart vented, and the patient cooled.
- The pulmonary artery, branch arteries, aorta, SVC, and RA are completely dissected, and the azygos vein is ligated.
- If a left SVC is present it is dissected free, so that a left cavopulmonary anastomosis can be performed.

Procedure

Bidirectional Glenn shunt

- This is the first stage of a TCPC (📖 p592), and the second stage of the Norwood sequence of operations (📖 p588). It prepares the way for the final completion of the total cavopulmonary connection, the currently favored form of Fontan circulation (📖 p592).
- As a stage II Norwood operation, the B-T or RV-PA shunt must be ligated and divided, any residual pulmonary artery stenoses or hypoplasia must be augmented, and lesions such as restrictive ASD, recoarctation or AV valve regurgitation addressed.
- If this is the first stage of a Fontan operation for other anomalies, lesions such as AV valve regurgitation or outflow tract obstruction must be repaired. If the MPA has forward flow it may be ligated at this stage.

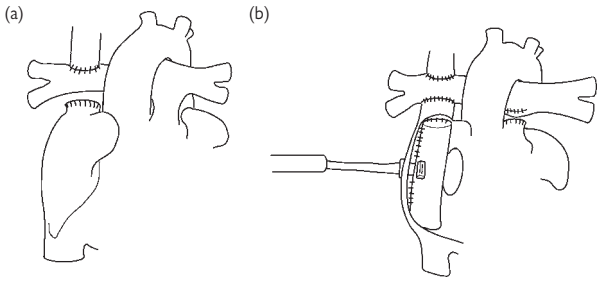


Fig. 12.12 Superior cavopulmonary anastomosis. (a) Bidirectional Glenn shunt. (b) Hemi-Fontan.

- The SVC is transected at its junction with the RA; the RA is oversewn.
- The SVC is then anastomosed to a longitudinal incision in the RPA, establishing continuity between the SVC and both PAs (Fig. 12.12).
- The PAP measured in the SVC must be <20mmHg postoperatively, and the transpulmonary gradient should be <10mmHg (PAP-CVP).

Hemi-Fontan

- If deep hypothermic circulatory arrest is used CPB is discontinued at 20°C and the patient exsanguinated into the venous reservoir.
- The dissection is completed.
- The main PA is divided, and the proximal end oversewn.
- The RA is opened towards the IVC, and an ASD created.
- The superior cavoatrial junction is opened and extended up the SVC.
- The right PA is incised inferiorly and superiorly so that the opening in the cavoatrial junction can be anastomosed to these incisions.
- A piece of homograft is used to close the SVC orifice into the RA.
- If a left SVC is present this is divided and anastomosed to the left PA.

Closure

- Atrial and ventricular temporary epicardial pacing wires are placed.
- Ventilation is resumed, the heart is de-aired and weaned from bypass.
- After placing mediastinal and pleural drains, and securing hemostasis, and patch closure of the pericardium, the chest is closed.

Classic Glenn shunt

The classic Glenn shunt is no longer used, but it is still encountered in adult patients. It was performed via a right–left posterolateral thoracotomy in the fourth intercostal space. The right PA was divided along the medial border of the SVC, and the proximal PA stump oversewn. The distal end of the right PA was anastomosed to the SVC.

Fontan circulation

Key facts

- A '***Fontan circulation***' is the physiological end result of staged procedures for univentricular hearts where the systemic veins are anastomosed directly to the pulmonary circulation, in an attempt to separate the systemic and pulmonary circulations which are otherwise mixed.
- The circulation completely depends on blood crossing a low-gradient pulmonary vasculature.

Indications for a Fontan

It is indicated for any lesion with only one well-developed ventricle:

- Tricuspid atresia/absent right AV connection (📖 p584) or mitral atresia/absent left AV connection.
- Double inlet left ventricle/double inlet right ventricle.
- Hypoplastic left heart syndrome (📖 p586).
- Unbalanced AVSD (📖 p568).
- Straddling left or right atrioventricular valves.
- Pulmonary atresia with intact ventricular septum and small right ventricle (📖 p614).

Principles of the Fontan circulation

- In a normal biventricular heart, the systemic and pulmonary circulations are in ***series***: each one is supported by a ventricle.
- In patients born with a single ventricle, the two circulations are in ***parallel***: systemic and pulmonary venous blood mixes.
- In the Fontan circulation, the systemic and pulmonary circulations are ***separated*** and placed in ***series*** without the interposition of a ventricle (***Fontan operation***): there are three types:
  - Atriopulmonary connection (📖 p590), e.g., ***hemi Fontan***.
  - Intracardiac total cavopulmonary connection (📖 p594).
  - Extracardiac total cavopulmonary connection (📖 p594).

Technical overview

The Fontan circulation is achieved in two stages (in hypoplastic left heart (📖 p586) a ***Norwood procedure*** precedes the two other stages):

- ***Bidirectional Glenn shunt,*** bidirectional superior cavopulmonary shunt (📖 p601): the systemic venous drainage from the ***upper body***, i.e., via the superior caval vein(s), is disconnected from the RA and anastomosed to the right PA (in the case of bilateral SVCs, bilateral CP shunts are constructed, with the LSVC to the LPA).
- ***Fontan operation,*** completion of total cavopulmonary connection (📖 p594): the systemic venous return from the ***lower body***, via the inferior caval vein, is redirected to the pulmonary circulation by disconnecting it from the RA and directing it to the right PA via an extracardiac conduit (usually Gore-Tex tube), or from within the RA using native tissue or a patch to baffle flow (lateral tunnel). If a connection is made between systemic venous circulation and pulmonary venous circulation, this is called a fenestration. The need for a fenestration in the Fontan circuit is controversial (📖 p522).

Physiological considerations

In the Fontan circulation, pulmonary blood flow is dependent on the passive return of systemic venous blood into the pulmonary vasculature, rather than by the RV. This is only possible if:

- Pulmonary vascular resistance is low (mean PAP≤15mmHg).
- Ventricular function is good.
- Atrioventricular valves are competent.
- There are no obstructive lesions in the circulation.
- There are no shunts (e.g., large collaterals, previous B-T shunt).

Requirements for a Fontan circulation to work

The original '10 commandments' indicating suitability for Fontan procedure (now not so rigidly applied) are listed here: the two current mandatory criteria are asterisked.

- Age >4.
- Sinus rhythm.
- Normal caval drainage.
- Normal volume RA.
- *Mean PAP≤15mmHg.
- PVR <4 units/m^2.
- PA diameter: aorta diameter ≥0.75.
- *LV ejection fraction ≥0.6.
- Competent atrioventricular valve(s).
- No shunts.

One-stage versus two-stage procedures

Two stages have lower mortality and morbidity than if the cavopulmonary connections are completed in a single operation; this may be because the first operation reduces the volume load on the single ventricle, allowing some of the resulting hypertrophy to regress, and diastolic function to improve, before the final operation. The bidirectional superior cavopulmonary connection can only be undertaken after 3 months of age when the neonatal pulmonary vascular resistance will definitely have fallen—with a higher pulmonary artery pressure the cavopulmonary shunt will fail. Thus, many neonates require a palliative operation before they can undergo staged Fontan.

Systemic-pulmonary versus cavopulmonary shunts

- Cavopulmonary shunts redirect venous, desaturated blood into the pulmonary circulation; systemic-pulmonary shunt redirects arterial, saturated blood into the pulmonary circulation.
- CP shunts divert 30–50% of systemic venous blood from the heart to the lungs, thus ***reducing*** the volume load on the heart; systemic-pulmonary shunts add to the pulmonary circulation, thus ***increasing*** the volume load on the heart.

Fontan operation

Key facts

- The Fontan operation has been described as, 'the surgical endpoint for patients whose cardiac anomalies do not allow two ventricle circulation.' It is the final stage of the three-stage repair of hypoplastic left heart syndrome (📖 p586).
- All forms of the Fontan operation aim to divert systemic (with or without coronary sinus) venous return to the pulmonary circulation.
- There are many modifications, but currently the extracardiac total cavopulmonary connection is most widely used.
- Only pulmonary venous (with or without coronary sinus) blood returns to the common atrial chamber and single ventricle.
- The entire cardiac output is provided by one ventricle, with the separated pulmonary and systemic circulations in series rather than in parallel.
- Pulmonary blood flow is dependent on CVP and outflow resistance.
- The lateral tunnel Fontan is usually performed with cardioplegic arrest, whereas the external conduit Fontan can be performed on the warm beating heart or off-pump.

Setup

- Standard invasive monitoring and general anesthetic, with antifibrinolytic.
- Median sternotomy, heparinization at 300U/kg.
- An atrial catheter is inserted directly into the common atrium via the right atrial wall or RSPV.
- Standard CPB with distal ascending aortic cannula, bicaval cannulation with the caval purse-strings as far distal as possible, and patient cooled to 32°C.
- The right and left PA are completely dissected free before CPB.
- If required the aorta is cross-clamped and cold blood cardioplegia given.

Procedure

Lateral-tunnel fenestrated Fontan operation

- The pulmonary trunk is transected and the ends oversewn, if this has not already been performed in previous surgery (hemi-Fontan 📖 p590, or Norwood 📖 p588).
- The RA is opened.
- The SVC is divided, and the distal portion anastomosed to the superior aspect of the right PA, and the proximal portion anastomosed to the inferior aspect of the right PA.
- If the Fontan procedure follows a previous Glenn shunt procedure the proximal SVC will have been oversewn: this is opened up, and enlarged if necessary onto the roof of the RA.
- A large ASD is created avoiding the conducting tissues.
- A PTFE baffle is sewn to channel IVC blood into the SVC and right PA.

- Typically a fenestration is placed on the edge of the lateral tunnel using a 4mm aortic punch: a purse string may be placed around this and brought out through the RA so that the size can be adjusted.
- The right atriotomy is closed.

Extracardiac conduit fenestrated Fontan operation

- May be performed off-pump, with CPB on the beating heart, or on the arrested heart.
- Usually the previous operation of a bidirectional Glenn also includes any other repairs/modifications needed, so that this final stage need only establish continuity between the inferior caval vein and the pulmonary circulation.
- A 20mm Gore-tex tube is first anastomosed to the inferior RPA.
- The IVC is transected at its junction with the RA, the RA is oversewn.
- The other end of the Gore-tex tube is anastomosed to the IVC.
- A fenestration may be formed by anastomosing a punch-hole in the Gore-tex tube to the RA wall. The need for a fenestration in the Fontan circuit is controversial—many authorities feel it is unnecessary if patient selection is optimal, however many centers continue to add a fenestration as routine.
- Chest drains are placed in both pleural cavities and the mediastinum.
- An atrial pressure monitoring line must be placed through the RA wall to allow fluid management ('CVP' lines no longer measure atrial pressure).

Closure

- An atrial catheter and atrial and ventricular temporary epicardial pacing wires are placed.
- Ventilation is resumed, the heart is de-aired and weaned from bypass.
- After placing mediastinal and pleural drains and securing hemostasis, the chest is closed with Ethibond sutures to the sternum.

Care after Fontan operation

Key facts

Monitoring of the common atrial pressure and the PAPs is mandatory in the first 24h. The difference between these is the transpulmonary gradient. The PA pressure and the transpulmonary gradient usually fall after extubation.

In the immediate postoperative period

- Blood loss causes profound hemodynamic instability—hemostasis must be meticulous.
- May need aggressive filling to maintain BP.
- Avoid the use of vasoconstrictors: elevated PVR will cause Fontan circulation to fail as minimal resistance across pulmonary bed is required given absence of chamber functioning as RV.
- Maintain sinus rhythm.
- Aim to extubate early to improve Fontan hemodynamics (positive pressure ventilation reduces systemic venous filling of pulmonary circulation).
- Respiratory failure and acidosis cause pulmonary vascular resistance to rise, therefore 'early' extubation must be rationally performed.
- The chest drains should not be removed in the first 48h.
- Atrial line can be removed on day 1 postoperatively.
- Heparinize when bleeding has stopped.

Hospital course

- The patient should be anticoagulated after the acute postoperative period, once the intracardiac lines and pacing wires have been removed.
- The chest drains will often continue to drain for several days, and may drain for weeks. Effusions are usually serous, occasionally chylous. If they drain for >2 weeks, the patient may be carefully assessed with cardiac catheter and any pathway obstructions corrected. Longer drainage may necessitate:
  - Cardiac catheter intervention: enlarge/create fenestration, stent PA stenoses.
  - Total parenteral nutrition.
  - Regime of fluid restriction, diuretics, pulmonary arterial vasodilators, ACE inhibitors, continuous oxygen administration, aggressive mobilization.
  - Intravenous ocreotide—only anecdotal evidence.
- Reversal of the Fontan circulation—this is high risk, particularly if the decision to undertake this management is delayed.

Early complications

- Elevated Fontan pathway pressure.
- Ascites, hepatomegaly, pleural effusions from high CVP required to generate pulmonary blood flow.
- Poor pulmonary blood flow secondary to ↑PVR, positive pressure ventilation, loss of sinus rhythm.

- LV dysfunction.
- Extremely poor hemodynamics requiring surgical reversal or fenestration of the circuit.

Long-term outcomes

- Atrial or Fontan pathway thrombosis.
- Patient should be maintained on anticoagulation for life.
- 5- and 10-year survival with modern techniques of Fontan completion are in the order of 95% and 90%.

Late complications

- Atrial arrhythmias: probably lower incidence with lateral tunnel and extracardiac conduit TCPC.
- Protein-losing enteropathy: loss of protein from the GI tract—unknown cause—debilitating and life threatening, occurs in up to 15%.
- Thromboembolic complications—up to 20%.
- Development of collateral vessels.
- Ventricular failure.
- Plastic bronchitis with bronchial casts—these may cause airway obstruction.
- 'Failing' Fontan—inability of the circulation to deliver adequate cardiac output—if heart transplantation undertaken for this, mortality is up to 65%.

Pathophysiologic sequelae

- Fontan circulation has the following pathophysiological sequelae:
  - Acute increases in PVR can decrease systemic venous return, particularly impeding portal venous flow.
  - High venous return causes increased lymph production, and contributes to impaired lymph reabsorption.
  - Chronic lymphatic hypertension contributes to protein losing enteropathy and plastic bronchitis.
- Aortopulmonary collaterals develop and may result in large left-to-right shunts contributing to ventricular volume overload.

Pulmonary artery banding

Key facts

- This is a palliative operation that aims to limit pulmonary blood flow by reducing the circumference of the main PA. Its indications have reduced over the years.
- A band is placed around the PA trunk, via either a left thoracotomy or median sternotomy. The band is sequentially tightened while PA pressures are monitored, until the PA pressure is <50% of the systolic pressure, or the lowest possible PA pressure with acceptable SaO_2 is achieved.

Indications

Pulmonary artery banding was first used for children with single ventricles, or large left-to-right shunts, e.g., VSD, AV canal, and truncus arteriosus. Improvements in surgical techniques and myocardial preservation mean that most of these disorders can be corrected, and PA banding is now reserved for a few defects:

- ***VSD***, multiple VSD, coarctation aorta with large VSD, large VSD with major non-cardiac malformation.
- In ***late presentation TGA*** (📖 p624) to prepare the LV for arterial switch.
- After ***Mustard/Senning***, to prepare the LV prior to converting to arterial switch (📖 pp626–629).
- Double inlet ventricle and no PS.
- Tricuspid atresia with no PS.
- AVSD unsuitable for repair (pulmonary hypertension, AV discordance, recent RSV bronchiolitis).
- HLHS—bilateral PA bands in combination with arterial duct stenting and atrial septostomy.
- ccTGA with TR.

Contraindications

- In single-ventricle anomalies where the aorta arises from muscular infundibulum, (e.g. DILV, TA with TGA), there is a potential for subaortic obstruction, and PA banding may hasten this. These patients should have Damus-Kaye-Stans Norwood (📖 p588).
- Not recommended in common arterial trunk.

Setup

- Standard invasive monitoring and general anesthesia, avoid inotropes to reduce acute changes in SVR and PVR.
- Median sternotomy or left thoracotomy with pericardium opened anterior to the phrenic nerve, and LA appendage retracted with a stay suture. Rarely, anterior extrapleural approach.

Procedure

The band may be a 3mm cotton tape, thick braided silk, 2mm strip of Gore-Tex, silk tape, or siliconized silk tape.

Placement of PA band

- As the PA may be dilated and extremely thin-walled, the safest method of encircling the PA is to place the band around both PA and proximal aorta, then develop the plane between aorta and PA, through which the free edge of the band may be pulled.
- The band is sequentially tightened and may be secured with Ligaclips so that it can be released later with angioplasty.

Degree of banding

- This is a clinical judgement based on:
- Systemic pressure.
- PAP.
- Echo appearance of the septum.

The degree of banding depends on the underlying lesion—the principle is to balance the systemic and pulmonary blood flow. Trusler and Mustard calculated optimal length of the band using the patient's weight:

- Biventricular heart: 20mm + weight in kg.
- Complex malformations with mixing: 24mm + weight in kg.
- Malformations proceeding to Fontan circulations: 22mm + weight in kg.

This formula can be used as a guide when banding, but must be adjusted according to the conditions found at operation, e.g., in VSD aim to reduce PAP to near normal, i.e., 30mmHg, with near normal O_2 sats.

Removal of PA band

- This is normally performed through a median sternotomy with CPB as dictated by the planned surgical repair (Fontan or arterial switch).
- All the band must be removed as adherent fragments cause strictures.
- The area banded is either opened and patched anteriorly or excised, anastomosing the PA trunk in an end-to-end fashion.
- The manner in which the chest is closed depends on the nature of the surgical correction performed at the same time.

Complications

These patients may be very unstable postoperatively, and some centers advocate deep sedation and paralysis for the first 24h. If there is sudden hemodynamic instability with loss of cardiac output, the safest option on the PICU is reopening of the chest and release of the band.

- RVOT obstruction.
- Severe right PA stenosis and left PA hypertension if the band migrates.
- Congestive cardiac failure if the band is too loose.
- Hypoxemia and shunt reversal if the band is too tight.
- Pulmonary valve damage.

Shunts

Key facts

- Systemic–pulmonary shunts provide a source of blood flow to the pulmonary arterial system from the systemic circulation, thus improving the oxygen saturations in cyanotic heart disease. They may facilitate pulmonary artery growth where these are hypoplastic.
- Indications include tricuspid atresia, tetralogy of Fallot, hypoplastic PAs, anomalous coronary artery with associated complex cardiac anomalies, pulmonary atresia with intact ventricular septum, pulmonary atresia with VSD, Ebstein anomaly, single-ventricle anomaly with pulmonary or aortic atresia.
- The ideal systemic–pulmonary artery shunt has several features:
  - Technically simple to perform.
  - Provide adequate but not excessive pulmonary blood flow.
  - Good long-term patency.
  - Causes no cardiopulmonary abnormalities.
  - Technically simple to close.

Historical

- ***Blalock–Taussig–Thomas shunt*** involved connecting the subclavian artery directly to the PA, thus sacrificing subclavian artery flow. Pulmonary blood flow was limited by the size of the subclavian artery, and the shunt grew with the child.
- ***Waterston shunt*** consisted of anastomosis of the ascending aorta directly to the right PA. Shunt size was difficult to regulate, with the risk of pulmonary overflow and the development of pulmonary vascular disease. The right PA was often distorted by this anastomosis. It may be difficult to close these shunts.
- ***Potts shunt*** connected the descending aorta directly to the left PA. This also raises the risk of over-shunting, and led to left PA distortion

Commonly used shunts in the modern era

Modified Blalock–Taussig shunt (mBT shunt)

- These are now the most commonly performed systemic-pulmonary artery shunts, with a resurgence in their use as part of the Norwood stage I procedure.
- An ePTFE (Gore-tex) tube is anastomosed end-to-side to the subclavian artery without dividing it, and end-to-side to the PA.
- Approach: either via lateral thoracotomy on the side of the innominate artery (i.e., on side opposite arch), or via a midline sternotomy in very small neonates, or when cardiopulmonary bypass may be required (if the child is hemodynamically unstable)if the duct has to be closed.
- 3.5–4mm shunts in neonates; flow regulated by size of subclavian artery.

Central shunt

- A short PTFE tube is used between the ascending aorta and the main PA, both anastomoses performed end-to-side.
- Particularly useful with small branch pulmonary arteries, encouraging their growth, useful in small infants
- Prevents distortion of the branch pulmonary arteries, equal blood flow to both lungs, low occlusion rate, easily closed.
- Performed through a median sternotomy.

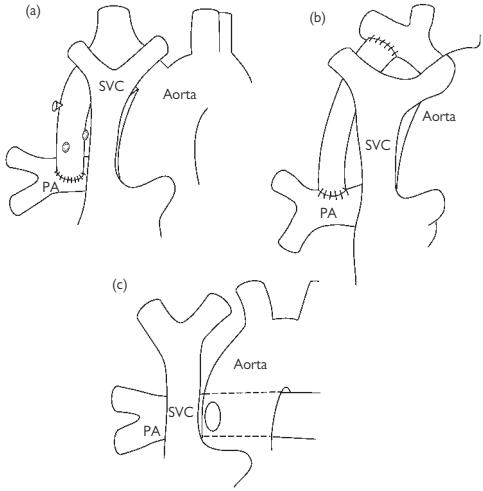


Fig. 12.13 Aortopulmonary shunts: (a) Blalock–Taussig. (b) Modified Balock–Taussig. (c) Cooley.

Melbourne/Mee shunt

The anastomosis of diminutive main PA end-to-side to ascending aorta to promote growth of PA in pulmonary atresia/VSD/MAPCAs.

Cavopulmonary shunt/bidirectional Glenn shunt

- This is the anastomosis of the superior caval vein to the right PA (and/or LSVC to LPA) (see also 📖 p590).
- PVR must be low, therefore contraindicated under 3–6 months.
- Does not increase volume load on the heart, commonly used as the first stage of a total cavopulmonary connection, or Fontan circulation.

Sano shunt

- PTFE tube connecting the RV to the left PA as part of the Norwood stage I operation, instead of a modified BT shunt.
- Prevents the low diastolic pressure and poor coronary perfusion seen with 'classical' Norwood stage I with mBT shunt
- Appears to lead to more stable immediate postoperative hemodynamics, and may reduce the mortality 'between stages'.
- 4mm shunt in neonates <2.5 kg, 5mm >2.5kg.
- Requires a right ventriculotomy, becomes obstructive over 3 months.
- When the shunt is taken from the RV to the RPA (Birmingham modification), it is easier to access at the second stage.

Complications of shunts

Sudden death, occlusion—patients should remain on aspirin ± dipyridamole until shunt takedown, infection, PA distortion/stenosis up to 30%, false aneurysm formation with rupture, phrenic nerve injury (left mBT shunt), Horner's syndrome, seroma formation from leak through the porous PTFE tube.

TAPVD and cor triatrium

Key facts

- In total anomalous pulmonary venous drainage (TAPVD) (1–3% of congenital malformations) there is no direct communication between the pulmonary veins and the LA: neonates rely on mixing of the pulmonary and systemic circulation through an ASD for survival.
- Total obstruction of pulmonary venous return is a neonatal emergency
- In cor triatrium, (0.1% of congenital cardiac malformations), the pulmonary veins drain into an antechamber separated from the true LA by a diaphragm.

Morphology

In TAPVD pulmonary veins converge to form a confluence behind the pericardial sack that connects to the systemic venous system in three ways (Fig. 12.14).

- ***Supracardiac*** (45%): pulmonary venous return via innominate or SVC.
- ***Cardiac*** (25%): pulmonary return via the RA or coronary sinus.
- ***Infracardiac*** (25%): via the hepatic or portal vein and hence IVC.
- ***Mixed type*** (5%): with features of >1 of the types listed.

Pathophysiology

This depends on the presence and degree of pulmonary obstruction.

- Non-obstructed TAPVD is effectively a large left-to-right shunt, with greatly increased pulmonary blood flow, and RV volume overload, resulting in ↑ PVR, RVH, CCF.
- Cyanosis occurs because fully oxygenated pulmonary venous blood mixes with desaturated systemic venous return, the resultant mix reaches the left side of the heart via an ASD or PFO, which is mandatory for survival, and the ductus arteriosus while patent.
- Obstruction to pulmonary venous flow in TAPVD, which is a common feature of infracardiac TAPVD results in pulmonary edema, ↑↑PVR, and severe cyanosis, although RV volume overload is less marked.
- It is common for the blood in all four chambers to have the same oxygen saturations because of mixing.

Presentation

Infants with non-obstructed TAPVC (uncommon) present a few months after birth with mild cyanosis and signs of heart failure, a systolic murmur, and loud second heart sound with fixed splitting. Infants with obstructed TAPVC, in contrast, present with profound cyanosis and signs of severe heart failure within hours to days of birth.

Investigations

- The CXR in non-obstructed TAPVD shows increased pulmonary vascularity, a prominent pulmonary border, enlarged right heart border and after 6 months in supracardiac TAPVD a ‘figure of 8’.
- In obstructed TAPVD the cardiac and pulmonary shadows are normal in size, but the lung fields show signs of severe pulmonary venous obstruction.
- TTE and TEE are diagnostic.

- Cardiac catheterization is only required if there are associated cardiac anomalies, or to examine individual pulmonary vein anatomy and identify obstructions.

Surgical management

TAPVD repair is described on 📖 p604.

Indications for surgery

- Obstructed TAPVD is a surgical emergency, even within the first few weeks of life.
- Non-obstructed TAPVD is an indication for urgent surgery.

Outcome

The hospital mortality of TAPVD repair is 5–10%.

Cor triatrium

In cor triatrium the pulmonary venous collecting chamber is ***intrapericardial*** and lies superior and posterior to the LA, from which it is separated by a ***fibromuscular diaphragm***. There may or may not be a communication between the two chambers; the RA may communicate with one or both chambers. The true LA contains the LA appendage and the mitral valve. Pulmonary venous flow reaches the left side of the heart by passing across the diaphragm into the LA, and thus is subject to obstruction at this site. If the collecting chamber communicates with the RA but not the LA, then the results are similar to TAPVC.

Presentation and investigation

Depending on the size of communication between the chamber, LA, and RA this may present like a large ASD, or like obstructed TAPVD. Echo is diagnostic: should delineate the pulmonary venous anatomy, as well as differentiating between cor triatrium and supramitral membrane, where the membrane lies between the mitral valve and the LAA: in cor triatriatum, the mitral valve and the LAA are on the same side of the diaphragm.

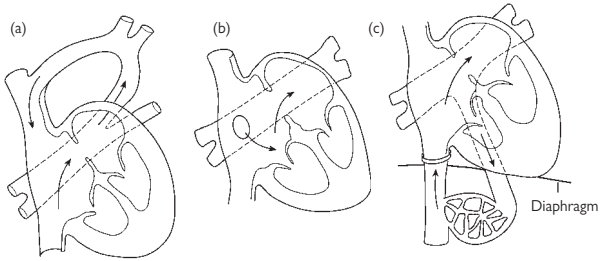


Fig. 12.14 The three types of anomalous pulmonary venous connection. (a) Supracardiac. (b) Cardiac. (c) Infracardiac.

Repair of TAPVD

Key facts

- Pulmonary venous obstruction is a surgical emergency in most neonates with TAPVD.
- Surgical technique is dictated by the anatomy (📖 p602).

Set-up

- Median sternotomy.
- Standard aorto-atrial cardiopulmonary bypass and DHCA.

Supracardiac TAPVC

- The heart is elevated out of the mediastinum to the right, to expose the LA and the pulmonary venous confluence behind the posterior pericardium (Fig. 12.15).
- The vertical vein is identified and either ligated and divided to provide mobility to the confluence, or left intact as a pressure vent for the non-compliant LV after surgery.
- Incisions are made in the pulmonary venous confluence and the base of the LA appendage, and the two anastomosed with 7/0 Prolene.

Infracardiac TAPVC

- This usually presents in the neonatal period with a shocked patient.
- The confluence behind the heart is exposed as previously described.
- The confluence and the LA appendage are incised and anastomosed together (Fig. 12.15).
- The vertical vein may be ligated or left open as previously described.

Intracardiac TAPVC

- This is performed under cardioplegic arrest.
- The RA is opened.
- If the pulmonary venous confluence opens directly into the RA, if the ASD is large enough, a patch is sutured around the defect and the pulmonary venous orifice, thus channelling pulmonary venous blood to the LA through the ASD.
- If the ASD is too small, it can be enlarged before this patch is applied.
- If the pulmonary veins drain to the RA through the coronary sinus, a cutback is made from the PFO/ASD to the coronary sinus, partially unroofing it. The defect and the coronary sinus are then patched over.

Mixed type

Repair depends on the anatomy. Most often there is a mixture of veins draining into the coronary sinus and elsewhere.

Postoperative management

In neonates following emergency surgery, it is advisable to leave the chest open. The pulmonary vasculature may be treated with milrinone, sodium nitroprusside, or inhaled nitric oxide to reduce the PAPs. The LV will be restrictive initially, and LA and PA pressures should be monitored. To reduce the risk of pulmonary hypertensive crises, patients should be paralyzed.

Complications

Early: pulmonary hypertensive crises, pulmonary edema, phrenic nerve injury. ***Late***: pulmonary venous obstruction, pulmonary vein stenosis, anastomotic stricture.

Outcomes

Pulmonary venous stenosis usually develops within the first postoperative year, and occurs in 10–20% of patients in reported series. The mortality after surgical reintervention is in the order of 25%, with risk factors being early representation and bilateral stenotic disease.

Management of cor triatrium

If the patient presents acutely with pulmonary venous obstruction, the aim of management is to stabilize the patient with a view to performing surgery urgently. When there is good communication across the diaphragm, elective repair is sufficient. Surgical repair is performed using cardiopulmonary bypass and cardioplegic arrest. The RA is opened, and the PFO or ASD enlarged to allow access to the LA. The diaphragm is visualized, and the LA appendage and mitral valve identified. The diaphragm is then resected, and the position of the pulmonary veins ascertained. The ASD is closed either directly or with a patch if necessary. The outcome from surgery depends on associated defects, and therefore ranges from 5 to 25% in published series.



Fig. 12.15 Correction of total anomalous pulmonary venous drainage. (a) Intracardiac. (b) Infracardiac.

Truncus arteriosus

Key facts

- There is a single arterial trunk arising from the ventricles, with a single valve with semilunar leaflets, and a subvalvar VSD. The PAs and the aorta both arise from this single arterial trunk (Fig. 12.16).
- This comprises 3% of congenital cardiac anomalies. 30–40% of patients with common arterial trunk have microdeletions in chromosome band 22q11.2 and have DiGeorge syndrome or velocardiofacial syndrome.

Morphology

The valve between the ventricles and the trunk is called the ***truncal valve*** and may be tricuspid (60%), bicuspid (25%), quadricuspid (10–15%), or, rarely, unicuspid. This valve is usually dysplastic. There is always continuity between the truncal valve and the mitral valve. The truncal valve overrides the VSD and in 50% is equally committed to both ventricles, in the remainder usually arises predominantly from the RV. The LV is relatively normal, but the RV is nearly always hypertrophied. The ***VSD*** is subarterial, with its roof made from the truncal valve, and its posterior border muscular (80%) or perimembranous (20%). The coronary arteries are often displaced caudally, and occasionally arise from the PA. The aortic arch may be left (60%) or right-sided. There may be arch interruption (10%), usually type B (between LCCA and LSCA). There may be coronary artery anomalies, such as single coronary artery or intramural coronary arteries. The most commonly used classification system, ***Collett and Edwards***, is based on the pattern of the origin of the PAs:

- Type I (60–70%): a single PA arises from the trunk and gives rise to the left and right branches.
- Type II (20%): the left and right PAs originate adjacent to each other from the trunk.
- Type III (10%): the left and right PAs arise separately from the lateral aspects of the trunk.
- Type IV: where the pulmonary arteries do not arise from the common trunk; now recognized as a form of pulmonary atresia with VSD.

Pathophysiology

There is complete mixing of systemic and pulmonary venous blood with a left-to-right shunt at the ventricular level leading to cyanosis, and pulmonary volume overload. AS the PVR falls in the weeks after birth pulmonary overload worsens, resulting in CHF, and eventual pulmonary hypertension in survivors by 6 months as a result of irreversible vascular changes. Truncal valve insufficiency exacerbates pulmonary overload and adversely affects survival, whereas pulmonary stenosis has a survival benefit. Coronary perfusion is jeopardized by run-off into the PA. A PDA is mandatory for survival in patients with an interrupted aortic arch. Untreated, most do not survive beyond 6 months.

Presentation

Patients present with symptoms of CHF (failure to thrive, poor feeding, tachypnea, sweating) in the first few weeks of life, with mild cyanosis in



Fig. 12.16 Truncus arteriosus.

1/3 of cases which increases in frequency and severity with age. Older children may present with worsening cyanosis as a result of Eisenmenger syndrome (📖 p558). On examination a collapsing arterial pulse, left parasternal systolic murmur and single second heart sound are features.

Investigations

- EKG shows RVH and LVH with right axis deviation and P pulmonale.
- CXR shows cardiomegaly, absent pulmonary trunk, and increased pulmonary vascularity.
- Echo is diagnostic. The pulmonary artery origins should be elucidated and there should be an attempt to characterize the truncal valve and its function. The coronary pattern may be determined. The arch must be investigated for hypoplasia or interruption.
- Cardiac catheterization is not necessary unless there is late presentation, in which case the aim is to measure PVR of both pulmonary vascular beds (they may differ). Pulmonary vascular resistance of >10U/m^2 is a contraindication to surgical repair.

Management

In patients who present subacutely with CHF, initial treatment is with diuretics. In patients presenting with circulatory collapse, there is usually associated aortic arch interruption, and a prostaglandin infusion is commenced to maintain the PDA, along with other supportive measures. Surgery should be performed soon after presentation, and urgently in the patient with uninterrupted arch.

Outcome

Early mortality after surgery is now in the order of 5–10%. Risk factors include arch interruption, severe truncal valve regurgitation, and coronary artery anomalies. In the longer term, the RV–PA conduit requires intervention as the child grows.

Truncus arteriosus repair

Key facts

- Truncus arteriosus is an indication for urgent operation to prevent pulmonary vascular disease.
- The PAs are removed from their origin from the arterial trunk, which is committed to the LV when closing the VSD.
- The PAs are then connected to the RV either via a valved conduit such as a homograft, or by anastomosing the PAs directly to the right ventriculotomy and completing the repair with a patch.

Set-up

- Standard invasive monitoring and general anesthesia.
- Asymmetrical pericardiotomy so that a large patch of pericardium can be harvested: this is placed in 0.6% glutaraldehyde.
- Median sternotomy, heparinization at 300U/kg.
- PDA, left and right PA and aorta dissected free. The RPA and LPA are taped so that they can be occluded as soon as CPB is commenced to prevent pulmonary flooding and systemic hypoperfusion.
- Standard CPB with distal ascending aortic cannula, two-stage venous cannula, RSPV vent and patient cooled to 20°C if DHCA planned.

Procedure

Division of the pulmonary arteries from the truncal root

- The right and left PA are excised from the truncal root, taking care not to damage the coronary arteries (Fig. 12.17).
- The defect in the truncal root is closed with a pericardial patch, or if small, by primary anastomosis.

VSD repair

- A vertical right ventriculotomy is made and the VSD is closed with a patch of bovine pericardium/Gore-tex/glutaraldehyde fixed autologous pericardium.
- The right ventriculotomy will be used to recreate the RVOT later.

ASD repair

- The ASD can be repaired wither via a right atriotomy or through the right ventriculotomy and tricuspid valve.
- It is closed with a pericardial patch, and the atriotomy is closed.

Reconstruction of the RVOT

The RVOT is reconstructed by anastomosing a valved conduit (homograft or bovine jugular) to the PA bifurcation distally, and then to the right ventriculotomy proximally. The PAs may be directly anastomosed to the RV with anterior augmentation with a patch if a valved conduit is unavailable.

Closure

- The heart is de-aired in the usual fashion (📖 p114).
- The patient is rewarmed and weaned from bypass, PA, LA, and RA catheters are placed.
- TEE may be used to assess residual shunts
- The chest is closed in the standard fashion.



Fig. 12.17 Truncus arteriosus repair. (a) Division of pulmonary arteries from the truncal artery. (b) Closure of the defect in the trunal artery primarily, and with a patch. (c) VSD patch closure, and reconstruction of the RVOT with a pulmonary homograft.

Specific complications

- Truncal valve regurgitation and LV failure.
- Persistent left-to-right shunts.
- Conduit failure.
- Pulmonary hypertension and RV failure.

Aortopulmonary window

Key facts

- This is a rare malformation: <1% of congenital cardiac anomalies.
- It consists of an abnormal communication between aorta and pulmonary artery distal to the semilunar valves.

Morphology

Associated defects include interrupted aortic arch, anomalous origin of right or left coronary artery, VSD (📖 p564), PDA (📖 p572), aortic atresia, tetralogy of Fallot (📖 p616). Divided into three types (Fig. 12.18):

- Type I: proximal.
- Type II: distal, close to RPA.
- Type III: anomalous origin of either PA from the ascending aorta.

Pathophysiology

There is a high-flow, left-to-right shunt at the arterial level with pulmonary over-circulation, resulting in rapidly progressive CHF and early development of high PVR and Eisenmenger syndrome. Survival past 1 year is rare if untreated.

Presentation

Heart failure secondary to left-to-right shunt. Wide pulse pressures, systolic flow murmur. There is pulmonary hypertension, loud pulmonary component to second heart sound.

Investigation

- EKG: right axis deviation, biventricular hypertrophy.
- CXR: cardiomegaly, plethoric lung fields, pulmonary edema.
- Echo should be diagnostic, but may be difficult to see. The differential diagnoses of VSD, PDA, and common arterial trunk should be excluded.

Management

Infants in heart failure should be treated medically, and if this fails to stabilize them, early surgery is warranted. If heart failure can be controlled, surgery can be performed electively, but as the development of pulmonary vascular disease is unpredictable, still early.

Surgery

Set-up

- Median sternotomy, pericardium opened towards the right.
- Anatomy of the window, right PA and coronary arteries identified.
- Systemic heparinization.
- Aortic cannula placed close to innominate artery origin.
- Bicaval cannulation if there is no VSD or ASD, otherwise aorto-bicaval cannulation, caval snares, RSPV vent.
- Complete dissection of ascending aorta, and pulmonary arteries so they can be snared with silastic loops immediately after CPB commenced to prevent flooding the pulmonary circulation.
- Clamp aorta distal to the defect, arrest heart with cardioplegia.

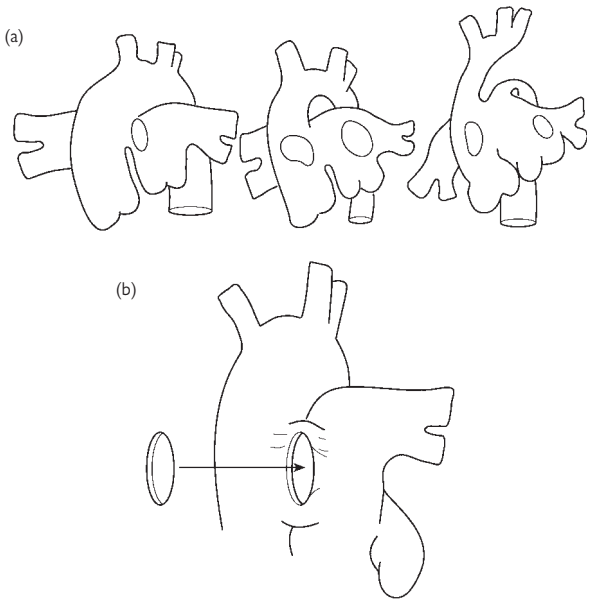


Fig. 12.18 Aortopulmonary window. (a) Spectrum of abnormalities. (b) PTFE sutured to posterior, superior, and inferior walls of the defect and then incorporated into the suture line closing the incision.

Procedure

- Vertical incision in the anterior portion of the window or aorta.
- Identify the defect, the coronary arteries and the origin of the branch PA.
- Close defect with a patch of Gore-tex, close aortotomy, rewarm, de-air, reperfuse, and come off bypass.
- Other less favoured techniques include the double sandwich technique with a patch of Gore-tex, ligation, clamping and division, and division and suture. There are a few case reports of catheter occlusion.
- Bypass is weaned using hypocarbia and pulmonary vasodilators such as SNP, milrinone, and inhaled nitric oxide to reduce PVR.
- The chest is closed in the usual fashion with a mediastinal drain.

Outcomes

Repair of simple AP window has low mortality, but in association with other complex defects, surgical mortality is up to 20%.

Specific complications

- Pulmonary artery stenosis.
- Residual aortopulmonary defects.

Pulmonary stenosis

Key facts

This defect makes up 8–10% of all congenital cardiac defects, and usually involves valvar PS, but includes subvalvar and supravalvar PS.

Morphology

The ***pulmonary valve is stenotic***, and usually is tricuspid with fusion of the commissures, thus producing a dome-like structure with a variable central opening. There is often a ***PFO or ASD*** (40%). There may be ***RV hypertrophy***, even at birth, and ***subvalvar stenosis*** from infundibular muscle hypertrophy. There may be ***poststenotic dilatation*** of the pulmonary arteries.

Pathophysiology

The stenotic pulmonary valve impedes blood flow to the pulmonary arteries, leading to RV hypertrophy and ***cyanosis***. If there is an atrial communication, there is ***right-to-left shunting***. The level of stenosis is assessed by measuring the systolic pressure gradient between the RV and the main PA; a gradient >50mmHg is considered significant.

Presentation

- ***Neonatal presentation***: neonates with severe pulmonary stenosis usually present critically unwell within a week of birth with severe cyanosis and tachypnea, a systolic murmur with an ejection click at the ULSE, hepatomegaly due to severe TR.
- ***Infants/children***: these may be largely asymptomatic, with a murmur picked up incidentally. However, moderate stenosis may lead to RV hypertrophy, with increasingly inadequate pulmonary blood flow leading to exertional dyspnea.

Investigation

- EKG: RA dilatation and RV hypertrophy.
- CXR: there may be moderate cardiomegaly, prominent main PA from poststenotic dilatation, and, if severe PS, oligemic lung fields.
- Echo is diagnostic, showing a stenotic, dysplastic, sometimes hypoplastic pulmonary valve, RV hypertrophy, and tricuspid valve incompetence. The gradient across the RVOT may be measured, as well as RV pressures. The size of the tricuspid valve, the RV cavity must be assessed in order to plan management.
- Cardiac catheterization is not indicated in mild stenosis, but is an essential prelude to interventional balloon dilatation of the stenotic valve. An accurate peak-to-peak gradient can be measured, the size of the RV assessed, and the distal pulmonary vasculature imaged. The RV pressure may be measured.

Management

Trivial (<25mmHg) or mild (<50mmHg) gradients across the pulmonary valve do not require intervention, but must be followed up regularly.

Balloon valuloplasty

Pulmonary stenosis that is ***moderate (50–79mmHg) or severe (>80mmHg***) both in the critically ill neonate and in the less symptomatic child is largely managed with interventional cardiological techniques. The critically ill neonate will require resuscitation, with the use of PGE_1 to maintain pulmonary blood flow, before transfer to the cath lab or operating room. ***Balloon pulmonary valvuloplasty*** has superseded the use of surgery to achieve an unobstructed RVOT in cases of isolated pulmonary stenosis with normal pulmonary valve annulus size. The aim is to achieve a gradient <30mmHg. If there is subvalvar stenosis secondary to muscle hypertrophy, this often recedes in the months following valvotomy. β-blockers may be used if infundibular stenosis is >50mmHg immediately after balloon valvuloplasty.

Surgical valvuloplasty

If the pulmonary valve annulus is small and the RV cavity appears small, surgery may be chosen in order to augment the annulus with a ***transannular patch***. Similarly, if there is fixed infundibular stenosis, the infundibulum may be augmented with this patch, having ***divided obstructing muscle bundles***, and extending the transannular incision further into the RVOT.

Adults

In adults with PS, the long-term outcomes may be good without intervention, but there is a trend towards intervention in those with moderate to severe PS in order to prevent RV dysfunction in the long term, due to lowered pre- and post-exercise cardiac indices and to prevent exercise-induced hemodynamic abnormalities.

Complications

- Residual/recurrent valve stenosis (10% at 6 months, but late recurrence is 1–2%).
- Persistent postoperative hypoxia.
- Pulmonary valve regurgitation (up to 80% at 10 years).
- The need for reintervention is higher after balloon valvuloplasty in neonates, with up to 25% of patients requiring reintervention for PS or associated defects compared to 8–10% of older children.

Outcomes

Mortality from balloon or open valvotomy is in the region of 1–3%, with RV size approaching normal in 90% of patients. 5- and 10-year freedom from reintervention is in the order of 88% and 84% respectively.

Pulmonary atresia with intact ventricular septum

Key facts

- Pulmonary atresia with an intact ventricular septum is a rare defect characterized by variable RV development in association with an imperforate pulmonary valve.
- There may be extensive right ventriculocoronary connections.
- It makes up about 1% of congenital cardiac defects.

Morphology

The pulmonary valve is ***completely atretic***, replaced by an ***imperforate fibrous membrane***, with ***thick RV hypertrophy*** and ***infundibular stenosis*** or atresia. The main and branch pulmonary arteries are usually of good size with variable ***poststenotic dilatation***. The RV and tricuspid valve have variable degrees of hypoplasia. There is always an ***interatrial communication.*** A substantial number of patients (50–60%) have ***fistulous communications*** from the hypertrophied RV to the coronary arteries. The ***coronary arteries may be disconnected*** from the aorta, and intimal fibromuscular dysplasia may develop in these abnormal connections causing stenosis.

Pathophysiology and natural history

The atretic or stenosed pulmonary valve restricts blood flow into the pulmonary arteries, causing ***hypoxia***, with the arterial duct being the only source of blood. ***Cyanosis*** becomes worse as the duct closes. When the ***coronary circulation is RV-dependent*** (20%), coronary perfusion is dependent on the hypertensive RV, thus there may be coronary steal into the RV during diastole when the RV pressure falls. Stenoses within the fistulas may also lead to ***myocardial ischemia***. If pulmonary blood flow can be maintained by ductal patency until surgical intervention, the 1- and 5-year survival is up to 80% and 75%. Sudden death, angina, ventricular arrhythmias, congestive heart failure, and the consequences of prolonged hypoxia are all risks for these patients.

Presentation

Most of these patients present in the neonatal period with worsening cyanosis and tachypnea as the arterial duct closes. Examination reveals central cyanosis, normal arterial pulses, single first and second heart sounds, pansystolic murmur of tricuspid regurgitation.

Investigation

- EKG usually shows sinus rhythm, ↓RV forces, LV dominance.
- CXR shows mild cardiomegaly and oligemic lung fields.
- Echo is diagnostic. The diameter of the tricuspid valve, the size of the RV cavity, TR, and the adequacy of the interatrial communication must be assessed. Ventriculocoronary connections may be identified.
- Cardiac catheterization is required to identify ventriculocoronary connections, and RV-dependent coronary circulations. The RV size and morphology may be further studied, and pressure measured.

Management

Percutaneous intervention

In some cases, an attempt may be made to cross the atretic valve using laser or radiofrequency and perform balloon valvuloplasty. The rate of complications is high, and often these patients will require further intervention to establish adequate pulmonary blood flow.

Surgery

The aim of management is to restore flow through the RV to the pulmonary circulation in order to decompress the RV and allow it to grow. However, the adequacy of the RV for two-ventricle repair is difficult to assess, and is generally based on the size of the tricuspid valve, as this correlates well with an adequate ventricular cavity.

In general, a tricuspid valve measuring at least three standard deviations of the mean normal valve should be able to support the pulmonary circulation (z score up to −3). These patients can undergo treatment to restore pulmonary blood flow, either by interventional cardiological techniques or surgery, with the aim of a two-ventricle circulation.

If the RV is hypoplastic with a small tricuspid valve and ventriculocoronary connections, pulmonary blood flow should be improved with systemic-pulmonary artery shunt; a transannular patch may be performed if there is hope that the RV may grow. These patients will probably require transplantation or Fontan palliation in the future.

Outcomes

Neonatal palliation

This may be surgical, with pulmonary valvotomy and systemic–pulmonary shunt, or RVOT patch; or by catheter intervention, decompressing the RV by radiofrequency ablation of the pulmonary valve with balloon dilatation. Any of these initial palliations may be followed by further interventions, surgical or catheter based. Mortality rates for these procedures are high, with mortality approaching 40% for surgical palliation and 20% for percutaneous intervention. In selected patients, without RV-dependent coronary circulations, RVOT patch has a low mortality.

Repair

The largest multicenter studies suggest that about 50% of children presenting with this condition should be suitable for a two-ventricle repair, with the remainder requiring Fontan palliation (📖 p594) or transplant (📖 pp504–519), with survival rates of about 60% at 15 years.

Tetralogy of Fallot

Key facts

- ToF makes up 10% of congenital heart defects and is the commonest cyanotic lesion. 25% of patients have DiGeorge syndrome or deletion/abnormalities of chromosome 22.
- It is usually an isolated defect, and is characterized by four lesions caused by anterior deviation of the outlet septum:
  - Pulmonary stenosis (at the infundibulum, valve or the PA).
  - RV hypertrophy.
  - Overriding aorta.
  - VSD.
- Children present soon after birth with cyanosis, depending on the severity of the pulmonary stenosis, and are usually managed with surgical correction.

Morphology

- The underlying morphological abnormality causing the features of tetralogy is anterior and leftward deviation of the outlet septum.
- There is a large perimembranous outlet VSD.
- Hypoplasia of the RVOT causing obstruction at multiple levels (10% pulmonary valve stenosis, 50% infundibular stenosis, 30% combined).
- The pulmonary valve is usually involved, and is bicuspid in 60% of ToF, with a hypoplastic annulus; however, it is usually not the narrowest point of the outflow tract. In 7% of patients with ToF there is complete pulmonary atresia; in 5% the pulmonary valve is completely absent and the PAs are aneurysmal.
- In about 5% of patients the LAD arises from the right coronary artery, crossing the right outflow tract just below the pulmonary valve annulus to achieve its interventricular course.
- There is a right arch in 25%.

Pathophysiology

The RVOT obstruction ***reduces pulmonary blood flow*** and also increases the RV pressure causing ***a right-to-left shunt*** across the large, unrestricted VSD. Both these effects cause ***cyanosis***. Immediately after birth, the ductus arteriosus supplies blood to the pulmonary vasculature, but after it closes there may be a sudden onset of cyanosis, and the RV will start to hypertrophy. In a small subset, the RVOT obstruction is so mild as to avoid cyanosis: ***'acyanotic' or 'pink' tetralogy***. The pulmonary vasculature is protected from the effects of the VSD by the outflow obstruction, and some patients may be well balanced for years.

Presentation and natural history

In children with a very hypoplastic annulus, the presentation is usually soon after birth, with cyanosis due to the obstruction in pulmonary blood flow and thus increased flow across the VSD from right to left. In children with a larger pulmonary valve annulus, where obstruction is muscular at the infundibular level, presentation is usually later, and may be with 'tet spells'—cyanotic episodes that occur when there is muscular

spasm causing acute obstruction to RV outflow and reversal of the shunt (also due to falling SVR with exercise). 'Squatting' is a characteristic compensatory maneuver, seen in right-to-left shunts developing on exercise. Hemoptysis may be present late. Examination shows RV heave and a systolic ejection murmur. The degree of RVOT obstruction is the main determinant of outcome in untreated ToF: most children develop signs and symptoms by 6 months of age, and <10% survive to the age of 21 without surgical intervention. 25% of untreated patients die within 1 year of life, 40% of untreated patients die within 4 years, and 95% of untreated patients die within 40 years.

Investigations

- EKG shows right-axis deviation, RVH and RA hypertrophy.
- CXR shows a 'boot-shaped' heart as a result of RVH and the small or absent MPA; and decreased pulmonary vascularity, ↑RA.
- Echo is diagnostic, and defines the severity and extent of the RVOT obstruction, the size of the pulmonary valve annulus, the size and position of the VSD, the coronary artery arrangement, and any associated anomalies.
- Cardiac catheterization is used if there is a need to further image the coronary arteries, or if there are concerns about the size or arborization of the pulmonary arteries.

Surgical management

There is an increasing trend towards earlier complete repair, but there is still a place for early palliation with a ***systemic–PA shunt*** (📖 p600) followed by repair at a later stage. A sensible approach is to aim to repair all tetralogies between 3–6 months. If they present with cyanotic spells they can be repaired earlier, or if institutional policy dictates, undergo a ***B-T shunt*** (📖 p601) with elective repair when they are bigger.

Palliative procedures

- Modified B-T shunt (📖 p601): this is an interposition shunt, usually a Gore-tex tube, placed from the subclavian artery to the left or right PA. It may be performed through a thoracotomy or a sternotomy. Many centers no longer perform palliative surgery such as the modified B-T shunt on symptomatic infants with ToF, instead opting for complete repair.
- RVOT patch or conduit: if the PAs are too small to accept a B-T shunt, flow to the PAs may be secured with an outflow tract patch or with a limiting RV-PA conduit, e.g. 5mm Gore-tex tube.

Tetralogy of Fallot repair

Key facts

- To repair this defect, the VSD must be closed, and the RVOT obstruction must be relieved. The RVOT obstruction may be at several levels: muscular infundibular stenosis, pulmonary valvar stenosis, small pulmonary annulus, small main pulmonary artery.
- There is some evidence that the need for future reoperation may be less if the pulmonary valve annulus is preserved.
- A patch across the RVOT acts as an akinetic, dilated area in the RV that does not contribute to useful function: the aim should be to preserve the PV annulus where possible, and limit as much as possible the size of the RVOT patch.
- Variations in morphology which impact on surgical technique include:
  - Right, left, and bifurcation PA stenosis.
  - Anomalous origins of the coronary arteries.
  - Pulmonary atresia.
  - Absent pulmonary valve.

Set-up

- Standard invasive monitoring, general anesthesia.
- Median sternotomy, heparinization at 300U/kg.
- Asymmetrical pericardiotomy, pericardial patch harvested and treated in 0.6% glutaraldehyde.
- PDA and left and right PA are dissected completely free, and the aorta is freed from root to head and neck vessels.
- If the PDA is patent it is ligated.
- Standard aorto-bicaval CPB, patient cooled to 25°C.
- The aorta is cross-clamped and cold cardioplegia given into the root.

Procedure

To repair this defect, the VSD must be closed, and the RVOT obstruction must be relieved. The RVOT obstruction may be at several levels: muscular infundibular stenosis, pulmonary valvar stenosis, small pulmonary annulus, small main PA. There is some evidence that the need for future reoperation may be less if the pulmonary valve annulus is preserved. Similarly, a patch across the RVOT acts as an akinetic dilated area in the RV that does not contribute to useful function. In operating, the aim should be to preserve the PV annulus where possible, and limit as much as possible the size of the RVOT patch.

Right atriotomy

- VSD closure through RA (📖 p566). Care is taken to avoid the conduction tissues by placing sutures into fibrous tissue rather than muscle tissue.
- There are frequently multiple VSDs, most frequently in the muscular portion anterior to the septal band: these are closed with patches.
- Resection/division of muscular obstruction through the RA, and assessment of tricuspid valve.

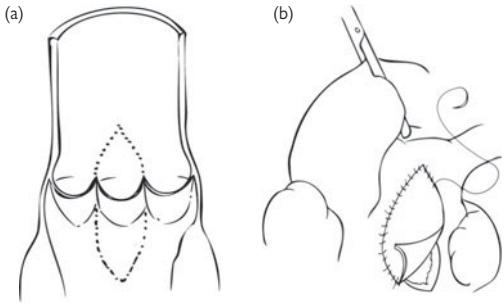


Fig. 12.19 Enlargement of the pulmonary annulus. (a) With a homograft monocusp. (b) With a pericardial patch.

RVOT reconstruction

- Open the MPA to inspect the valve and the RVOT through the valve.
- If PV annulus adequate, preserve the valve and patch the infundibulum and/or the MPA.
- If the PV annulus is too small, the PA incision is taken across the annulus and into the RVOT if necessary—the whole can be patched over with autologous pericardium, bovine pericardium, Gore-tex, 'monocusp' patch with some form of valvar substitute (Fig. 12.19).
- In principle, as small a patch as possible is used on the RV aspect in order to preserve RV function.
- In principle, leave the annulus intact if at all possible, even if this means going back on bypass to redo it if the RV pressures are too high at the end of the operation.

Closure

- After the VSD is closed the right heart is filled with saline, the atrial cannula replaced, and bypass recommenced.
- The patch on the RVOT can be sutured to the PA and right ventriculotomy during rewarming.
- RV pressure measurement can be performed at the end of the procedure with a pressure transducer. Generally, if RV pressure is 80% or less, this can be tolerated. Above this pressure, further RVOT obstruction relief may be necessary.

Specific complications

The operative mortality from repair of ToF is 1–2%. The early outcomes for repair of ToF are now excellent. However, in the long term, pulmonary insufficiency causing RV dilatation and dysfunction may cause symptoms, and increase the risk of sudden death due to malignant arrhythmias. Early pulmonary valve replacement, before the development of symptoms, may protect against irreversible RV dilatation.

- Early complications: bleeding, heart block, 1%, restrictive RV physiology and RV dysfunction requiring ITU support.
- Late complications: residual/recurrent RVOT obstruction, pulmonary regurgitation, RV dilatation and dysfunction.

Absent pulmonary valve syndrome

Key facts

This is a variant of ToF, with rudimentary or absent pulmonary valve causing severe PR.

Morphology

The pulmonary annulus is narrowed with vestigial leaflets that are only minimally stenotic, the pulmonary arteries are often dilated placing pressure on the underlying tracheobroncial tree, and the VSD is similar to that in TF with PS.

Pathophysiology

The absence of a pulmonary valve leads to severe PR, and contributes to the massive PA dilatation. There may be airway compression from the dilated branch pulmonary arteries, but even deeper in the lung there may be small airway obstruction from abnormal pulmonary vasculature. Although there is usually some pulmonary stenosis, in this condition there is usually a net left-to-right shunt.

Presentation

These patients can be broadly categorized into two groups:

- Very sick infants or neonates with severe bronchial compression, with signs and symptoms of respiratory distress: tachypnea, wheeze, cyanosis, air-trapping, respiratory failure.
- Older children with less severe respiratory symptoms where the intracardiac manifestations of ToF are the dominant pathology.

There may be mild cyanosis. There is a hyperactive precordium. The first heart sound is normal, the second single due to absent pulmonary component, and there is a to-and-fro murmur over the pulmonary area. There may be atelectasis, hyperinflation, or consolidation of the lungs.

Investigations

- Echo is diagnostic, but may be difficult due to distortion of the cardiac anatomy by the huge PAs. Other anomalies must be identified.
- CT scan may delineate airway compression by the PAs, and identify intrapulmonary vascular abnormalities.
- Bronchoscopy or bronchogram are often necessary in those patients with a long postoperative ICU course due to difficulties weaning from respiratory support.

Management

In asymptomatic patients, surgery is deferred until later childhood. If there are significant airway problems, earlier surgery is necessary, but outcomes may be worse in this subgroup. Preoperative intubation and ventilation is a risk factor for poor outcome, and repair in the critically ill neonate has a high mortality rate.

Surgical strategies

Closure of the VSD and relief of RVOT obstruction as for isolated ToF (📖 p616).

- It may be necessary to plicate or resect pulmonary arterial wall in order to reduce their size and propensity to compress the airways.
- Alternatively, the ***Lecompte maneuver*** (📖 p626), in which the pulmonary arteries are brought anterior to the aorta, may be performed.
- There is debate about how to reconstruct the RVOT, with many centres advocating the insertion of a valved conduit to prevent the severe PR seen with the relatively high pulmonary resistances seen in this condition.

Complications

Morbidity arises mainly from the effects of airway compression. Many of these children have a prolonged postoperative course on the ICU, requiring invasive and non-invasive positive pressure ventilation. If a valved conduit is used in the reconstruction of the RVOT, then further intervention will be mandatory as the child grows. In neonates with APV, the postoperative mortality is 20–60% in historical series, although in older children the mortality is closer to that for ToF repair. There may be continuing respiratory symptoms in spite of adequate surgery. As with ToF, there must be continuing surveillance.

Pulmonary atresia with VSD

Key facts

- This cyanotic congenital cardiac anomaly is characterized by an underdeveloped RVOT, atretic pulmonary valve, a VSD with variable aortic override, and variable development of the PAs.
- It makes up about 3% of all congenital cardiac anomalies.
- There is a spectrum of severity dictated by the degree of PA development. 35% of these patients have 22q11.2 deletion, while 65% of patients with PA/VSD/MAPCAs have this anomaly.

Morphology

There is ***absence of continuity between the RV and the PA***, and the pulmonary artery development varies from normal-sized to entirely absent. The RV may be entirely disconnected from the MPA, connected by an atretic valve, or obstructed by infundibular muscle. Blood flow to the lungs may be from a ***PDA***, or from ***MAPCAs***. These patients are more likely to have more severe PA hypoplasia, and pulmonary vascular arborization abnormalities. The ***VSD*** is outlet, and the aortic override may be more than 50% (i.e., DORV) in 26–50%. There is a right aortic arch in 25–50%. RV is usually hypertrophied; LV and coronaries normal.

Pathophysiology

Good-sized, confluent pulmonary arteries supplied by a PDA

Intrapulmonary vascular patterns should be normal; when the duct closes, an alternative pulmonary inflow is needed, e.g., a systemic–PA shunt.

Hypoplastic pulmonary arteries or absent PDA

Intrapulmonary arborization is abnormal. Collateral supply to lungs may come from MAPCAs from the descending thoracic aorta, systemic arteries such as the subclavian or intercostals, bronchial arteries, or coronary arteries. These collaterals have a tendency to stenose: ***there may be different sources of blood for different segments of lung, all with different pressures***. The intrapulmonary vasculature may have abnormal distribution and distal stenoses. Even after reconstruction of the pulmonary arteries and reconnection to the RV, the distal vasculature may be so abnormal that if the VSD is closed, RV pressure would be too high to be sustainable, causing RV failure and circulatory collapse.

Presentation

Most patients present in the neonatal period with cyanosis and hypoxia as the duct closes. If there are well-developed collateral arteries, presentation may be delayed as these children may have good SaO_2, and hypoxia only occurs as growth outstrips pulmonary blood supply. Rarely, there may be CHF in the case of a large PDA or multiple large MAPCAs.

Examination

There may be central cyanosis, worse after ductal closure. There may be signs of ***DiGeorge/velocardiofacial syndrome***, developmental delay, or

growth retardation. First heart sound is normal, single second heart sound, continuous murmurs in the case of PDA or MAPCAs.

Investigation

- CXR: normal sized, boot-shaped heart, absent MPA shadow.
- Echo: this is diagnostic and quantifies size of the PAs and their blood supply. It is unable to define the course of collateral arteries adequately, and the intrapulmonary vascular arborization cannot be defined.
- If PAs are large and confluent, supplied by a PDA, catheterization is unnecessary. There are hypoplastic PAs with undefined or variable blood supply. It is also required after systemic–PA shunt to define the PA anatomy and diagnose damage or distortion by the shunt.
- MRI/CT: these define collateral arteries within the chest, and the size and distribution of the PAs, allowing the surgeon to plan incorporation of MAPCAs into the native pulmonary circulation.

Management

Good-sized, confluent pulmonary arteries supplied by a PDA

The adequacy of the pulmonary arteries may be determined using the ***McGoon ratio or Nakata index***. Repair is performed by closing the VSD and placing a valved conduit from the RV to the PA, or it may be staged by performing a palliative systemic–pulmonary artery shunt and then a repair when the child is bigger, allowing placement of a more satisfactory conduit from the RV to the PA. Mortality is 15%, and 5-year survival is 75%.

PA/VSD with hypoplastic central pulmonary arteries, no MAPCAs

Systemic–pulmonary artery shunt from the ascending aorta to the PAs via a direct connection (***Watterston shunt***) or a Gore-tex tube—this promotes growth of the central PAs. An RVOT patch may be used, or a limiting RV-PA conduit. Repair may only be possible after reconstruction of the central PAs, either with patches or with an interposition tube in the case of non-confluent PAs. The VSD can be closed once the PA reconstruction is adequate; however, if it is left open, a restrictive RV to PA conduit may be used to prevent overflow of blood to lungs.

PA/VSD/MAPCAs

Many children are stable, with good SaO_2, and repair may require several operations and reinterventions. However, MAPCAs and dependent lung territories may change, with stenoses in the MAPCAs and PVR changes in the lungs, leading to deterioration. The aim of surgery is to unifocalize the sources of supply to each lung, centralize those to the native central pulmonary arteries if possible, and connect these to the RV via a valved conduit. If the reconstructed pulmonary arteries supply adequate segments of lung, the VSD can be closed. This can be done as a single procedure, with a thoracotomy on the side of the descending aorta to mobilize the MAPCAs, then a sternotomy to perform the repair. VSD closure may be performed if there is perfusion of 15 lung segments.

Outcomes

10-year survival is roughly 80%, and freedom from reintervention at 10 years is <50%.

Transposition of great arteries

Key facts

- TGA is a congenital cardiac anomaly in which the aorta arises from the RV and the pulmonary trunk arises from the LV.
- Complete TGA is the most common cyanotic lesion presenting in the neonatal period, and makes up 5% of congenital cardiac lesions. In 90% it is isolated, with no associated extracardiac malformations.
- Infants present with cyanosis and in the absence of an ASD or VSD infants die within a few hours of closure of the ductus arteriosus.
- A corrective arterial switch procedure can be performed in neonates with low mortality.

Morphology

In simple TGA, the ventricular septum is intact (70%); however, a VSD is present in 30%. In 1/3 the coronary anatomy is abnormal; the commonest aberrant pattern is the left circumflex arising from the right coronary (22%). However, the coronary arteries arise from the 'facing' sinuses in 99%. There is LVOTO (pulmonary or subpulmonary) in 10% of simple TGA and 30% of TGA with VSD.

Pathophysiology and natural history

- TGA results in the systemic and pulmonary circulations operating in parallel rather than in series: i.e., non-oxygenated venous blood passes through the RV into the aorta, while oxygenated pulmonary blood passes through the LV back to the pulmonary circulation.
- Mixing between the two circulations is mandatory for survival: this may occur through an ASD, PFO, VSD, or PDA.
- Inadequate mixing leads to cyanosis, acidosis, and death in the neonatal period.
- The LV ejecting to the low-resistance pulmonary bed does not develop its muscle mass in the normal manner, and after a few weeks loses the ability to produce normal cardiac output against systemic resistance.
- Patients with a VSD maintain this ability, as the LV pressure and volume load is closer to systemic values, and they also have better SaO_2 because of greater mixing of cardiac blood, but may have CHF, and in the long term develop irreversible pulmonary vascular disease.
- Untreated TGA causes death in 30% in the first week, 50% in the first month, and 90% by the first year. In long-term survivors, the complications of cyanosis develop: polycythemia, hyperviscosity, thrombocytopenia with bleeding complications.

Presentation

Male to female ratio is 2:1. Neonates present with cyanosis: if a VSD is present cyanosis is not so severe, but there may be signs of heart failure. Unless there is a shunt at the atrial or ventricular level, infants collapse within a few hours of ductus arteriosus closure. In the first 3 weeks as pulmonary resistance falls leading to overcirculation, signs of CHF dominate

(failure to thrive, tachypnea, sweating). Examination in TGA without VSD may be normal apart from cyanosis. In TGA/VSD: mild cyanosis, increased RV impulse, prominent grade 3–4/6 holosystolic murmur, third heart sound, mid-diastolic murmur, gallop rhythm, with hepatomegaly.

Investigations

- EKG shows RV hypertrophy in isolated TGA, and LVH if patent VSD.
- CXR: with intact septum there may be classic '***egg-on-a-string***' cardiac silhouette (narrow mediastinum and enlarged heart), with VSD present, there may be cardiomegaly and increased pulmonary vascularity.
- Echocardiography is diagnostic demonstrating the bifurcating pulmonary artery arising from the LV. The echo should further delineate the relationship of the great arteries, the presence and adequacy of shunts such as ASD, VSD, PDA, the coronary anatomy, and other malformations, such as coarctation.
- Cardiac catheterization is occasionally useful to characterize VSDs, or to improve mixing, e.g., for balloon atrial septostomy. In older children, it may be required to assess PVR, or to test the adequacy of left ventricular preparation for a late arterial switch operation.

Surgical management

Surgical repair (📖 pp626–629) is well established, effective and mandatory for TGA, and ideally performed within the first few weeks of life.

Congenitally corrected transposition

Congenitally corrected transposition of the great arteries (ccTGA) is a very different problem to TGA, with a more insidious course, and no well-defined surgical solution.

- The main abnormality is reversed morphology of the RV and LV. As in TGA the aorta arises from the RV and the pulmonary trunk from the LV, but there is atrioventricular discordance, i.e., the RA connects with the LV and the LA connects with the RV. The pulmonary and systemic circulations are therefore in series (rather than in parallel as in TGA).
- In the long-term the RV is unable to sustain a normal cardiac output against systemic vascular resistance and RV failure occurs in the third or fourth decade.
- Associated lesions which may pose additional problems include VSD (📖 p564), pulmonary atresia (📖 p612), and tricuspid regurgitation.
- Patients usually present late in their first decade with exercise intolerance. Bradycardia caused by complete heart block may be present.
- EKG may show AV block. Cardiac catheterization quantifies shunts and delineate anatomy, particularly when associated with pulmonary atresia.
- Management is controversial: PA banding early in life may preserve morphological LV function, restore septal symmetry and reduce systemic AV valve (tricusid) regurgitation. Results of the double switch or Senning-Rastelli operation to "correct" the anatomy are best in the first decade of life.

Arterial switch (Jatene)

Key facts

- This is the operation of choice for neonates presenting with TGA/IVS or TGA/VSD, and the related DORV with subpulmonary VSD.
- It is a single-stage anatomical correction consisting of transection of the aorta and PA, anastomosis of the PA to the aortic root, anastomosis of the ascending aorta to the pulmonary root, with both coronary arteries reimplanted into the neo-aorta, and closure of the ASD (Fig. 12.20).
- After balloon atrial septostomy, if the neonate is well palliated, the operation is performed electively in the first 2 weeks of life. If BAS does not solve hypoxia, prostaglandin is restarted and the operation should be performed within the next 24–48h.

Contraindications

- Intramural coronary arteries were historically a contraindication.
- LV outflow obstruction.
- Abnormal pulmonary valve (may require Ross + switch).
- Late presentation with deconditioned LV.

Set-up

- Standard invasive monitoring and general anesthesia.
- Median sternotomy, heparinization at 300U/kg.
- Asymmetrical pericardiotomy so that a large patch of pericardium can be harvested: this is placed in 0.6% glutaraldehyde.
- Aorta, head/neck vessels, and PAs mobilized to hilum.
- Aortic cannulation high, RA cannula; if VSD, bicaval cannulation.
- Duct ligated and divided after establishment of bypass.
- PA branches snared with silastic loops.
- RSPV vent, patient cooled to 20–25°C.

Procedure

Division of PDA and aorta

- PDA ligated and divided and both ends are oversewn.
- Distal ascending aorta is clamped and cold blood cardioplegia given into the root.
- The aorta is transected about 3–5mm above coronary ostial bulge.

Coronary transfer and reimplantation of coronaries

- Coronary ostia excised (right first) with surrounding button of aortic wall, mobilized, and the resultant deficiency in the aortic sinuses repaired with harvested pericardium to form the neo-MPA.
- MPA transected just proximal to bifurcation.
- 'Trapdoor' incisions made in the facing sinuses of this neo-aorta, to receive the coronary buttons.
- Coronary buttons sutured to the trapdoor incisions (left first).

Lecompte manoeuvre and aortic anastomosis

- ***Lecompte manoeuvre*** brings the neo-MPA anterior to the neo-aorta. This maneuver avoids the need for an interposition graft to connect the new pulmonary root to the pulmonary bifurcation, by bringing the pulmonary bifurcation anterior to the ascending aorta.
- The neo-aortic root is anastomosed to the ascending aorta.

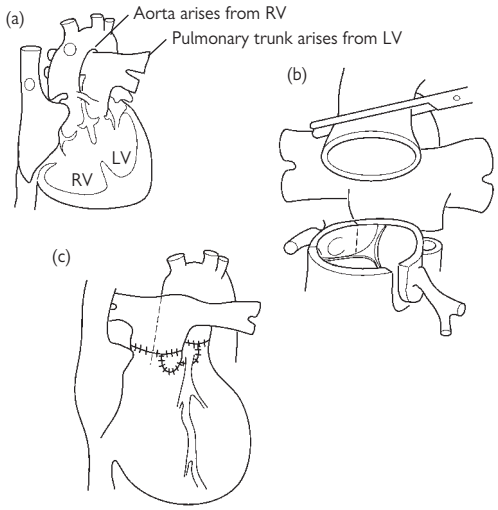


Fig. 12.20 Arterial switch (Jatene) operation. (a) Anatomy of TGA. (b) Division of the pulmonary trunk, aorta and coronary buttons. (c) Completed operation.

ASD repair and reconstruction of RVOT

- Bypass is discontinued, the atrial cannula is removed, and the ASD closed primarily through the right atriotomy which is then closed.
- The venous cannula is replaced, bypass resumed at full flow, rewarming commenced, the cross-clamp removed and the RSPV vent turned on.
- The neo-PA trunk is anastomosed to the PA bifurcation.
- TEE or epicardial US is mandatory post CPB, LA catheter is placed via LAA or RSPV.

Specific complications

- LV failure due to ischemia (due to coronary kinking/malposition) or inability to support the systemic vascular resistance: any evidence on post-bypass TEE of ischemia must be addressed, with a coronary bypass graft if the coronary implantation cannot be revised.
- Renal failure requiring peritoneal dialysis.
- Supravalvar pulmonary and/or aortic stenosis.
- Aortic root dilatation and regurgitation.

Outcomes

Surgical mortality for simple TGA with the commonest coronary patterns is 2%. Intramural coronaries, single coronary, or a long tortuous LMS are risk factors for a higher mortality. Overall mortality for all arterial switch operations is 10%. There is concern about the function of the neo-aortic valve in the long-term: incidence of AI is still 5–7%.

Other procedures for TGA

Key facts

- A number of alternatives to the single-stage Jatene arterial switch (📖 p624) may be indicated for TGA according to anatomy, associated lesions and presentation.
- Palliative ***Blalock–Taussig shunt*** (📖 p600) and ***PA banding*** (📖 p598) which may be followed later by complete repair by the Rastelli operation.
- Atrial switch operations such as the ***Mustard*** and ***Senning procedures.***
- The ***Nikaido operation*** and the ***Rastelli operation.***

Atrial switch: Mustard or Senning operations

Indications

- Previously for TGA ± VSD.
- Now as part of the double switch operation for ccTGA (📖 p625).
- Late presentation TGA unsuitable for LV preparation (see 📖 p624), e.g., pulmonary vascular disease.

Technique

In the Mustard operation, the atrial septum is excised, and a large patch of autologous pericardium is used to form an intra-atrial baffle, directing systemic venous flow to the mitral valve and the pulmonary venous flow to the tricuspid valve (Fig. 12.21). In the Senning operation, there is redirection of systemic and pulmonary venous pathways by cutting and folding of the native atrial tissues, usually without the use of extraneous material.

Complications

- Mustard: constriction of the venous pathways, baffle leaks, sinus node dysfunction, atrial arrhythmias.
- Senning: systemic venous pathway obstruction.

Outcomes

Surgical mortality is <5% for both these procedures. 20-year survival in the order of 80–90% for both these operations; VSD at time of surgery is a risk factor for late reoperations. Senning probably has a higher reintervention-free survival at 20 years.

PA band for TGA

In infants presenting late with TGA, where there is involution of LV muscle, the LV can be 'prepared' to support the systemic circulation with a PA band. This is combined with a systemic-pulmonary shunt, e.g., modified B-T shunt (📖 p600), in order not to limit pulmonary blood flow. Although the LV may be prepared within 2 weeks, it may be more practical to perform the PA band and shunt with elective switch operation in the following 3–6 months.

Nikaido operation

This operation is used for TGA with pulmonary stenosis or pulmonary annular hypoplasia. The aortic root is excised from the RVOT, requiring explantation, mobilization, and reimplantation of the coronary arteries. The outlet septum is excised and the aortic root is translocated posteriorly to lie over the morphologic LV. The VSD is closed, and an RV-PA conduit used.

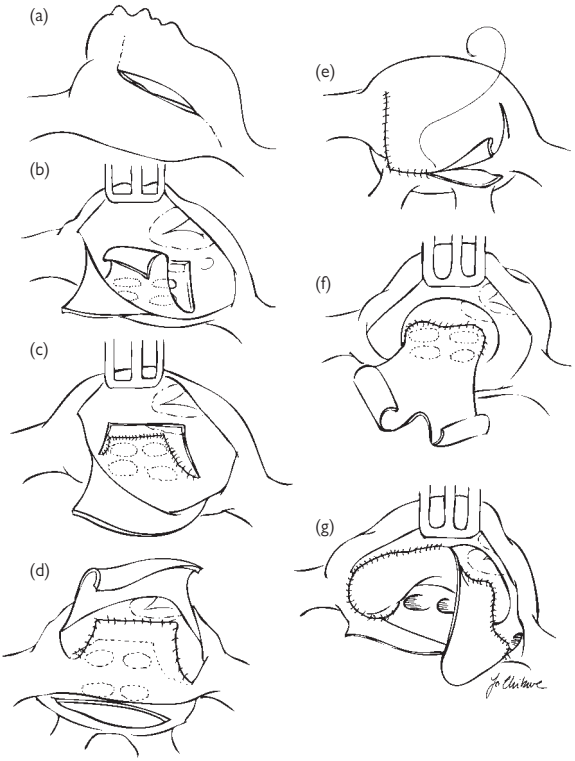


Fig. 12.21 Atrial switch operations. (a–e) Completed Senning procedure showing baffle created by septal flap redirecting IVC and SVC blood via the enlarged ASD into the left atrium, and hence LV and pulmonary arteries. Pulmonary vein inflow is directed across the tricuspid valve into the RV and aorta. (f) Mustard procedure showing baffle sutured via enlarged ASD around pulmonary veins. (g) Completed baffle will direct IVC and SVC blood in the atrium and hence pulmonary arteries. Pulmonary vein inflow is directed across the tricuspid valve into the RV and aorta.

Rastelli operation

This is a single-stage repair of d-transposition of the great arteries (d-TGA) which is used instead of the ***Jatene arterial switch*** (📖 p626) when the pulmonary valve is hypoplastic and there is a VSD. The VSD is not closed—rather a pathway from the LV to the aortic valve is baffled through the VSD with a patch of Dacron. This is easiest when the VSD is a large outlet perimembranous VSD. The RV is then connected to the PA with a valved conduit (homograft, bovine jugular vein, Hancock valved conduit). The RV-PA conduit will is changed when the child outgrows it, or earlier if there is conduit stenosis.

Double outlet right ventricle

Key facts

- This term is used when >50% of the circumference of each great artery arises from the morphologically RV.
- It is therefore a very heterogeneous group of hearts, particularly where the ventriculoarterial connections are not the dominant lesion.
- Here we will consider patients with DORV with normal atrial arrangements, AV connections, and balanced ventricles
- Pulmonary stenosis occurs in around 50% of hearts with DORV. Other commonly associated lesions are ASD, AVSD, LVOTO, and PDA.

Morphology

The ***VSD*** is the key lesion in these hearts and its position defines the pathophysiology. There is a spectrum of DORV from ***tetralogy of Fallot***-type physiology to ***transposition of the great arteries***-type physiology that is dependent on the position of the VSD. If both great arteries arise from the RV, closing the VSD would disconnect the LV from the circulation, as the VSD is the outlet from the left heart. Therefore, it is useful to classify DORV by the position of the VSD, as this is applicable to surgical repair:

- ***Subaortic VSD***: if the VSD is in a subaortic in position, with >50% of the aorta arising from the RV, the circulation is similar to a VSD, or in the presence of pulmonary stenosis, similar to ToF. Subaortic VSD is the most common arrangement in DORV.
- ***Subpulmonary VSD***: if the VSD is in a subpulmonary position, with >50% of the pulmonary artery arising from the RV (and usually all of the aorta), then the hemodynamics are similar to TGA with VSD, as blood from the LV is directed into the pulmonary arteries, and flow from the RV is directly into the aorta.
- ***Non-committed VSD***: when both great arteries arise exclusively from the RV, each one is usually supported on its own muscular infundibulum. The VSD therefore does not lie beneath either of the great arteries, and is called a non-committed VSD. The pathophysiology is that of a VSD, but if there is pulmonary stenosis, there is right-to-left shunting causing cyanosis.
- ***Doubly committed VSD***: when there is an absence of RV outlet septum, the VSD is committed to both great arteries and is said to be doubly committed. The pathophysiology is then similar to a doubly committed VSD. If there is pulmonary stenosis, there is right-to-left shunting causing cyanosis.

Presentation and natural history

This depends on the anatomic arrangement. If there is no pulmonary stenosis, there is heart failure and pulmonary hypertension. If pulmonary stenosis is present, there is cyanosis. Due to the heterogeneity of this condition, it is difficult to compare series of cases and surgical approaches. Overall 15-year survival for DORV with or without repair is approximately 60%, with 15-year survival after repair of non-complex DORV of 95%, and after repair of DORV/subpulmonary VSD of 90%.

Investigation

- EKG: RV hypertrophy.
- CXR: cardiomegaly. If there is pulmonary stenosis, there are diminished pulmonary vascular markings; if not, these may be increased.
- Echo is diagnostic, and additional information necessary for surgical decision-making includes location and margins of the VSD, straddling of the atrioventricular valves, the sizes of the great arteries and their relationship, distance of the tricuspid valve from the pulmonary valve, the presence of pulmonary stenosis, chordal connections of the tricuspid valve to the outlet septum, coronary anatomy, and arch abnormalities.
- Cardiac catheterization is useful at the more complex end of the spectrum to delineate the relationship of the VSD with the great arteries, the coronary artery anatomy, and the exact hemodynamics.

Management

Medical management depends on the anatomy. If there is inadequate pulmonary blood flow, the PDA must be kept open with PGE_1; conversely, if there is CHF, diuresis and management of the pulmonary overflow with manipulation of ventilation may be needed. Indications for surgery depend on anatomy. Across the spectrum of DORV, from VSD to ToF to TGA, indications for surgery are similar to these conditions.

Taussig–Bing anomaly

This is a form of DORV with ***subpulmonary VSD*** and ***malposed great arteries*** that is associated with ***subaortic stenosis, aortic arch abnormalities, and unusual coronary patterns***. (Taussig–Bing anomaly is often used to describe any DORV with subpulmonary VSD).

- The principles of repair are as previously described, with arterial switch and intracardiac baffle to direct ventricular flow into the neo-aorta.
- Outcomes are not as good as for uncomplicated DORV, with 15-year survival after surgery in the region of 90%.

Repair of DORV

- Surgical approach to ***DORV*** varies according to the location of the VSD:
- ***Subaortic VSD***: the intraventricular shunt should be closed early with an intraventricular baffle if there is a large shunt with pulmonary hypertension, CHF, failure to thrive. In the presence of pulmonary stenosis, decision-making is similar to ToF (📖 p616).
- ***Subpulmonary VSD***: this requires an arterial switch operation with closure of the VSD. There may be arch abnormalities, which can be corrected at the same operation. If there is pulmonary stenosis, intraventricular tunnel repair with ***Rastelli operation*** (📖 p629).
- ***Doubly committed VSD***: intraventricular baffle to close the intraventricular shunt and connect the LV to the aorta.
- ***Non-committed VSD***: often, palliation with PA band is performed to allow growth and re-evaluation of the intracardiac anatomy. If possible, the LV is baffled to the aorta with an intraventricular baffle, with or without an arterial switch operation. The baffle will be long and large, and the VSD may need to be enlarged. It may not be possible to repair this defect, and some of these are palliated with a Fontan circulation.

LVOT obstruction

Key facts

- LVOT obstruction occurs at the subvalvular, valvular, and supravalvular level, and repairs must be tailored to the specific morphology.
- Generally operation is performed in the presence of symptoms, or an asymptomatic but significant (>50mmHg) gradient.
- LVOT obstruction may be associated with obstructive lesions of the left heart such as coarctation or interruption of the aorta, mitral stenosis, or parachute mitral valve; it is also associated with VSD, AVSD, pulmonary stenosis, PDA.

Subvalvar stenosis

This is an acquired lesion (it rarely occurs at birth or in the first year of life, is usually associated with other lesions, and has a high recurrence rate after surgery) and may be caused by abnormal flow patterns resulting from abnormal angulation of the LVOT. Exceptions include patients with ***Shone complex*** or interrupted aortic arch, where it may be present at birth. It makes up about 15% of fixed LVOTO lesions. There are four anatomic forms: the first two account for 80% of subaortic stenoses and may involve the AMVL or the base of the aortic valve cusps:

- A ***discrete*** membrane of endocardial fold and fibrous tissue.
- A ***fibromuscular ridge*** at the crest of the septum.
- A ***diffuse***, tunnel-like narrowing of the LVOT consisting of circumferential fibromuscular hypertrophy with bulging of the septum and dynamic obstruction due to abnormal systolic anterior motion of the AMVL.
- ***Accessory or anomalous mitral valve*** tissue (this form of LVOTO is usually progressive, and leads eventually to aortic and mitral valve abnormalities, with 65% of patients developing AR).

Valvar aortic stenosis

This is the result of valvular maldevelopment. There are three types:

- ***Bicuspid*** aortic valve (70%): there are two leaflets instead of the usual three, they are usually thickened, with one leaflet having a fibrous ridge or raphe. There is usually some degree of commissural fusion, although this is not necessary for stenosis to occur.
- ***Tricuspid*** valve: there are three leaflets, they may be thickened and unequal, with variable degrees of commissural fusion. Usually this type of valve 'domes' in systole, with a central orifice, and is most amenable to valvotomy.
- ***Unicuspid*** valves are rare, and usually very stenotic.

Supravalvar stenosis

This comprises <10% of congenital LVOT obstruction and involves the aorta distal to the STJ. Most commonly it occurs sporadically (>50%), but it may occur as a part of Williams syndrome or in a familial form inherited in an autosomal dominant fashion. It is associated with coronary abnormalities such as diffuse narrowing of the left coronary, obstruction due to

leaflet fusion with the ridge, and narrowed left coronary ostium. There are two types:

- ***Discrete*** (75%): stenosis is usually due to an annular ridge within the lumen of the aorta, with a normal or reduced outer diameter. The aortic cusps are abnormal in about 30% of cases, and may adhere to the ridge, thus impairing filling of the coronary arteries. Other associations include peripheral pulmonary artery stenoses; when part of Williams syndrome it is associated with elfin facies, hypercalcemia, and severe developmental delay.
- ***Diffuse*** (<25%): the ascending aorta and origins of the head and neck vessels may be involved, with an abnormally thick-walled aorta due to intimal thickening and medial hypertrophy with increased fibrous and elastic tissues.

Presentation and management

Supra- and subvalvar stenosis

- Catheter peak-to-peak gradients of >50mmHg across the LVOT is an indication for surgical repair.
- Gradient of 30–50mmHg with symptoms, progression, ST/T wave changes at rest or exercise, or the Doppler peak systolic gradient is more than 50mmHg may indicate need for earlier surgery.

Valvar stenosis

In 10% this presents in the first year of life, usually as critical stenosis requiring urgent treatment, and with worse lifelong outcomes. In older children and adults the mortality arising from this diagnosis is low, with less need for urgent intervention. The neonate with critical aortic stenosis may present collapsed, with severe LV dysfunction, and will require a period of resuscitation with PGE_1, inotropes, and mechanical ventilation before intervention on the valve. In older patients, valve intervention may be performed in line with guidelines developed in the adult population.

LVOT obstruction repair

Valvar stenosis

Balloon valuloplasty

This is the mainstay of treatment of critical AS in neonates and infants, the goal being to relieve obstruction without causing AI. Balloon diameters of 80–100% of the valve annulus are used, aiming to reduce the peak-to-peak gradient by 50%.

Surgical valvuloplasty

Some centres continue to prefer surgical valvotomy, hoping to achieve a better result with fewer incidences of AI and longer reintervention-free survival. Where there are associated lesions requiring surgery or there is concern about the adequacy of the LV, surgery is preferred. Surgery involves CPB, an oblique aortotomy, and valvuloplasty consisting of ***commissurotomies*** and thinning of the leaflets where necessary.

Aortic valve replacement

In children this is reserved for those in whom balloon valvotomy and surgical valvuloplasty has failed, or where intervention has caused significant aortic incompetence. Where there is residual significant stenosis there may be an inadequate aortic annulus, and valve replacement (📖 p378) may be combined with root enlargement (📖 p392 and p635). Valve substitutes in small children are not available, thus making the ***Ross operation*** (📖 p388) the procedure of choice. In older children and young adults, usually a ***mechanical valve*** is the appropriate choice, due to the rapid degeneration of tissue valves in the young.

Supravalvar stenosis

- Hypothermic aorto-bicaval CPB and cardioplegic arrest.
- Incise across the localized stenosis into the non-coronary sinus and enlarge incision with a patch.
- ***Brom aortoplasty***: transect ascending aorta, excise the whole ridge, enlarge all three sinuses with patches.
- If supravalvar stenosis is diffuse and extending into head and neck vessels, extensive reconstruction may be warranted, using DHCA, and reconstructing the aorta and arch with homograft, bovine pericardium or Gore-tex.

Subvalvar stenosis

- Hypothermic aortocaval CPB and cardioplegic arrest.
- Antegrade cardioplegia into root, or if AI, directly into coronaries.
- Oblique aortotomy into non-coronary sinus.
- RCC retracted anteriorly to expose subvalvular area.
- Fibrous shelf resected with knife, or peeled off with help of Watson–Cheyne dissector.
- Peel off the underside of the valve leaflets, release the fibrous trigones.
- Avoid area of membranous septum, below commissure between non-coronary and right coronary cusps.
- If necessary perform septal myotomy/myectomy—from below the commissure between RCC and LCC to below the body of the RCC.

Small aortic roots and tunnel-like LVOT obstruction

Where there is a small aortic root and diffuse narrowing of the LVOT, the outflow tract can be enlarged prior to inserting an aortic valve substitute:

Nick's aortic root enlargement (📖 p392)

Oblique aortotomy incision into the non-coronary sinus is extended across the aortic annulus, and a patch sutured into this defect to enlarge the root.

Manougian aortic root enlargement (📖 p393)

The aortotomy is directed towards the commissure between the LCC and NCC, and extended across the fibrous continuity between the aortic and mitral valves—patch inserted.

Konno aortic root enlargement

The aortotomy is made longitudinally towards the commissure between the RCC and the LCC, avoiding the RCA. A transverse incision is made proximal to the pulmonary valve in the RVOT; then with scissors, one blade in the LV the other in the RV, the interventricular septum is opened. The aortic valve is removed, then a patch used to reconstruct the septum to the desired width. An aortic valve substitute is then inserted. This method of enlargement may be easily performed during a Ross operation, the ***Ross–Konno operation***.

Extracardiac valve conduit

Extracardiac valved conduit from LV apex to ascending, descending or abdominal aorta have been used to bypass very narrow tunnel-like LVOTs in the past.

Outcomes

Valvular stenosis

- Neonatal aortic valve stenosis: mortality for balloon aortic valvotomy is ~10%, with risk of significant AI ~15%. The mortality risks are similar with surgical valvotomy, with less incidence of AI. 25–40% need reintervention by 5 years.
- Early degeneration/calcification/structural failure of heterograft substitutes have lead to discontinuation of their use in children. Homografts are slightly better, with 10-year freedom from reoperation about 40–50%, and are also not commonly used.
- Mechanical valves have low postoperative mortality, 2–3%, and good long-term outcomes, but in smaller children may have to be up-sized. At 20 years, freedom from thromboembolism is ~93%, from reoperation ~87%.
- The early mortality from the Ross operation is <2%, freedom from reoperation of 90% at 7 years.

Subaortic stenosis

Low early mortality, but recurrence rates up to 35% have been reported.

Supravalvar stenosis

Early mortality is up to 10%, with recurrence rates up to 25% reported, with no differences between the surgical techniques.

Ebstein anomaly

Key facts

- This rare malformation accounts for <1% of congenital cardiac anomalies, and is characterized by abnormalities of the tricuspid valve and RV.
- 10–20% have Wolff–Parkinson–White (WPW) syndrome, 25–30% have accessory pathways on EPS, and up to 25% have pre-excitation on the surface EKG.

Morphology

Very variable. Posterior and septal tricuspid leaflets are displaced inferiorly into the RV with associated RV hypoplasia and 'atrialization,' tricuspid incompetence, and ASD. The displaced leaflets are usually hypoplastic and tethered. The functional RV lies beyond the apically displaced coaptation point, and proximal to this the RV is 'atrialized.' The anterior leaflet is a '***sail-like***' structure (Fig. 12.22), and tethering of this leaflet can cause RVOT obstruction. Lungs are hypoplastic in severe cases. WPW syndrome is present in 14%.

Pathophysiology

Pathophysiology depends on the severity of the RV hypoplasia, tricuspid valve abnormalities, and WPW syndrome. At birth the PVR is close to systemic pressures and as a result of tricuspid regurgitation and functionally small TRV, right-to-left shunting occurs resulting in cyanosis. The PVR falls after birth. If the RV is adequate then shunting and cyanosis reduce. If TR and RV pressures are high, cyanosis persists.

Presentation

Depending on the severity of the anomalies, patients may present with any of the following symptoms and signs at any stage of childhood:

- Cyanosis.
- Cardiomegaly and congestive cardiac failure.
- Malar flush, systolic murmur along LLSE, diastolic murmur.
- Hepatomegaly, ascites and peripheral edema.
- Paroxysmal supraventricular and ventricular tachyarrhythmias.
- Paradoxical embolus, cerebral abscess, sudden death.

Investigations

- EKG shows right bundle branch block, RA hypertrophy, and type B WPW in 14%.
- CXR shows marked cardiomegaly.
- Echocardiography is diagnostic and shows apical displacement of the septal leaflet >8mm/m^2, abnormalities of the leaflets and their septal attachments, dilated RA, dilated RV with reduced contractile function, delayed closure of the tricuspid valve (>65ms after MV) paradoxical septal motion: echo is also used to assess functional RV size (<35% of total RV size is poor prognostic indicator), degree of leaflet displacement, tethering, deformity, severity of TR and RVOT.
- Cardiac catheterization is sometimes used to quantify shunts and delineate anatomy.

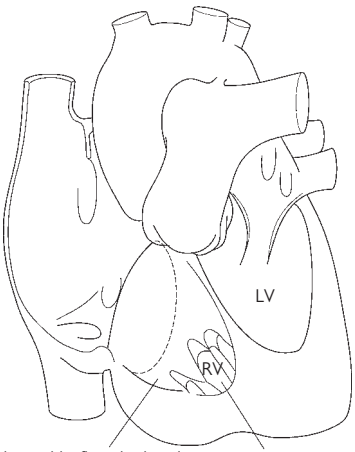


Fig. 12.22 Ebstein anomaly.

Surgical management

Neonates

Just over a half of neonates will need surgical intervention. Neonates dependent on PGE_1 require early intervention. If stable with good SaO_2 and no CHF, a systemic–pulmonary shunt may be adequate palliation. If on PGE_1, oxygenation is adequate but CHF is present, then tricuspid repair or replacement is required if the RV is adequate, or tricuspid valve closure (right ventricular exclusion) and B-T shunt if not. In severe cases, transplant may be the only option (📖 pp504–519).

Older children and adults

May become symptomatic with onset of arrhythmias. Subtle decline in exercise tolerance must be anticipated with regular exercise testing. Tricuspid repair for all but the most severe forms should be attempted; surgical outcome is dependent on the size and function of the right RV. Techniques include: detachment of the anterior and some of the inferior leaflet to allow mobilization of the tethered leaflet tissue, plication of the atrialized chamber and right atrium, annuloplasty (with a ring if adequate siize), repositioning of the anterior and inferior leaflets at the repaired annulus. If repair fails, replacement with a biological prosthesis has acceptable results. If the RV of inadequate size or function, tricuspid repair may be combined with a cavopulmonary shunt to ensure off-loading of the right heart. In severe RV hypoplasia, total cavopulmonary connection may have to be considered

Outcomes

In neonates, survival 1 year after surgery is about 65%. In older patients, postoperative mortality is about 10%.

Congenital mitral anomalies

Key facts

- Congenital anomalies of the mitral valve are rare, and may result in stenosis or regurgitation.
- Mitral valve reconstruction is favoured wherever possible over replacement, to the extent that suboptimal repairs may be accepted as a temporizing measure until the infant is big enough to accept an adequately sized prosthesis which is usually mechanical.

Mitral valve stenosis

Anatomy

- Normal papillary muscles:
  - Commissural fusion with short chordae.
  - Excess valvar tissue and double mitral orifice.
  - Supravalvar ring.
  - Small annulus.
- Abnormal papillary muscles:
  - Parachute mitral valve.
  - Absent papillary muscles.
  - Hammock valve.

Pathophysiology and presentation

Results in pulmonary venous hypertension and presents with features of pulmonary hypertension, e.g., RV heave, prominent S2, mid-diastolic flow murmur.

- EKG: LA enlargement and RV hypertrophy.
- CXR: enlarged LA, increased pulmonary vascular markings.
- Echo: this is diagnostic and some time and effort should be spent in analysing the mechanism of valve dysfunction. Associated lesions must be sought.

Management

Surgery is best not undertaken in the first 6 months of age unless absolutely necessary. However, severe pulmonary hypertension may necessitate early surgery, which is high risk. Techniques employed include papillary muscle splitting, commissurotomies, resection of secondary chordae, chordal fenestration, resection of supravalvar fibrous rings. Mitral valve replacement can often be avoided.

Mitral valve regurgitation

Anatomy

This classified according to Carpentier's classification (📖 p400):

- Type I (normal leaflet motion):
  - Annular dilatation and deformation.
  - Cleft leaflet.
  - Leaflet defect.
- Type II (excessive leaflet motion):
  - Elongated chordae.
  - Elongated papillary muscle.
  - Absent chordae.

- Type III restricted leaflet motion:
  - Normal papillary muscles: commissural fusion or short chords.
  - Abnormal papillary muscles: parachute or hammock mitral valve.

Pathophysiology

Results in progressive pulmonary venous hypertension, LV volume overload, and eventually CHF (📖 p401).

Management

Techniques to ***repair*** these defects include the use of annular rings, triangular resections, annular plication, commissuroplasties, leaflet extension, suture closure of clefts, chordal transposition, chordal shortening, papillary muscle sliding plasty, papillary muscle plication.

If ***mitral valve replacement*** is required, this usually necessitates the use of mechanical valves in small sizes, with significant morbidity in children aged <6. Outcomes are better when the prosthesis is implanted in the anatomical position, rather than in the supra-annular position, and when there is less mismatch in size between the native annulus and the replacement. A strategy of palliative mitral valve repairs accepting a suboptimal result until the patient is big enough to undergo replacement with a reasonable sized prosthesis may have to be considered.

Outcomes

Mortality for surgery for true congenital mitral valve anomalies is 5% for mitral regurgitation and 25–30% for mitral stenosis, with 10-year survival in the order of 50–60%. Mortality of mechanical valve replacement in children <2 years old is >30–40%.

ALCAPA

Key facts

- Anomalous origin of the left coronary artery from the pulmonary artery (ALCAPA) is rare condition in which the whole or one branch of the left main coronary artery (LMCA) arises from the main pulmonary artery, or occasionally from the right pulmonary artery.
- It leads to early symptoms in infancy, with heart failure although it can present for the first time in adulthood.
- Coronary artery reimplantation is the procedure of choice.

Morphology

The left coronary artery usually arises from the left-sided sinus of the main pulmonary artery, then assumes the normal course of the LMCA.

Pathophysiology

In the neonate with high PVR, the pressure within the coronary artery is adequate for coronary perfusion. As PVR falls, perfusion pressure within the coronaries falls as well, and myocardial oxygenation depends on collaterals. Thus during increased myocardial oxygen demand, the low-pressure, low-oxygen blood supply may lead to anterolateral LV free wall ischemia and ***infarction***—the latter causes mitral valve papillary muscle dysfunction and ***MR***.

Presentation

Symptoms may begin soon after birth or within the first 3 months as the PVR falls. There is heart failure with dyspnea; poor feeding, failure to thrive, sweating, and tachycardia are common. 85% of patients present within the first 2 months of life. When symptoms are delayed, presentations may be in later years, and adult patients may remain asymptomatic for many years. Sudden cardiac death may be the mode of presentation at any stage.

Investigation

- EKG may show Q-waves and ST elevation in the anterolateral leads.
- CXR: will show cardiomegaly.
- Echo shows a dilated and poorly contracting LV, with mitral regurgitation, and the appearances must be distinguished from dilated cardiomyopathy. The anomalous coronary artery should be demonstrable on echo, with colour-flow Doppler showing retrograde flow from the coronary into the PA.
- Cardiac catheterization is unnecessary in infancy, but show the anomalous origin, a step up of oxygen saturations in the PA, and only the RCA arising from the aorta.

Surgical management

Surgery is indicated in the infant as they are at risk of sudden cardiac death. Infants in heart failure with diffuse myocardial fibrosis and severe mitral regurgitation may require transplantation. In all other patients, surgery should be undertaken to provide a coronary supply arising from the systemic circulation.

Mitral regurgitation improves after coronary artery reimplantation, but may require further intervention if it remains significant. Usually, mitral valve repair is not performed at the first operation. Surgical techniques include ligation of LMCA if there is a good collateral circulation, subclavian-coronary anastomosis, CABG, intrapulmonary tunnel by creating an AP window and baffling the coronary through the MPA to the aorta. The operation of choice currently is coronary artery reimplantation.

Coronary artery reimplantation

- Standard aortocaval bypass via a median sternotomy.
- Snare the PAs to prevent 'steal' from the LCA.
- Cardioplegia into the aortic ***and*** pulmonary roots, and LCA origin once the MPA opened.
- Button of PA around the LMCA ostium excised, mobilized, anastomosed to aortic root.
- If far from aorta, posterior PA wall harvested with the coronary, and this is used as a tube to extend the LCA towards the aorta.
- A minority of patients require postoperative mechanical support.

Outcomes

Early postoperative mortality is in the order of 5–10%. Late deaths are rare since the adoption of coronary artery reimplantation. Late patency of the reimplanted coronary artery has been shown to be excellent.

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